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DSGE Models in Systematic Trading Strategies

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Supervisor declaration

I declare that this Master's thesis has been prepared under my guidance and that it meets the criteria for submission with the goal of obtaining a Master's degree certificate.

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Abstract

DSGE modelling and systematic trading strategies have long been explored separately. This paper creates a new framework for applying DSGE modelling in the context of low-frequency systematic trading strategies. DSGE models are calibrated to a particular economy and can generate relatively long-term impulse response functions of economic variables. Comparing such forecasts with equivalent realised values in a novel way provides grounds for signal generation, the result of which can be applied to assets associated with the forecast variable. The framework is run on US equity indices and produces healthy performance. It also provides insights into the link between macroeconomic forecasts and financial assets.

Keywords

DSGE, systematic, trading, strategy, momentum, mean-reversion

(DSGE, systematyczna, strategia, trading, momentum, mean-reversion)

Field of work (Erasmus code)

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Thematic Classification

Tytuł pracy w języku polskim

Modele DSGE w zastosowaniu w systematycznych strategiach tradingowych

CONTENTS

1. INTRODUCTION

The search for trading strategies is extensive and spans multiple asset classes. Both discretionary and systematic strategies have been researched, tested and implemented by various practitioners and professionals associated with the field of finance. Within the early days of the systematic category, simple investment decisions otherwise performed by analysts and traders were automated (Chan, 2013; Narang, 2013). This gradually became more sophisticated and soon took advantage of increases in computing power and reduced latency to target market phenomena at high frequency (Aldridge, 2013). Such strategies generally attempted to exploit market microstructure nuances often at tick level (Hasbrouck, 2007; O'Hara, 1995). Systematic strategies have become more complex over time, most recently taking advantage of ever-increasing volumes of data and further leaps in computing ability that can help forecast asset price changes, often departing from economic theory in favour of raw statistical accuracy (Gu, Kelly, & Xiu, 2020; López de Prado, 2018).

In a separate world, modern macroeconomics has largely been shaped by a bottom-up style of modelling economic interactions between agents in the economy, characterised as Dynamic Stochastic General Equilibrium (DSGE) modelling (Kydland & Prescott, 1982; Woodford, 2003; Galí, 2015). In one of its basic forms in the Real Business Cycle (RBC) model, households interact with firms. A large number of households seek to maximise their lifetime utility by balancing consumption and leisure. They supply labour and save by investing in capital, which they then rent to firms. Firms maximise profit. They hire labour and rent capital to produce goods based on the available technology and prevailing competitive wages and interest rates (King, Plosser, & Rebelo, 1988). This relationship has its equilibrium and can be affected by stochastic shocks, for example to total factor productivity (TFP) (Kydland & Prescott, 1982). The modelling framework allows for an elegant depiction of the interaction between entities in the economy and to quantify the impact of macroeconomic shocks. Furthermore, DSGE models can be calibrated to suit the characteristics of particular economies (Smets & Wouters, An Estimated Dynamic Stochastic General Equilibrium Model of the Euro Area, 2003). As is broadly the case in economics, its use case is primarily for inference with the added benefit of forecasting impulse response functions (IRF) of macroeconomic variables under aforementioned shocks (Christiano, Eichenbaum, & Evans, 2005).

On the face of it, it might appear that systematic trading strategies and DSGE models have little overlap, as systematic strategies generally rely on explicit forecasts of asset returns in the hope of generating a signal and opening a position, whilst DSGE models explain the broader economy and are useful for drawing inferences. These models operate at a relatively low frequency in line with the rarer evolution and publishing of economic data which stems from only being measured at such frequency (often quarterly). Nevertheless, the fact that DSGE models offer the ability to forecast macroeconomic variables means that there exist datapoints to which realised values can be compared with the potential of triggering a signal. What remains on the DSGE modelling side is determining the magnitude of shocks, for which estimation techniques can be used, as well as calibrating parameters that resemble a particular economy. Appropriate assets must then be linked to the modelled macroeconomic variables for any signal to hold robustly.

Each of these assumptions and interactions can be explored further. **The intention of this paper is to lay the foundation of a new type of low-frequency systematic trading strategy based on DSGE modelling.** It offers an alternative to established long-term strategies, such as buy-and-hold.

It also does not have major barriers to entry. Coupling low dependence on data and being programmed in Python in a specific environment, it ensures reproducibility, short implementation time and ease of modification.

This framework offers further insights as a byproduct: it assesses DSGE model effectiveness by comparing IRF forecasts with realised macroeconomic values in the signal-generation phase and checks the implicit correlation between lagged asset prices and macroeconomic variables via performance of the strategies over time.

2. LITERATURE REVIEW

The development of systematic trading strategies has closely mirrored advances in technology, data availability and quantitative methods. Early systematic approaches largely automated pre-existing discretionary investment rules, allowing traders and institutions to remove emotional biases and improve execution consistency (Chan, 2013; Narang, 2013). These strategies often relied on relatively simple signals derived from technical indicators, statistical arbitrage or trend-following methodologies. Improvements in computing infrastructure and electronic market access gradually enabled higher-frequency implementation, particularly from the late 1990s onwards. This led to the growth of algorithmic and high-frequency trading strategies seeking to exploit temporary inefficiencies in market microstructure, order flow and liquidity provision (Aldridge, 2013; Hasbrouck, 2007; O'Hara, 1995).

The increasing sophistication of systematic trading has subsequently been characterised by a movement away from explicitly theory-driven models and towards empirical optimisation. Modern quantitative investment frameworks frequently rely on large datasets, machine learning techniques and statistical pattern recognition to forecast returns or classify market states (Gu, Kelly, & Xiu, 2020; López de Prado, 2018). Such approaches often prioritise predictive performance over economic interpretability, reflecting a broader trend in finance toward data-centric modelling. This has been facilitated by the expansion of alternative datasets and advances in computational power. However, despite their forecasting capabilities, many machine learning-based strategies remain vulnerable to overfitting, regime dependence and a lack of structural interpretability, particularly during periods of macroeconomic instability.

In parallel, macroeconomic modelling has evolved along a largely separate trajectory. Modern DSGE models emerged following weaknesses of aggregate metrics in macroeconomics. These models sought to explain aggregate economic fluctuations through optimising behaviour of representative agents under rational expectations and stochastic shocks. RBC models demonstrated that relatively simple economies subject to productivity shocks could reproduce several stylised features of business cycles (Kydland & Prescott, 1982; King, Plosser, & Rebelo, 1988). However, the inability of early RBC models to adequately explain nominal rigidities and monetary transmission mechanisms motivated the development of New Keynesian DSGE frameworks.

Subsequent DSGE models incorporated price stickiness, wage rigidities, monopolistic competition and monetary policy rules, substantially improving empirical performance and policy applicability (Woodford, 2003; Galí, 2015). Estimated DSGE models, particularly those of Smets & Wouters, 2003 and Smets & Wouters, 2007, became central tools within central banks and academic macroeconomics due to their ability to jointly model macroeconomic variables while maintaining structural consistency. Such models offered an internally coherent framework through which macroeconomic shocks could be identified, quantified and propagated throughout the economy. Their widespread use in monetary policy analysis reinforced DSGE modelling as a dominant paradigm in modern macroeconomics.

Despite their prominence in economics, DSGE models have rarely been integrated directly into systematic trading strategies. The majority of macro-driven systematic investment approaches instead rely on reduced-form econometric models, factor investing frameworks or purely statistical forecasting techniques. Traditional macro trading strategies often exploit relationships between interest rates, inflation, growth and asset prices using discretionary interpretation or econometric forecasting rather than structurally derived equilibrium models. Similarly, quantitative asset pricing literature has historically focused on factor models such as the Capital Asset Pricing Model (CAPM) (Sharpe, 1964) and Fama-French extensions (Fama & French, 1993; Fama & French, 2015), which explain expected returns through empirically observed risk premia rather than fully specified macroeconomic equilibria.

There exists, however, a significant body of literature linking macroeconomic conditions to financial market performance. Research has demonstrated that equity returns, bond yields and foreign exchange dynamics respond systematically to inflation surprises, monetary policy shocks and broader business cycle fluctuations (Chen, Roll, & Ross, 1986; Cochrane, 2005). Furthermore, DSGE models have increasingly been used for forecasting macroeconomic variables and analysing policy transmission, particularly through IRFs (Christiano, Eichenbaum, & Evans, 2005). This suggests the possibility that DSGE-implied forecasts or deviations between forecast and realised macroeconomic outcomes may contain exploitable information relevant to asset pricing.

Nevertheless, relatively little literature explores the use of DSGE-generated forecasts as direct trading signals within systematic trading frameworks. Where DSGE models are applied in finance, they are more commonly used for valuation, stress testing, scenario analysis or academic asset pricing applications rather than executable systematic strategies. Existing barriers likely stem from several factors. Firstly, DSGE models operate at comparatively low frequency due to the periodicity of macroeconomic data. Secondly, calibration and estimation procedures are computationally intensive and sensitive to modelling assumptions. Thirdly, the translation of macroeconomic variables into tradable asset signals is non-trivial and may vary substantially across economies and market regimes.

This gap in the literature motivates the present study. Whilst systematic trading strategies have increasingly prioritised predictive accuracy without economic structure, DSGE models provide economically interpretable forecasts grounded in equilibrium relationships between agents in the economy. Combining these domains offers the possibility of constructing low-frequency systematic strategies that retain macroeconomic interpretability while producing investment signals upon which to trade. Furthermore, such an approach enables simultaneous evaluation of DSGE model forecasting performance and the strength of relationships between macroeconomic dynamics and financial asset returns.

The contribution of this paper therefore lies not only in proposing a novel systematic trading framework based on DSGE modelling, but also in bridging two largely disconnected strands of literature: structurally grounded macroeconomic modelling and systematic quantitative trading.

3. DATA

The DSGE trading framework in this paper is centred around the US economy and its assets due to availability of data and the prevalence of this country as a benchmark in research. The components of the framework fall into three categories in terms of data: the DSGE model, realised macroeconomic data and assets used in systematic trading strategies.

The DSGE model used in this paper is a basic RBC model. Whilst the RBC model is largely instantiated with preprepared parameters calibrated to the US economy, it requires a variable component over time which contributes towards signal generation. This is achieved by estimating an autoregressive process of order 1 (AR1) using TFP data. *Nonfarm Business Sector: Labor Productivity (Output per Hour) for All Workers* in the US (OPHNFB) from the Federal Reserve Bank of St Louis (FRED) is a quarterly seasonally adjusted metric used as TFP.

IRF forecasts of output are one of the results of running the RBC model. These are compared to the realised equivalent variable, which is quarterly seasonally adjusted *Real Gross Domestic Product* in the US (GDPC1) from FRED.

Finally, the assets sourced are the daily *S&P 500* index (^GSPC) and the daily *Dow Jones Industrial Average* index (^DJI) from Yahoo Finance, two indices composed of US equities and that are largely reflective of the US economy. In both cases, close prices are used.

FRED and Yahoo Finance data are pulled using the Fred class from the fredapi library and the yfinance library in Python, respectively. The limiting factor in terms of time horizons of the data is GDPC1 which is limited to dates after 1989-12-29 in the API, which still ensures a plethora of observations to test multiple strategy variations. All data is pulled once ahead of running any model and is instantiated using an independent data class that spans the maximum possible training and test periods. This prevents spillover between data and its handling, plus avoids having to make multiple slow connections to retrieve data that could be subject to throttling.

There is minimal data cleaning required due to the low frequency and the scrutiny applied by reputable institutions ahead of publishing. Null values are dropped and series indices are formatted to dates. US holidays as per the USFederalHolidayCalendar class in the pandas library are sourced to offset dates to nearest trading days during signal generation.

The steps performed in this section can be observed in Appendix B: DSGE Strategy Code.

3.1. Reproducibility

The design of this framework has kept ease of implementation and reproducibility at its core. The entirety of the framework has been built using Python, including data sourcing for which dedicated APIs are used.

Running the main framework script in a virtual environment with the required libraries installed is sufficient for reproducing the results in this paper. Full instructions are outlined in Appendix A: Framework Setup and in the DSGE strategy repository (Foster, 2026).

4. METHODOLOGY

The steps performed in this section can be observed in Appendix B: DSGE Strategy Code.

Signals in this strategy are fundamentally dependent on the divergence between RBC model forecasts of output and realised output. Producing forecasts requires running a particular RBC model to begin with.

4.1. RBC Model

A basic RBC model is used: it depicts the interaction between households and firms at a foundational level and is available for execution in the dolo Python library. It comes from EconForge (Winant, rbc_model.ipynb, 2019) and is defined in YAML code as per Appendix C: Basic RBC Model YAML Code. Dolo is a tool to assist researchers in solving several types of DSGE models using either local of global approximation methods (Winant, Welcome to dolo's documentation!, 2026).

In this model, households maximise expected lifetime utility as per Equation 1: Household objective:

Equation 1: Household objective

$$\max_{c_t,n_t} E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, n_t)$$

where β is the discount factor, U is utility, c is consumption, n is labour and t is time.

Firms maximise profit Π as per Equation 2: Firm objective:

Equation 2: Firm objective

$$\max_{k_t,n_t} \Pi_t = y_t - w_t n_t - r_t^k k_{t-1}$$

where y is output, w is wage, r is rental rate on capital, k is capital.

These are subject to the constraints outlined in Equation 3: Budget constraint, Equation 4: Production function, Equation 5: Capital accumulation constraint and Equation 6: Exogenous TFP shock process.

Equation 3: Budget constraint

$$c_t + i_t = y_t$$

Equation 4: Production function

$$y_t = e^{z_t} k_t^{\alpha} n_t^{1-\alpha}$$

Equation 5: Capital accumulation constraint

$$k_{t+1} = (1-\delta)k_t + i_t$$

Equation 6: Exogenous TFP shock process

$$z_{t+1} = \rho z_t + e_{z,t+1}$$

where i is investment, z is TFP, α is elasticity of capital, δ is depreciation of capital, ρ is persistence of the TFP shock.

Resulting equilibrium conditions and pre-calibrated parameters are coded in Appendix C: Basic RBC Model YAML Code.

For the framework to retain trading applicability, time series data entering the model remains sequential and is used for next-period forecasting. Iterating this approach along a moving window continually fine-tunes model results and applies the latest shock magnitude. This also enables strategy backtesting.

Various training lengths are employed, whereby OPHNFB productivity data is sliced in preparation for the model. This data is converted to natural logarithm form for percentage deviations and undergoes HP filtering assuming a lambda value of 1600 for quarterly data to extract the cyclical component of the time series.

The modified time series z_t is regressed against its lagged values as per the AR1 process in Equation 6: Exogenous TFP shock process using the OLS estimator. The resulting coefficient is set as the calibrated TFP persistence parameter ρ in the model. For the model to be solved quickly and given the basic RBC model type with small shocks, a local perturbation solution with a first-order Taylor approximation of the policy functions is used. An IRF is then run, assuming the latest TFP shock value $e_{z,t}$ from the residuals in the AR1 estimation.

The goal is to obtain a next-period GDP change forecast. IRF forecast values are multiplicative deviations from the normalised steady state level of 1 (as per Figure 1: Output IRF assuming rho=0.771 and e=0.0054) and are therefore converted to percentage deviations. The output IRF array is assumed to correspond to GDP. The response at the second horizon period is recorded as the output forecast for this run of the model: the initial response corresponds to the impact period, the response at the first horizon period reflects only the immediate labour and consumption reallocation, whilst the response at the second horizon period acknowledges capital adjustment and reflects the intended impact direction of a productivity shock. It should reflect the timing of quarterly national accounts.

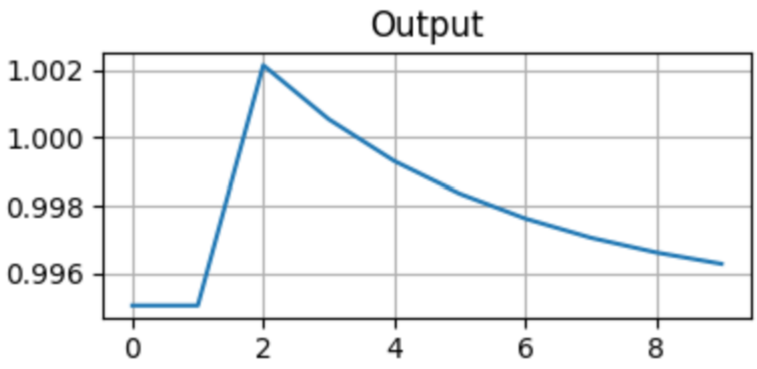

Figure 1: Output IRF assuming rho=0.771 and e=0.0054

The process is repeated for data windows shifted quarterly within an overall horizon of the strategy backtest, as illustrated in Figure 2: Signal generation illustration. The result is a time series of next-period output forecasts.

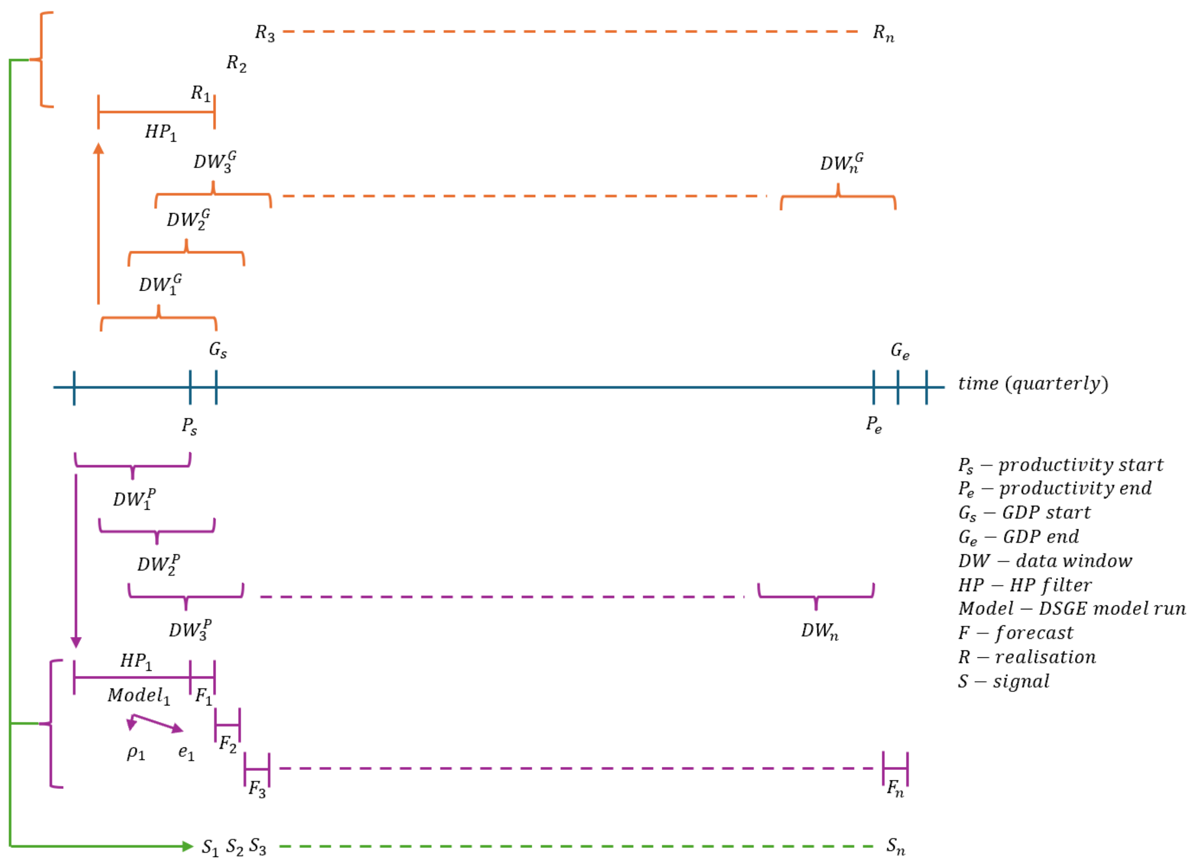

Figure 2: Signal generation illustration

4.2. Realised GDP

Realised values of GDP are needed in addition to output forecasts for subsequent signal generation. GDPC1 data undergoes HP filtering on natural logarithms to ensure a like-for-like comparison with output forecasts. It is therefore initially sliced similarly to OPHNFB data, aligning data lengths but setting the end date one quarter later than OPHNFB corresponding to the date for which the next-period forecast is made. The last element of the array is selected as the realised output value accordingly.

The process is repeated for data windows shifted quarterly, forming a time series of realised output values. This is illustrated in Figure 2: Signal generation illustration.

4.3. Signal Generation

Upon joining output forecasts with realised output, signals can be generated. Two types of signals are implemented: momentum and mean-reversion.

The momentum signal takes a long position when output forecast is greater than realised output, a short position when output forecast is less than realised output and no position otherwise. The intuition in this scenario is that the DSGE model anticipates higher output growth than had been observed, thus a reference asset could be expected to increase in value as output catches up to close the gap.

Conversely, the mean-reversion signal takes the inverse of the momentum position with a view that new information driving divergence in realised output from the forecast implies a pending correction in equity markets and an expectation for them to price in this new information.

Ahead of signal implementation, asset data is prepared for trading. ^GSPC and ^DJI are used in this framework. The two indices are composed of US equities and are largely reflective of the US economy and are therefore natural candidates for the strategy. The daily series of each are restricted to the aforementioned strategy horizon, whereby the last observation ends one quarter after the last signal becomes available. This is due to a period of time required after trading a given signal for the strategy to generate profit and loss (PnL), plus the quarterly frequency of signal changes is in line with the macroeconomic data supplying the model. The series is then converted to percentage returns.

The final modification of the signal series is the crucial delay in its enforcement. In practice, macroeconomic data is released with a lag. Therefore, various lag times are tested in this strategy for the implementation to be feasible and the results to be realistic. For example, a lag of 20 allows 20 business days for GDPC1 data to be released and 20 business days plus one month for OPHNFB data to be released. Signals are further lagged to the nearest trading day if needed for trade execution to be possible.

At this stage signals are joined onto the processed asset data and filled forwards. This effectively sets the position once a quarter and keeps it until the next position is established as shown in Figure 3: Positioning assuming lag=10 over one quarter. The frequency makes it a low-frequency systematic trading strategy with clear room for the assessment of whether the economic phenomena behind the momentum and mean-reversion signals hold.

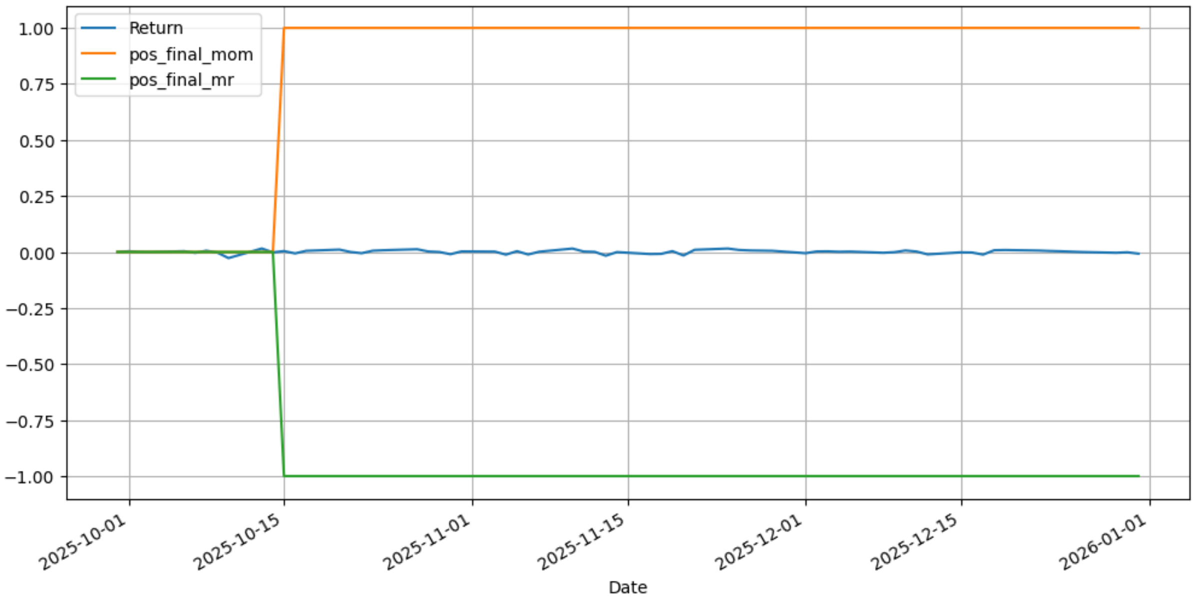

Figure 3: Positioning assuming lag=10 over one quarter

4.4. Extensions and Variations

The strategy is extended beyond the aforementioned setup to potentially improve performance.

Firstly, there exists a doubling down feature. This sets a signal scalar of 2 if two consecutive signals are the same at the quarterly signal generation phase, rewarding signal persistence in each of the momentum and mean-reversion approaches. For example, if output forecasts exceed realised output for two quarters, this doubles the momentum position.

Secondly, leverage can be employed. This is a multiplicative scalar passed down directly to the raw signal.

Coupling these extensions with variations mentioned throughout this section, there are numerous combinations of parameters in this strategy. These are summarised in Table 1.

Table 1: Strategy hyperparameters

Hyperparameter	**Values**
Model type	Basic RBC
Data length (years for model training)	5, 10, 20
Horizon	12/1999 – 9/2025 12/1999 – 12/2007[1]

[1] 20-year data lengths are not available for these horizons, as the earliest observation available is as of 12/1989

	12/2007 – 12/2009 12/2009 – 12/2019 12/2019 – 12/2022 12/2022 – 9/2025
Signal lag (days after quarter-end)	10, 20, 30
Leverage	1, 2
Double down	False, True
Asset	S&P 500, Dow Jones Industrial Average
Strategy type	Momentum, mean-reversion

4.5. Costs

Trading costs are considered for a fair and realistic assessment of the performance of the strategy.

Equation 7: Trading cost

$$c_t = \frac{\kappa}{2}|p_t - p_{t-1}|$$

In line with the literature (Grinold & Kahn, 2000) and as per Equation 7: Trading cost, cost at time t, c_t, is the cost of one full portfolio turnover, κ, scaled by the change in position, p, between time periods. The approach therefore generates a non-zero cost only when a position changes and a trade is required to achieve this. This is further scaled by $\frac{1}{2}$ to avoid double counting trades, for example one unit is charged when changing from a unit long position to a unit short position. The scaling continues to work for larger or levered positions, whose cost grows proportionally to position magnitude.

Throughout the process κ is assumed to be a very low value around 1bp: large-cap US equities have extremely tight spreads and deep order books and their equity indices that are used in this paper further benefit from significant liquidity and brokerage competition (Frazzini, Israel, & Moskowitz, 2014).

Trading costs are expected to have a minimal impact regardless, due to the anticipated infrequent trading activity over the time horizons considered stemming from quarterly signals. This is an advantage of the DSGE strategy that few systematic strategies can afford.

4.6. PnL Calculation and Performance Metrics

Gross PnL is calculated daily after incorporating the asset return of the current day and the position of the previous day.

Net PnL is calculated by subtracting cost, c_t, from gross PnL every day.

In both cases, PnL reflects the gain or loss of a unit position in a given day. It does not consider the effect of compounding an investment were to have if it were placed at a given point in time and remained in place, reinvesting proceeds with every time period. It is therefore a cumulative arithmetic PnL which scrutinises returns every period and displays lower PnL values than those that would actually be achieved via compounding. For example, a value of 2 indicates a 200% return. This has the additional benefits of comparing signals and strategies more effectively, computing Sharpe ratios and visualising whether a strategy adds value over time.

Sum, mean and standard deviation are calculated across the strategy horizon for both gross PnL and net PnL. In addition, two corresponding variants of the Sharpe ratio (SR) are calculated for a risk-adjusted view of the returns of the strategy. Daily SR is converted to annualised SR, assuming 252 trading days per year as in Equation 8: Annual Sharpe ratio.

Equation 8: Annual Sharpe ratio

$$SR_{annual} = SR_{daily} \times \sqrt{252}$$

4.7. Buy-and-Hold Benchmark

As the DSGE model-inspired strategy is of low frequency, pertains to equity markets and is expected to have a simple trading profile, a buy-and-hold strategy was selected as the benchmark strategy. This is also a well-established benchmark in the literature (Malkiel, 2019).

The benchmark is set up in alignment with the DSGE model strategy in terms of assets, data, and performance metrics.

5. RESULTS

The framework runs seamlessly and produces a wide range of results. Performance varies by each of the hyperparameters with a general skew towards mean-reversion, longer signal lags and leverage for improved results.

PnL and annualised SR are the key metrics used to assess performance of the strategy. Of all horizons, the longest one spanning 12/1999 – 9/2025 is most useful in determining the long-term feasibility of challenging the buy-and-hold portfolio whose position does not change by design and favours long-term holding periods. It should be noted that whilst both gross and net performance metrics are available, more attention is devoted to net PnL and net SR. This is for real-world applicability and due to the their alignment with gross equivalents in this framework in line with very few position changes over the horizon. The mean holding period in the unlevered 10 year-calibrated 30 day-signal lagged 12/1999-09/2025 horizon (full horizon) mean-reversion strategy is 3.15 quarters.

The labelling of strategies throughout this section follows this convention:

Strategy example: rbc_10cal_1999-12-31_2025-09-30_30lag_2lev_dd_mrn

- Model: *rbc*
- Data length (years): *10cal*
- Horizon (date range): *1999-12-31_2025-09-30*
- Signal lag (business days): *30lag*
- Leverage (2x or none): *2lev*
- Double down (binary): *dd*
- Strategy type (momentum or mean-reversion): *mr*
- Gross vs net: *n*

Table 2: Results using S&P500 asset between 12/1999-09/2025

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	30b	2	Yes	MR	3.97	3.98	0.24	0.24
RBC	5y	1999-12-31_2025-09-30	30b	2	Yes	MR	3.69	3.7	0.22	0.22
RBC	10y	1999-12-31_2025-09-30	20b	2	Yes	MR	3.2	3.21	0.19	0.19
RBC	5y	1999-12-31_2025-09-30	20b	2	Yes	MR	2.89	2.89	0.17	0.17
RBC	5y	1999-12-31_2025-09-30	10b	2	Yes	MR	2.63	2.64	0.16	0.16
RBC	10y	1999-12-31_2025-09-30	10b	2	Yes	MR	2.47	2.47	0.15	0.15

RBC	10y	1999-12-31_2025-09-30	30b	2	No	MR	2.32	2.33	0.23	0.23
BH	N/A	1999-12-31_2025-09-30	N/A	1	No	N/A	2.03	2.03	0.4	0.4
RBC	10y	1999-12-31_2025-09-30	30b	1	Yes	MR	1.98	1.99	0.24	0.24
RBC	5y	1999-12-31_2025-09-30	30b	2	No	MR	1.96	1.97	0.2	0.2
RBC	5y	1999-12-31_2025-09-30	30b	1	Yes	MR	1.84	1.85	0.22	0.22
RBC	10y	1999-12-31_2025-09-30	20b	2	No	MR	1.84	1.84	0.18	0.18
RBC	10y	1999-12-31_2025-09-30	20b	1	Yes	MR	1.6	1.6	0.19	0.19
RBC	5y	1999-12-31_2025-09-30	20b	1	Yes	MR	1.44	1.45	0.17	0.17
RBC	5y	1999-12-31_2025-09-30	10b	1	Yes	MR	1.32	1.32	0.16	0.16
RBC	5y	1999-12-31_2025-09-30	20b	2	No	MR	1.27	1.28	0.13	0.13
RBC	10y	1999-12-31_2025-09-30	10b	1	Yes	MR	1.23	1.24	0.15	0.15
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.16	1.16	0.23	0.23
RBC	10y	1999-12-31_2025-09-30	10b	2	No	MR	1.01	1.02	0.1	0.1
RBC	5y	1999-12-31_2025-09-30	30b	1	No	MR	0.98	0.98	0.2	0.2
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MR	0.92	0.92	0.18	0.18
RBC	5y	1999-12-31_2025-09-30	10b	2	No	MR	0.74	0.74	0.07	0.07
RBC	5y	1999-12-31_2025-09-30	20b	1	No	MR	0.64	0.64	0.13	0.13
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MR	0.51	0.51	0.1	0.1
RBC	5y	1999-12-31_2025-09-30	10b	1	No	MR	0.37	0.37	0.07	0.07
RBC	5y	1999-12-31_2025-09-30	10b	1	No	MOM	-0.37	-0.37	-0.07	-0.07
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MOM	-0.51	-0.51	-0.1	-0.1
RBC	5y	1999-12-31_2025-09-30	20b	1	No	MOM	-0.64	-0.64	-0.13	-0.13
RBC	5y	1999-12-31_2025-09-30	10b	2	No	MOM	-0.75	-0.74	-0.07	-0.07
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MOM	-0.92	-0.92	-0.18	-0.18
RBC	5y	1999-12-31_2025-09-30	30b	1	No	MOM	-0.99	-0.98	-0.2	-0.2
RBC	10y	1999-12-31_2025-09-30	10b	2	No	MOM	-1.02	-1.02	-0.1	-0.1
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MOM	-1.17	-1.16	-0.23	-0.23
RBC	10y	1999-12-31_2025-09-30	10b	1	Yes	MOM	-1.24	-1.24	-0.15	-0.15
RBC	5y	1999-12-31_2025-09-30	20b	2	No	MOM	-1.29	-1.28	-0.13	-0.13
RBC	5y	1999-12-31_2025-09-30	10b	1	Yes	MOM	-1.32	-1.32	-0.16	-0.16
RBC	5y	1999-12-31_2025-09-30	20b	1	Yes	MOM	-1.45	-1.45	-0.17	-0.17
RBC	10y	1999-12-31_2025-09-30	20b	1	Yes	MOM	-1.61	-1.6	-0.19	-0.19
RBC	10y	1999-12-31_2025-09-30	20b	2	No	MOM	-1.85	-1.84	-0.18	-0.18
RBC	5y	1999-12-31_2025-09-30	30b	1	Yes	MOM	-1.85	-1.85	-0.22	-0.22
RBC	5y	1999-12-31_2025-09-30	30b	2	No	MOM	-1.97	-1.97	-0.2	-0.2
RBC	10y	1999-12-31_2025-09-30	30b	1	Yes	MOM	-1.99	-1.99	-0.24	-0.24
RBC	10y	1999-12-31_2025-09-30	30b	2	No	MOM	-2.33	-2.33	-0.23	-0.23
RBC	10y	1999-12-31_2025-09-30	10b	2	Yes	MOM	-2.48	-2.47	-0.15	-0.15
RBC	5y	1999-12-31_2025-09-30	10b	2	Yes	MOM	-2.64	-2.64	-0.16	-0.16

RBC	5y	1999-12-31_2025-09-30	20b	2	Yes	MOM	-2.9	-2.89	-0.17	-0.17
RBC	10y	1999-12-31_2025-09-30	20b	2	Yes	MOM	-3.21	-3.21	-0.19	-0.19
RBC	5y	1999-12-31_2025-09-30	30b	2	Yes	MOM	-3.7	-3.7	-0.22	-0.22
RBC	10y	1999-12-31_2025-09-30	30b	2	Yes	MOM	-3.98	-3.98	-0.24	-0.24

Table 3: Results using Dow Jones Industrial Average asset between 12/1999-09/2025

Model	**Data Length**	**Horizon**	**Signal Lag**	**Leverage**	**Double Down**	**Strategy Type**	**Net PnL**	**Gross PnL**	**Net SR**	**Gross SR**
RBC	10y	1999-12-31_2025-09-30	30b	2	Yes	MR	3.48	3.48	0.22	0.22
RBC	5y	1999-12-31_2025-09-30	30b	2	Yes	MR	3.32	3.32	0.21	0.21
RBC	10y	1999-12-31_2025-09-30	20b	2	Yes	MR	3	3.01	0.19	0.19
RBC	5y	1999-12-31_2025-09-30	20b	2	Yes	MR	2.81	2.82	0.18	0.18
RBC	5y	1999-12-31_2025-09-30	10b	2	Yes	MR	2.41	2.42	0.15	0.15
RBC	10y	1999-12-31_2025-09-30	10b	2	Yes	MR	2.18	2.19	0.14	0.14
RBC	10y	1999-12-31_2025-09-30	30b	2	No	MR	2.18	2.18	0.23	0.23
RBC	5y	1999-12-31_2025-09-30	30b	2	No	MR	1.88	1.88	0.2	0.2
BH	N/A	1999-12-31_2025-09-30	N/A	1	No	N/A	1.87	1.87	0.39	0.39
RBC	10y	1999-12-31_2025-09-30	20b	2	No	MR	1.87	1.87	0.2	0.2
RBC	10y	1999-12-31_2025-09-30	30b	1	Yes	MR	1.74	1.74	0.22	0.22
RBC	5y	1999-12-31_2025-09-30	30b	1	Yes	MR	1.66	1.66	0.21	0.21
RBC	10y	1999-12-31_2025-09-30	20b	1	Yes	MR	1.5	1.51	0.19	0.19
RBC	5y	1999-12-31_2025-09-30	20b	1	Yes	MR	1.41	1.41	0.18	0.18
RBC	5y	1999-12-31_2025-09-30	20b	2	No	MR	1.34	1.34	0.14	0.14
RBC	5y	1999-12-31_2025-09-30	10b	1	Yes	MR	1.21	1.21	0.15	0.15
RBC	10y	1999-12-31_2025-09-30	10b	1	Yes	MR	1.09	1.09	0.14	0.14
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.09	1.09	0.23	0.23
RBC	10y	1999-12-31_2025-09-30	10b	2	No	MR	1	1	0.1	0.11
RBC	5y	1999-12-31_2025-09-30	30b	1	No	MR	0.94	0.94	0.2	0.2
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MR	0.93	0.94	0.2	0.2
RBC	5y	1999-12-31_2025-09-30	10b	2	No	MR	0.76	0.77	0.08	0.08
RBC	5y	1999-12-31_2025-09-30	20b	1	No	MR	0.67	0.67	0.14	0.14
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MR	0.5	0.5	0.1	0.11
RBC	5y	1999-12-31_2025-09-30	10b	1	No	MR	0.38	0.38	0.08	0.08
RBC	5y	1999-12-31_2025-09-30	10b	1	No	MOM	-0.38	-0.38	-0.08	-0.08
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MOM	-0.5	-0.5	-0.11	-0.11
RBC	5y	1999-12-31_2025-09-30	20b	1	No	MOM	-0.67	-0.67	-0.14	-0.14
RBC	5y	1999-12-31_2025-09-30	10b	2	No	MOM	-0.77	-0.77	-0.08	-0.08
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MOM	-0.94	-0.94	-0.2	-0.2

RBC	5y	1999-12-31_2025-09-30	30b	1	No	MOM	-0.94	-0.94	-0.2	-0.2
RBC	10y	1999-12-31_2025-09-30	10b	2	No	MOM	-1.01	-1	-0.11	-0.11
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MOM	-1.09	-1.09	-0.23	-0.23
RBC	10y	1999-12-31_2025-09-30	10b	1	Yes	MOM	-1.1	-1.09	-0.14	-0.14
RBC	5y	1999-12-31_2025-09-30	10b	1	Yes	MOM	-1.21	-1.21	-0.15	-0.15
RBC	5y	1999-12-31_2025-09-30	20b	2	No	MOM	-1.35	-1.34	-0.14	-0.14
RBC	5y	1999-12-31_2025-09-30	20b	1	Yes	MOM	-1.41	-1.41	-0.18	-0.18
RBC	10y	1999-12-31_2025-09-30	20b	1	Yes	MOM	-1.51	-1.51	-0.19	-0.19
RBC	5y	1999-12-31_2025-09-30	30b	1	Yes	MOM	-1.67	-1.66	-0.21	-0.21
RBC	10y	1999-12-31_2025-09-30	30b	1	Yes	MOM	-1.74	-1.74	-0.22	-0.22
RBC	10y	1999-12-31_2025-09-30	20b	2	No	MOM	-1.88	-1.87	-0.2	-0.2
RBC	5y	1999-12-31_2025-09-30	30b	2	No	MOM	-1.89	-1.88	-0.2	-0.2
RBC	10y	1999-12-31_2025-09-30	30b	2	No	MOM	-2.19	-2.18	-0.23	-0.23
RBC	10y	1999-12-31_2025-09-30	10b	2	Yes	MOM	-2.19	-2.19	-0.14	-0.14
RBC	5y	1999-12-31_2025-09-30	10b	2	Yes	MOM	-2.43	-2.42	-0.15	-0.15
RBC	5y	1999-12-31_2025-09-30	20b	2	Yes	MOM	-2.83	-2.82	-0.18	-0.18
RBC	10y	1999-12-31_2025-09-30	20b	2	Yes	MOM	-3.02	-3.01	-0.19	-0.19
RBC	5y	1999-12-31_2025-09-30	30b	2	Yes	MOM	-3.33	-3.32	-0.21	-0.21
RBC	10y	1999-12-31_2025-09-30	30b	2	Yes	MOM	-3.49	-3.48	-0.22	-0.22

5.1. Gross and Net Performance

As illustrated in Figure 4: Gross vs Net PnL for mean-reversion rbc_10cal_1999-12-31_2025-09-30_30lag, Table 2: Results using S&P500 asset between 12/1999-09/2025 and Table 3: Results using Dow Jones Industrial Average asset between 12/1999-09/2025, net PnL does not depart meaningfully from gross PnL and net SR is aligned with gross SR. This is a key advantage of the DSGE strategy framework. There is a total of only 23 position changes for the unlevered 10 year-calibrated 30 day-signal lagged 12/1999-09/2025 horizon (full horizon) strategy. Every such position change is assumed to execute cheaply. This means it triggered a charge only 0.35% of the time out of 6540 daily observations throughout the horizon.

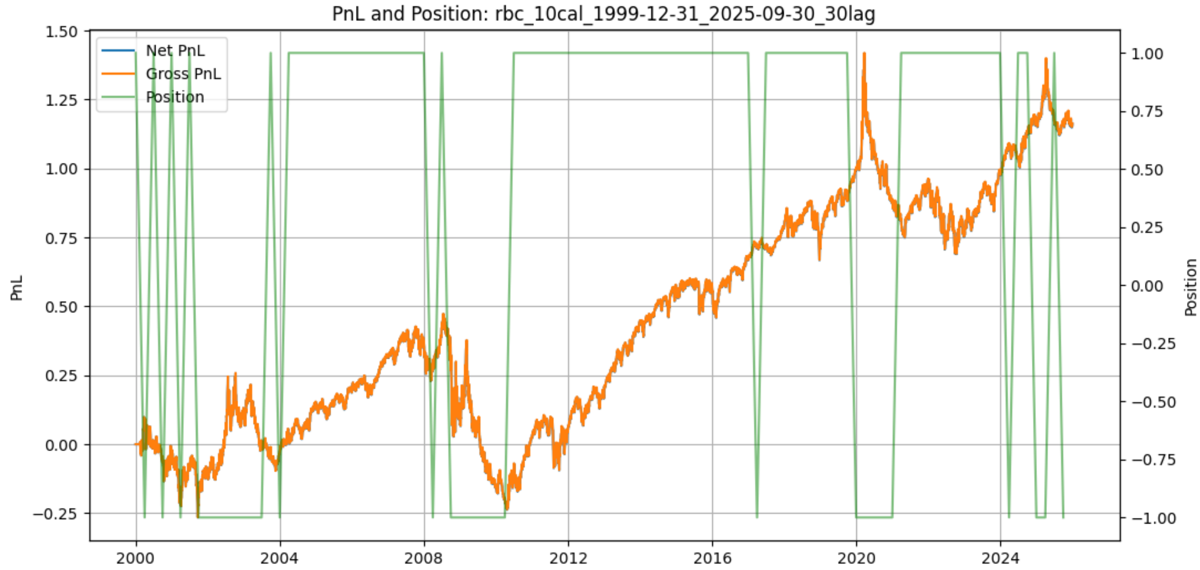

Figure 4: Gross vs Net PnL for mean-reversion rbc_10cal_1999-12-31_2025-09-30_30lag

In fact, all full horizon strategy variations ranged 23-37 trades as per Table 4: Position changes in 12/1999-09/2025 strategies. Therefore, there is little need to analyse net performance divergence from gross performance and the focus throughout the remainder of this section is on the more realistic net performance.

Table 4: Position changes in 12/1999-09/2025 strategies

Strategy	**Position Changes Count**	**Position Changes Percentage (%)**
rbc_5cal_1999-12-31_2025-09-30_10lag_dd	37	0.57
rbc_5cal_1999-12-31_2025-09-30_10lag_2lev_dd	37	0.57
rbc_5cal_1999-12-31_2025-09-30_20lag_dd	37	0.57
rbc_5cal_1999-12-31_2025-09-30_20lag_2lev_dd	37	0.57
rbc_5cal_1999-12-31_2025-09-30_30lag_dd	37	0.57
rbc_5cal_1999-12-31_2025-09-30_30lag_2lev_dd	37	0.57
rbc_10cal_1999-12-31_2025-09-30_30lag_2lev_dd	32	0.49
rbc_10cal_1999-12-31_2025-09-30_30lag_dd	32	0.49
rbc_10cal_1999-12-31_2025-09-30_20lag_2lev_dd	32	0.49
rbc_10cal_1999-12-31_2025-09-30_20lag_dd	32	0.49
rbc_10cal_1999-12-31_2025-09-30_10lag_2lev_dd	32	0.49
rbc_10cal_1999-12-31_2025-09-30_10lag_dd	32	0.49
rbc_5cal_1999-12-31_2025-09-30_10lag	25	0.38

rbc_5cal_1999-12-31_2025-09-30_30lag_2lev	25	0.38
rbc_5cal_1999-12-31_2025-09-30_30lag	25	0.38
rbc_5cal_1999-12-31_2025-09-30_20lag_2lev	25	0.38
rbc_5cal_1999-12-31_2025-09-30_20lag	25	0.38
rbc_5cal_1999-12-31_2025-09-30_10lag_2lev	25	0.38
rbc_10cal_1999-12-31_2025-09-30_10lag_2lev	23	0.35
rbc_10cal_1999-12-31_2025-09-30_20lag	23	0.35
rbc_10cal_1999-12-31_2025-09-30_20lag_2lev	23	0.35
rbc_10cal_1999-12-31_2025-09-30_30lag	23	0.35
rbc_10cal_1999-12-31_2025-09-30_30lag_2lev	23	0.35
rbc_10cal_1999-12-31_2025-09-30_10lag	23	0.35

In contrast to the DSGE strategy, buy-and-hold strategy results do not assume trading costs as the long position remains once initiated, thus there is no need to distinguish between gross and net performance.

5.2. Horizon

Whilst full horizon variants of the strategy are most useful in determining long-term feasibility of the DSGE trading framework, it is worth exploring how the strategies fared throughout this period, spanning various economic conditions and asset fluctuations.

Among the strategies that did not employ leverage, the most successful one for the S&P500 asset across the full horizon was the mean-reversion one with the RBC model and AR functions trained over 10 years and a signal lag of 30 business days. It produced net PnL of 1.16 and net SR of 0.23. Its comparison with the strategies parametrised the same way except for horizon is shown in Table 5: Strategy comparison across horizons for S&P 500 and Figure 5: Strategy comparison across horizons for S&P 500.

Table 5: Strategy comparison across horizons for S&P 500

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.27	0.27	0.19	0.19
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.5	-0.5	-0.67	-0.67
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.4	1.4	0.81	0.81
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25

RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.32	0.76	0.76
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.16	1.16	0.23	0.23

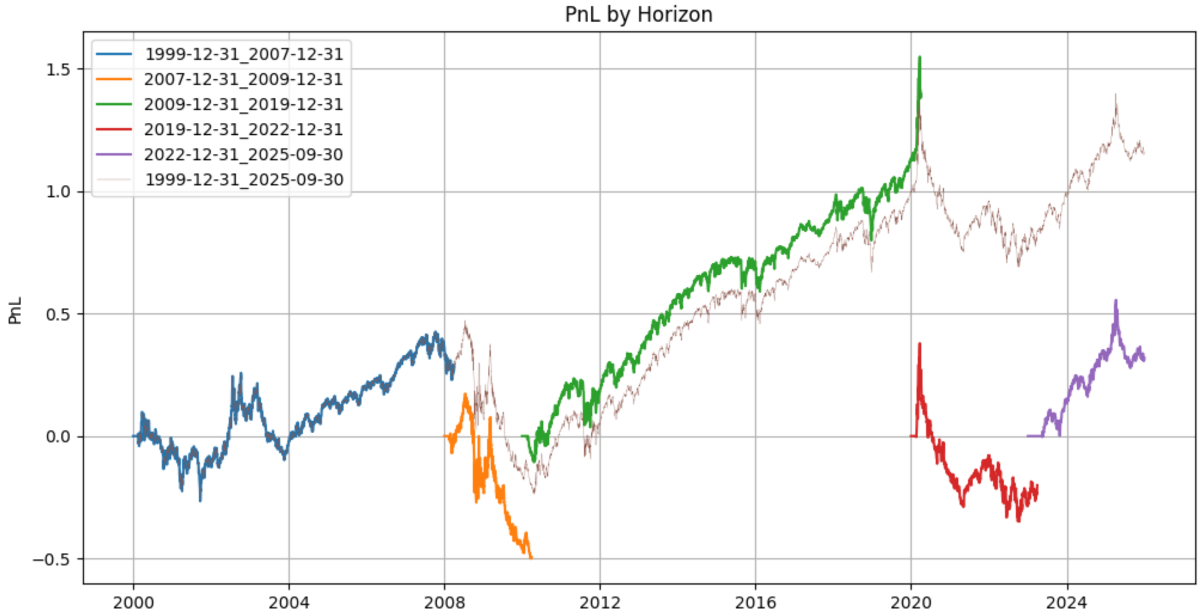

Figure 5: Strategy comparison across horizons for S&P 500

Despite overall Sharpe ratios not indicating high risk-adjusted performance, it can be observed that the strategies generally perform healthily and are consistent. In the mean-reversion (MR) scenario, a short position is adopted when output forecast exceeds realised output and a long position otherwise. Throughout most of the bull markets of the mid-2000s, 2010s and mid-2020s the signal captures output beating forecast expectations, suggesting better-than-expected economic growth and corresponding stock market rallies. It profitably holds long positions for most of those time periods accordingly. Impressively, it shorts the S&P 500 several times in the first half of the 2000s through a series of occasions when realised output does not meet forecast expectations and a bear market ensues. Conversely, it loses money in the early 2020s when it intuitively goes short into an economic downturn driven by the Covid pandemic. The stock market had quickly reacted positively following interest rate decreases and a technology firm-driven rally, thus it be argued there existed a greater decoupling of financial markets from the real economy. The strategy variant running throughout 2008-2009 also loses ground, although this is in line with the buy-and-hold strategy.

The strategies are somewhat stable in their respective horizons and are expected to be as such due to the slow-moving nature of economic data and especially its forecasts. Moreover, the simplicity of the basic RBC model used throughout these variations ensures consistency and is well adapted to the frequency of the data, preventing overtrading. This is particularly useful in extended bull runs, as observed in 2010s where the strategy is profitable over the long-run and remains committed to positions. The additional benefit is the ability to identify several-quarter periods of acyclical changes in output, as in the mid-2000s, to which equities appear to be correlated and from which the signal is able to generate positive PnL.

For the framework to be profitable throughout all periods, it would require selecting the momentum (MOM) strategy variations around the times of the Global Financial Crisis (GFC) and Covid. As per Appendix D: Strategy Results across Horizons, these returned a respectable 0.54 net PnL (0.72 net SR) and 0.51 net PnL (0.32 net SR), respectively. This suggests that the forecast becomes the more valuable source of information during a downturn and, with it anticipating output lower than the output observed following negative productivity shocks, it correctly signals that equity markets are due to decrease further.

Setting MOM variants in place of MR for the aforementioned horizons produces lucrative results whereby net PnL reaches 2.68, at which point it comfortably beats the buy-and-hold strategy. The relevance of this is that it offers the potential to beat the well established strategy in a like-for-like setting: there is no leverage employed, the strategy horizon remains long-term and there are minimal costs involved in trading. The only requirement for this to function is the existence of a trigger of regime change. This may not necessarily be trivial as a general exercise as has been documented in the literature (Ang & Timmermann, 2012), yet there are factors that improve the chances in a future application to this framework. Firstly, the time between successive position changes amounts to a full quarter which leaves ample time to initiate a change in strategy type based on financial or economic data, for example yield curve inversions, poor Purchasing Managers' Index (PMI) readings or sudden geopolitical crises. Secondly, the systematic trading strategy is long-term by design, thus as impending crises or regime changes become apparent to a wide audience, strategy type can be switched to reap the rewards for the several quarters for which positions remain fixed on average. As a result of the superior returns of this hybrid approach (yet noting the constraint to identify regime changes), it may be considered a realistic alternative to buy-and-hold. Figure 6: Strategy comparison across horizons for S&P 500 with strategy type switching visualises this.

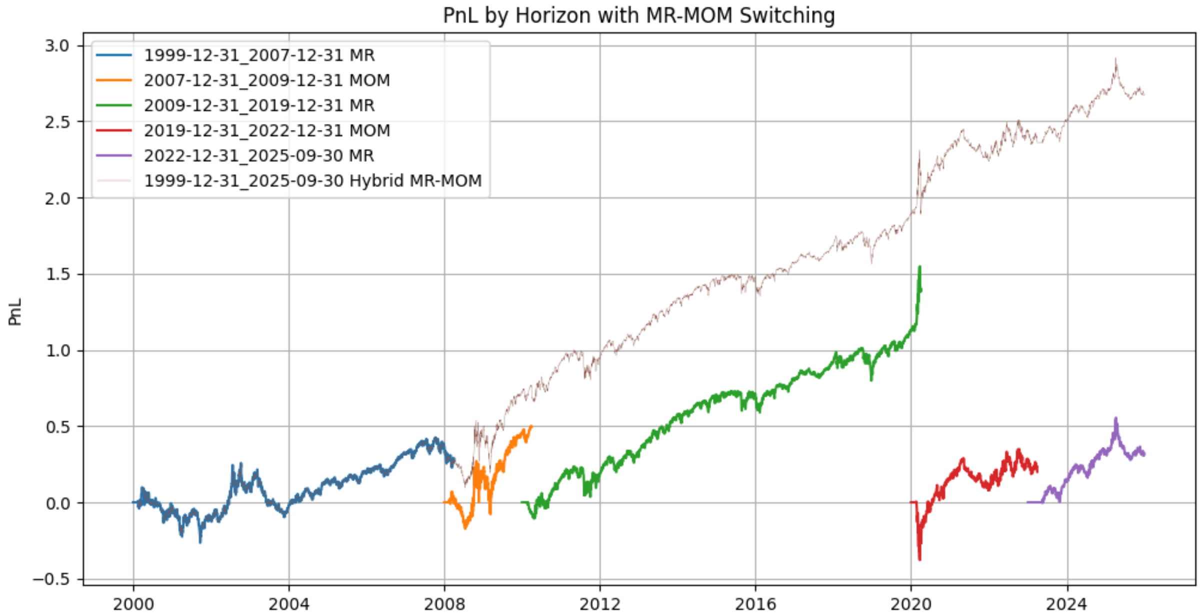

Figure 6: Strategy comparison across horizons for S&P 500 with strategy type switching

The results across strategy horizons of the Dow Jones Industrial Average (DJIA) asset are aligned with those of the S&P 500, however net PnL and net SR are lower with this asset. This is summarised in Table 6: Strategy comparison across horizons for Dow Jones Industrial Average. S&P 500 has slightly larger net SR in absolute value than DJIA for all horizons except 12/2022-09/2025. This due to greater PnL even when accounting for its standard deviation, except for the final horizon in which the S&P 500 strategy experiences greater volatility and a lower cumulative PnL, as evident in Figure 7: Strategy comparison across horizons for S&P 500 and Dow Jones Industrial Average.

Table 6: Strategy comparison across horizons for Dow Jones Industrial Average

Model	**Data Length**	**Horizon**	**Signal Lag**	**Leverage**	**Double Down**	**Strategy Type**	**Net PnL**	**Gross PnL**	**Net SR**	**Gross SR**
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.18	0.18	0.13	0.13
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.36	-0.36	-0.54	-0.54
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.27	1.27	0.74	0.74
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.19	-0.19	-0.23	-0.23
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.36	0.36	0.95	0.95
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.09	1.09	0.23	0.23

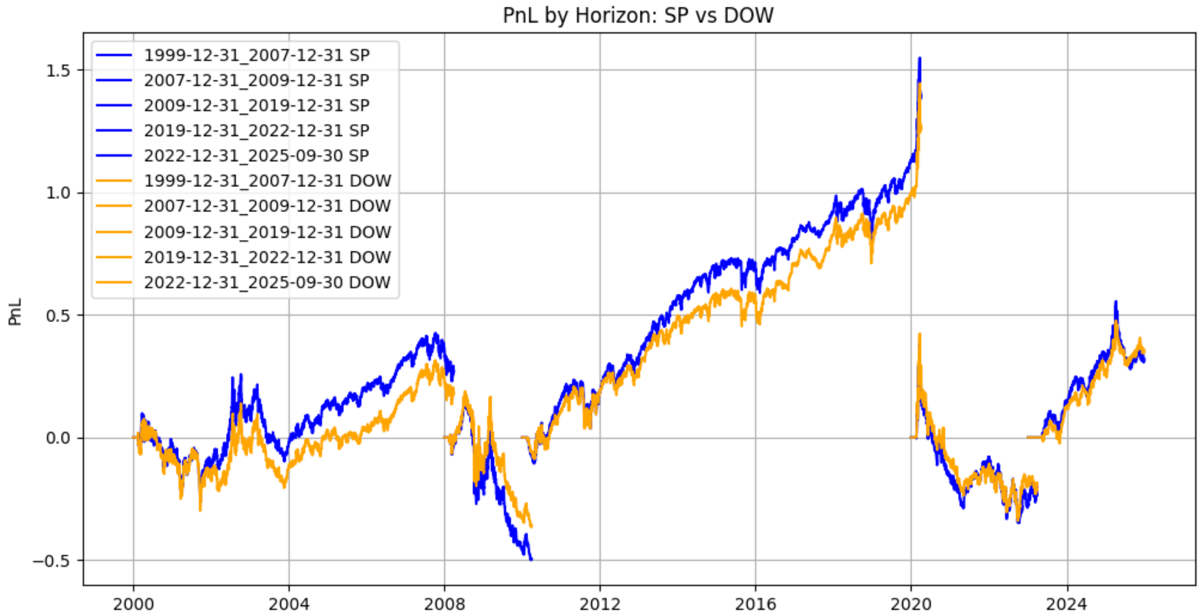

Figure 7: Strategy comparison across horizons for S&P 500 and Dow Jones Industrial Average

5.3. Strategy Type

Mean-reversion strategies perform better than momentum strategies in the DSGE trading framework. Table 2: Results using S&P500 asset between 12/1999-09/2025 and Table 3: Results using Dow Jones Industrial Average asset between 12/1999-09/2025 show the dominance of mean-reversion in the terms of PnL and SR, regardless of training data length, signal lag and other attributes. This suggests that positioning in line with how output beats or misses expectations is key in attempting to capture alpha in equity markets: the more output exceeds forecast values, the more it could indicate the economy is outperforming and that equities are undervalued and worth taking a long position in; equally the lower output turns out to be relative to the forecast, the more it could suggest the economy is underperforming and a stock market correction is due.

The association between output performance and subsequent equity market moves is evident through the long-term consistency of the strategy, yet not as strong as other potentially shorter-term variables, as net Sharpe ratios are below 1 for several unlevered strategy variants. This is illustrated in Figure 8: MR and MOM Sharpe ratios.

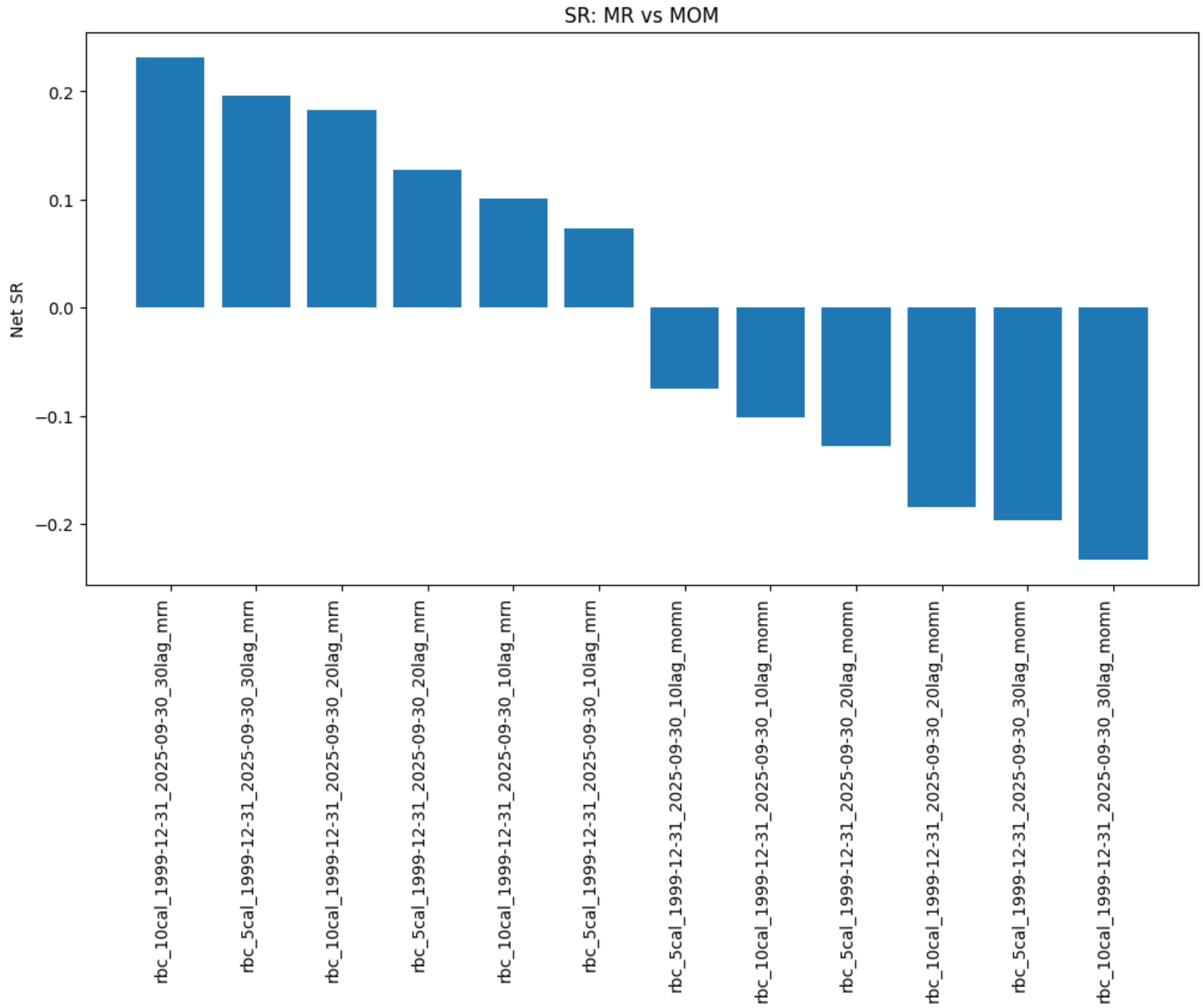

Figure 8: MR and MOM Sharpe ratios

Furthermore, mean-reversion strategies are unprofitable during some periods of time whilst momentum strategies generate positive PnL, as discovered in 5.2. Horizon. The association may therefore be the inverse during such periods, particularly during periods of increased volatility and financial crises. Over GFC and Covid periods, it can be seen that the forecast drives the direction for a profit-generating signal: as the economy encounters negative productivity shocks and resulting output forecasts are lower than realised output, generally taking a short position with a view that corresponding equity markets have more to lose earns positive PnL.

The overall mechanism appears to favour mean-reversion to realised output (which assumes more information about the state of the economy and reacts to short-term shocks) in the application of a trading signal to assets linked to the economy, which in this case is two major US equity indices. This reverses during volatile and troubling economic periods whereby pursuing the direction of the forecast is beneficial. Equity markets may be pricing in interest rate cuts too optimistically and departing from the worse actual reality of the state of the economy.

A separate observation is that the consistency with which momentum and mean-reversion variants all either generate positive PnL or negative PnL demonstrates the appropriateness of the strategy setup. The opposite directions of momentum and mean-reversion variants under the same remaining parameters indicate signals are being generated as intended and driving contrasting long and short positions.

5.4. Data Length

Three training data lengths are employed throughout the framework. In line with the long-term nature of the framework, these lengths span multiple years for AR equations to produce feasible results and meaningful productivity persistence parameters for the model.

The 5y strategy was less reliable, producing a larger net SR for the 12/1999-12/2007 period, yet performing noticeably worse in 12/2022-09/2025 as shown in Table 7: Unlevered S&P500 Data Length Results for 30b signal lag for all horizons. Its erratic nature stems from a shorter time period to form robust persistence parameters and the susceptibility to being swayed by several volatile quarters of TFP data. The HP filter also likely struggles to split cyclical and trend components for data this short. The 20y strategy fares slightly worse than the 10y strategy across variants, plus does not produce results up to 12/2009 observations due to insufficient data availability. 10 years are most effective in increasing net PnL and net SR across horizon and signal lag variants for unlevered mean-reversion strategies. Results for more signal lags can be found in Appendix E: Strategy Results across Data Lengths.

Table 7: Unlevered S&P500 Data Length Results for 30b signal lag for all horizons

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.16	1.16	0.23	0.23
RBC	5y	1999-12-31_2025-09-30	30b	1	No	MR	0.98	0.98	0.2	0.2
RBC	5y	1999-12-31_2007-12-31	30b	1	No	MR	0.56	0.57	0.39	0.39
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.27	0.27	0.19	0.19
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.5	-0.5	-0.67	-0.67
RBC	5y	2007-12-31_2009-12-31	30b	1	No	MR	-0.56	-0.56	-0.76	-0.76
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.4	1.4	0.81	0.81
RBC	20y	2009-12-31_2019-12-31	30b	1	No	MR	1.25	1.25	0.72	0.72
RBC	5y	2009-12-31_2019-12-31	30b	1	No	MR	1.18	1.18	0.68	0.68

RBC	5y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	20y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.32	0.76	0.76
RBC	20y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.32	0.76	0.76
RBC	5y	2022-12-31_2025-09-30	30b	1	No	MR	0.18	0.18	0.43	0.43

5.5. Signal Lag

Three signals lags are run in the DSGE framework. These delay the relation of output forecasts and realised output values to the actual triggering of a position.

As with the 10y data length conclusion, the 30b signal lag consistently outperforms for unlevered mean-reversion strategies: it typically ranks first in terms of net SR and, when second, it produces results close to that of the winner. A general rule of thumb appears to be that the longer the lag, the more profitable the resulting strategy. This implies that the factors underlying divergence of realised output from forecasts take at least a month to fully translate to corresponding equity valuations for mean-reversion variants. A longer lag length is also more practical in terms of signal generation: 30 business days allow more time for macroeconomic data providers to publish TFP and GDP data. Results are summarised in Table 8: Unlevered S&P500 MR signal lag results for 10y data length for all horizons.

Table 8: Unlevered S&P500 MR signal lag results for 10y data length for all horizons

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.16	1.16	0.23	0.23
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MR	0.92	0.92	0.18	0.18
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MR	0.51	0.51	0.1	0.1
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.27	0.27	0.19	0.19
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MR	0.16	0.16	0.11	0.11
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MR	-0.16	-0.16	-0.11	-0.11
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MR	-0.43	-0.43	-0.58	-0.58
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.5	-0.5	-0.67	-0.67
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MR	-0.54	-0.54	-0.72	-0.72
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MR	1.42	1.42	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.4	1.4	0.81	0.81
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MR	1.3	1.3	0.75	0.75
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32

RBC	10y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.32	0.76	0.76
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MR	0.27	0.27	0.64	0.64
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MR	0.16	0.16	0.38	0.38

The reverse is true for momentum strategies: 10b appears to produce higher net SR. This implies that forecast supremacy relative to realised output takes less time for the equities to price in for momentum variants. In other words, the sooner a position is taken following the realisation that output is not at the level its forecast suggests it should be, the more likely it is that a corresponding gain can be made in the corresponding asset. However, this phenomenon is not as clear as with the 30b signal lags in mean-reversion. Results are summarised in Table 9: Unlevered S&P500 MOM signal lag results for 10y data length for all horizons.

Table 9: Unlevered S&P500 MOM signal lag results for 10y data length for all horizons

Model	**Data Length**	**Horizon**	**Signal Lag**	**Leverage**	**Double Down**	**Strategy Type**	**Net PnL**	**Gross PnL**	**Net SR**	**Gross SR**
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MOM	-0.51	-0.51	-0.1	-0.1
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MOM	-0.92	-0.92	-0.18	-0.18
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MOM	-1.17	-1.16	-0.23	-0.23
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MOM	0.16	0.16	0.11	0.11
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MOM	-0.16	-0.16	-0.11	-0.11
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MOM	-0.27	-0.27	-0.19	-0.19
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MOM	0.54	0.54	0.72	0.72
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MOM	0.5	0.5	0.67	0.67
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MOM	0.43	0.43	0.58	0.58
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MOM	-1.3	-1.3	-0.75	-0.75
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.4	-1.4	-0.81	-0.81
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.42	-1.42	-0.82	-0.82
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MOM	0.29	0.29	0.35	0.35
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MOM	0.25	0.26	0.32	0.32
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MOM	0.2	0.2	0.25	0.25
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MOM	-0.16	-0.16	-0.38	-0.38
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MOM	-0.27	-0.27	-0.64	-0.64
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.32	-0.32	-0.76	-0.76

Overall, these observations and practical ease of implementation point to 30b being the preferred signal lag.

5.6. Leverage and Doubling Down

Table 2: Results using S&P500 asset between 12/1999-09/2025 and Figure 9: PnL with doubling down and 2x leverage of rbc_10cal_1999-12-31_2025-09-30_30lag show that doubling down and leverage have positive effects on net PnL and net SR.

The approach to signal generation and position setting in this framework is conducive to applying leverage. Positions change infrequently and, provided the right mean-reversion strategy type is selected, the portfolio is well prepared to earn excess returns and cover higher trading costs. However, the benefit is path-dependent and may not materialise under high volatility.

Doubling down is very similar to 2x leverage in the DSGE framework. This is due to long periods of time during which positioning remains unchanged, hence doubling down often sets in. Combining the impact of both scaling approaches results in an effective near 4x leverage and final net PnL of 3.97 in the 10 year-calibrated 30 day-signal lagged 12/1999-09/2025 horizon (full horizon) mean-reversion strategy for the S&P 500 asset. Net SR remains at 0.24 due to the increased volatility of PnL.

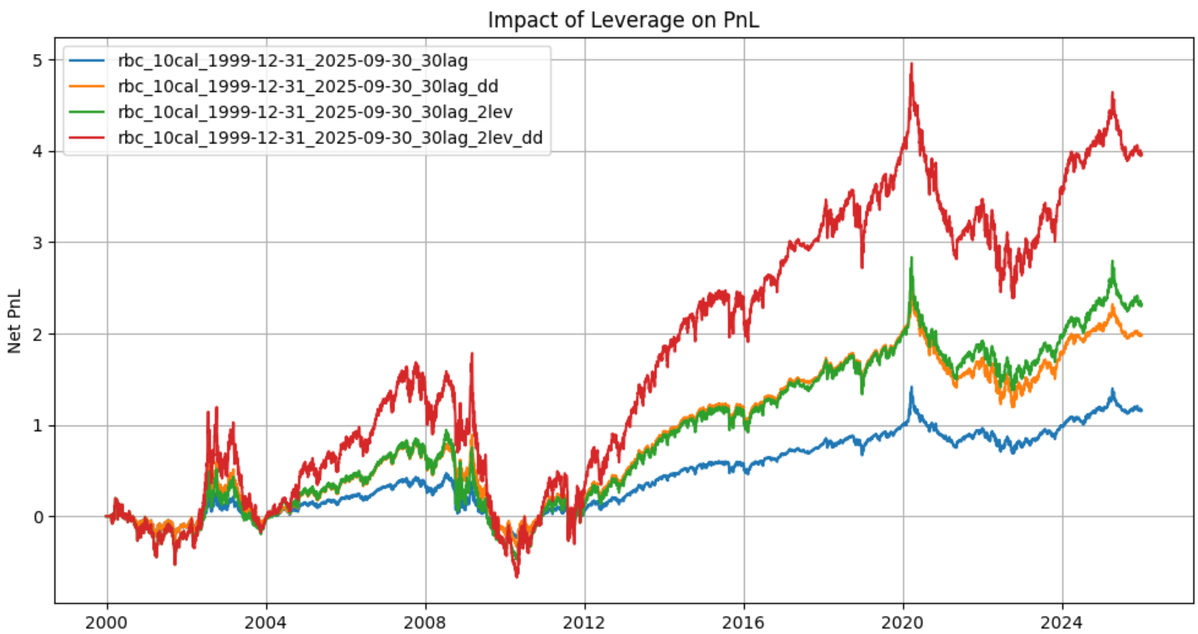

Figure 9: PnL with doubling down and 2x leverage of rbc_10cal_1999-12-31_2025-09-30_30lag

5.7. Buy-and-Hold Benchmark and Asset Differences

The buy-and-hold strategy prints a net PnL of 2.03 and net SR of 0.4. Being long one unit in an equity asset, it performs well in bull markets and poorly in bear markets.

The best unlevered DSGE strategy performs slightly worse than that over the full horizon, noting the aforementioned nuances described throughout this section. Adding leverage helps greatly and beats the buy-and-hold strategy. Net PnL progression over time is illustrated in Figure 10: DSGE strategy benchmark with buy-and-hold for S&P 500.

Results of the DJIA asset are comparable to those of the S&P 500 and conclusions are aligned between them. There are not any meaningful differences observed across strategy variants. Relevant information can be found in 5.2. Horizon.

Figure 10: DSGE strategy benchmark with buy-and-hold for S&P 500

6. LIMITATIONS AND IMPROVEMENTS

The DSGE trading framework is designed in a way to facilitate the interpretation of DSGE model forecasts and realised macroeconomic data in the context of signal generation in a systematic trading strategy. Results show that a basic RBC model and AR1 process estimation are capable of producing the data required. Nevertheless, these could be enhanced. There exist DSGE models that assume the existence of government, export sectors and even financial frictions, each of which could potentially make the model better at forecasting variables than the current RBC model. Estimating variables in addition to TFP may also yield improvements, as the DSGE model would benefit from updated model parameters most tailored to the economy in question at a given point in time. Furthermore, DSGE modelling allows the forecasting of variables other than output (such as consumption and investment) whose predictive power could be explored in the context of trading. Such variations may improve forecast quality, thereby producing signals that may be more profitable.

Mean-reversion and momentum are both tested in this framework and they each have their own strengths, particularly when isolating horizons. 5.2. Horizon demonstrates when each variant generates the larger PnL and hints at the idea of switching between them based on regime change. Incorporating a trigger for regime change into the framework may be worth investigating in the future.

Finally, assets intuitively associated with the US economy are tested. Broadening the assets tested beyond the S&P 500 and DJIA could result in more profitable scenarios. With more macroeconomic variables potentially being forecast by the DSGE model in the future, there could consequently be more asset candidates that might be linked to these. For example, a consumption-related equity basket could be traded in response to consumption forecasts.

7. CONCLUSION

The systematic trading strategy introduced in this paper lays the groundwork for a novel type of low-frequency trading based on DSGE modelling. The framework effectively combines DSGE modelling and its forecasts with realised macroeconomic data and systematic trading infrastructure. It processes macroeconomic input data via a functioning RBC model to produce next-period output forecasts and compares these to realised output for generating trading signals. Both momentum and mean-reversion views are tested and, alongside several parameter variations, a series of backtests are produced. The entirety of this pipeline is programmatic and reproducible.

The best strategy in terms of performance across the full horizon of data excluding leverage for the S&P 500 asset is the 10 year-calibrated 30 day-signal lagged mean-reversion strategy. It generates a net PnL and net SR slightly lower than the buy-and-hold benchmark, yet is still at healthy levels, is of a similar risk profile having traded infrequently and is designed for long-term holding (due to quarterly position reassessment and positions linked to US macroeconomic modelling). Results show that a 10-year training period for the TFP series and resulting DSGE model is suitable, as this decreases the chances that several volatile quarters skew regression results and make HP filtering less effective. They also show that a longer 30 business day lag from the reference date of comparison between forecast and realised values to signal implementation is conducive towards increasing PnL, thus reflecting a potential delay between factors driving realised output divergence and equity valuations. Mean-reversion proves to be more profitable overall, indicating output surprises rather than forecast excesses or misses drive the direction of subsequent corresponding equity moves. This does not hold for all subperiods and could be taken advantage of with additional regime switching identification techniques.

Leverage in its pure form and doubling down upon position consistency are both effective in increasing PnL and SR. Acknowledging timing constraints and challenges during volatile markets, strategies employing leverage generally outperform buy-and-hold overall.

The framework produces a healthy low-frequency systematic trading strategy which stands the test of time and is a good alternative to the buy-and-hold strategy, particularly when levered. It has the potential to be expanded further in terms of DSGE modelling assumptions, assets traded and signal generation.

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APPENDICES

Appendix A: Framework Setup

Please make sure you have all usual dependencies installed on your system. The most important ones include python, pip and ipykernel. If you're using *Visual Studio Code* for development, you should be prompted to install them automatically. Next, follow this process:

1. Create the virtual environment: python -m venv venv
2. Enter venv: source venv/bin/activate on Linux or venv\Scripts\activate.bat on Windows
3. Upgrade your pip: pip install --upgrade pip
4. Install dependencies from the requirements file: pip install -r requirements.txt
5. Navigate to the Model directory and open the Jupyter Notebook IDE of your choice. Select the Python version from inside the venv you just created when prompted by ipykernel package. Suggested command: python -m notebook

Appendix B: DSGE Strategy Code

```
## Setup and Dependencies

import numpy as np

import pandas as pd

from pandas.tseries.holiday import USFederalHolidayCalendar

from matplotlib import pyplot as plt

from fredapi import Fred

from dolo import *

import os

from statsmodels.regression.linear_model import OLS

from pandas_datareader import data as pdr

from statsmodels.tsa.filters.hp_filter import hpfilter

import yfinance as yf

import datetime as dt
```

```
## Model Setup

yaml_rbc = """

name: Real Business Cycle


symbols:


   exogenous: [e_z]

   states: [z, k]

   controls: [n, i]

   parameters: [beta, sigma, eta, chi, delta, alpha, rho, zbar, sig_z]


definitions: |

    y[t] = exp(z[t])*k[t]^alpha*n[t]^(1-alpha)

    c[t] = y[t] - i[t]

    rk[t] = alpha*y[t]/k[t]

    w[t] = (1-alpha)*y[t]/n[t]


equations:


    arbitrage: |

        chi*n[t]^eta*c[t]^sigma - w[t]                    ⊥ 0.0 <= n[t] <= inf

        1 - beta*(c[t]/c[t+1])^(sigma)*(1-delta+rk[t+1])   ⊥ -inf <= i[t] <= inf


    transition: |
```

```
    z[t] = rho*z[t-1] + e_z
    k[t] = (1-delta)*k[t-1] + i[t-1]

calibration:

  # parameters
  beta : 0.99
  phi: 1
  delta : 0.025
  alpha : 0.33
  rho : 0.8
  sigma: 5
  eta: 1
  sig_z: 0.016
  zbar: 0
  chi : w/c^sigma/n^eta
  c_i: 1.5
  c_y: 0.5
  e_z: 0.0
  n: 0.33
  z: zbar
  rk: 1/beta-1+delta
  w: (1-alpha)*exp(z)*(k/n)^(alpha)
  k: n/(rk/alpha)^(1/(1-alpha))
```

```
    y: exp(z)*k^alpha*n^(1-alpha)

    i: delta*k

    c: y - i

    V: log(c)/(1-beta)

    u: c^(1-sigma)/(1-sigma) - chi*n^(1+eta)/(1+eta)

    m: beta/c^sigma*(1-delta+rk)

    kss: 10


exogenous: !UNormal

    sigma: 0.01


domain:

    z: [-2*sig_z/(1-rho^2)^0.5,  2*sig_z/(1-rho^2)^0.5]

    k: [ kss*0.5, kss*1.5]


options:

    grid: !Cartesian

        n: [100, 100]

    discrete_choices: [n]

"""


file_path = "yamlDSGE/rbc.yaml"


os.makedirs(os.path.dirname(file_path), exist_ok=True)
```

```
with open(file_path, "w", encoding="utf-8") as f:
    f.write(yaml_rbc)


print(f"RBC model saved to {file_path}")
## Model Execution
class Modifications():
    def __init__(self):
        self.months_in_quarter = 3


    def convert_years_to_quarters(self, length_years):
        return length_years * 4


    def convert_years_to_months(self, length_years):
        return length_years * 12


    def get_shifted_date(self, date_input, date_offset_months, direction="backwards"):
        if direction == "backwards":
            date_offset_months *= -1


        shifted_date = (
            ((pd.tseries.offsets.BMonthBegin().rollback(date_input)) + pd.tseries.offsets.DateOffset(months=date_offset_months))
            + pd.tseries.offsets.BMonthEnd()
        ).date()
```

```
        return shifted_date


    def add_buffer_for_start_offset(self, offset_months):
        return offset_months + 3 # Productivity and GDP starts need a buffer of 1 quarter, as the quarterly index is as of the start of the quarter rather than the end of the quarter
class Data(Modifications):
    def __init__(self, data_length_years_max, horizon_start_in, horizon_end_in):
        Modifications.__init__(self)
        self.data_length_years_max = data_length_years_max
        self.data_length_quarters_max = self.convert_years_to_quarters(self.data_length_years_max)
        self.data_length_months_max = self.convert_years_to_months(self.data_length_years_max)


        self.horizon_start_in = horizon_start_in
        self.horizon_end_in = horizon_end_in
        self.horizon_start = self.convert_str_date_to_dt(self.horizon_start_in)
        self.horizon_end = self.convert_str_date_to_dt(self.horizon_end_in)
        self.date_start_gdp = self.get_shifted_date(self.horizon_start, self.add_buffer_for_start_offset(self.data_length_months_max)) # Data starts 10 years prior to window start
        self.date_start_prod = self.get_shifted_date(self.date_start_gdp, self.months_in_quarter) # Data starts one quarter prior to GDP data for next-period forecasting
        self.date_end_prod = self.get_shifted_date(self.horizon_end, self.months_in_quarter) # Data ends one quarter before window end as the final forecast must be of the window end date
```

```
        # self.horizon_start_prod = self.get_shifted_date(self.horizon_start, self.months_in_quarter) # Window starts one quarter prior to GDP data for next-period forecasting
        self.date_end_asset = self.get_shifted_date(self.horizon_end, self.months_in_quarter, "forwards") + pd.tseries.offsets.DateOffset(days=1) # end date in yahoo finance query is exclusive so extend by 1 day

        self.prod = self.pull_data_productivity()
        self.gdp = self.pull_data_gdp()
        self.asset_sp500 = self.pull_asset("^GSPC")
        self.asset_dow = self.pull_asset("^DJI")
        self.holidays = self.get_holidays()

    def convert_str_date_to_dt(self, date_in):
        return dt.date(int(date_in[:4]), int(date_in[5:7]), int(date_in[8:]))

    def convert_fred_index_to_qe(self, index_dates):
        return [date_i.date() for date_i in index_dates.date + pd.tseries.offsets.BQuarterEnd()]

    # Non-farm business labor productivity index
    def pull_data_productivity(self):
        prod = pdr.DataReader("OPHNFB", "fred", start=self.date_start_prod, end=self.date_end_prod)
        prod_index_name = prod.index.name
        prod.index = self.convert_fred_index_to_qe(prod.index)
        prod.index.name = prod_index_name
```

```
        prod = prod.dropna()

        return prod

    # Real GDP
    def pull_data_gdp(self):
        gdp = pdr.DataReader("GDPC1", "fred", start=self.date_start_gdp, end=self.horizon_end)
        gdp_index_name = gdp.index.name
        gdp.index = self.convert_fred_index_to_qe(gdp.index)
        gdp.index.name = gdp_index_name
        gdp = gdp.dropna()

        return gdp

    # Asset
    def pull_asset(self, ticker):
        asset_ser = yf.Ticker(ticker).history(start=self.horizon_start,
end=self.date_end_asset)["Close"]

        asset_ser_name = asset_ser.index.name
        asset_ser.index = asset_ser.index.date
        asset_ser.index.name = asset_ser_name

        return asset_ser
```

```
    def get_holidays(self):
        cal_us = USFederalHolidayCalendar()
        holidays = cal_us.holidays(start=self.horizon_start, end=self.date_end_asset) # holiday date range for the window considered for the country linked to the asset

        return holidays
class Model(Modifications):
    def __init__(self, model_type, prod, gdp, assets, asset_names, holidays):
        Modifications.__init__(self)
        self.model_type = model_type
        self.model = self.load_model(self.model_type)
        self.prod = prod
        self.gdp = gdp
        self.assets = assets
        self.asset_names = asset_names
        self.holidays = holidays

        self.latest_quarters_dict = {}
        self.output_forecasts_dict = {}
        self.latest_quarters_gdp_dict = {}
        self.outputs_realised_dict = {}

    def load_model(self, model_type):
        model = yaml_import(f"yamlDSGE/{model_type}.yaml")
```

```
        return model

    def generate_signals(self, data_length_years, window_gdp_start, window_gdp_end, leverage, double_down, show_irf=False):
        assert len(self.prod) == len(self.gdp)

        data_length_quarters = self.convert_years_to_quarters(data_length_years)
        data_length_months = self.convert_years_to_months(data_length_years)

        # Check the window and preceding data are in bounds of overall productivity and GDP data
        earliest_datapoint_gdp = self.get_shifted_date(window_gdp_start, data_length_months)
        assert earliest_datapoint_gdp >= self.gdp.index[0], f"Outside bounds: {earliest_datapoint_gdp} !>= {self.gdp.index[0]}"

        earliest_datapoint_prod = self.get_shifted_date(earliest_datapoint_gdp, self.months_in_quarter)
        assert earliest_datapoint_prod >= self.prod.index[0], f"Outside bounds: {earliest_datapoint_prod} !>= {self.prod.index[0]}"

        assert window_gdp_end <= self.gdp.index[-1], f"Outside bounds: {window_gdp_end} !<= {self.gdp.index[-1]}"

        latest_datapoint_prod = self.get_shifted_date(window_gdp_end, self.months_in_quarter)
        assert latest_datapoint_prod <= self.prod.index[-1], f"Outside bounds: {latest_datapoint_prod} !<= {self.prod.index[-1]}"
```

```
        # Generate output forecasts
        prod_index_name = self.prod.index.name
        window_prod_start = self.get_shifted_date(window_gdp_start, self.months_in_quarter)
        window_prod_end = self.get_shifted_date(window_gdp_end, self.months_in_quarter)
        window_prod = self.prod[(self.prod.index >= window_prod_start) & (self.prod.index <= window_prod_end)]
        print("\nwindow_prod:\n")
        print(window_prod)

        latest_quarters = []
        output_forecasts = []
        for q in window_prod.index:
            q_key = str(q) + "_" + str(data_length_years) + "y"
            if q_key in self.latest_quarters_dict.keys():
                latest_quarters.append(self.latest_quarters_dict[q_key])
                output_forecasts.append(self.output_forecasts_dict[q_key])
                print(f"Reading IRF forecast as of {q} with {data_length_years} years of calibration")
            else:
                prod_data = self.prod[(self.prod.index >= self.get_shifted_date(q, data_length_months)) & (self.prod.index <= q)]
                assert len(prod_data.index) == data_length_quarters + 1

                prod_cycle, prod_trend = hpfilter(np.log(prod_data), lamb=1600)
```

```
# prod_cycle.plot()

a_hat = prod_cycle
a = a_hat
a_lag = a.shift()

y = a.iloc[1:]
X = a_lag.iloc[1:].values

ar1 = OLS(y.values, X).fit()

rho_hat = ar1.params[0]
print(f"Estimated rho: {rho_hat:.3f}")

eps_hat = pd.Series(
    ar1.resid,
    index=y.index,
    name="tfp_shock"
)
# eps_hat.plot()

latest_quarter = eps_hat.index[-1]
latest_shock = eps_hat.iloc[-1]
```

```
print(f"Estimated TFP shock for {latest_quarter}: {latest_shock:.4f}")

# Set the estimated TFP persistence coefficient as a calibrated parameter in the model
self.model.set_calibration(rho=rho_hat)
# print(model.calibration.flat)

#dr = time_iteration(model)
dr = perturb(self.model, order=1)

irf = response(self.model, dr, 'e_z', T=10, impulse=float(latest_shock))

if show_irf == True:
    plt.figure(figsize=(8,4))
    plt.subplot(221)
    plt.plot(irf.sel(V='z'))
    plt.title('Productivity')
    plt.grid()
    plt.subplot(222)
    plt.plot(irf.sel(V='y'))
    plt.title('Output')
    plt.grid()
    plt.subplot(223)
    plt.plot(irf.sel(V='i'))
    plt.title('Investment')
```

```
            plt.grid()
            plt.subplot(224)
            plt.plot(irf.sel(V='c'))
            plt.title('Consumption')
            plt.grid()
            plt.tight_layout()
            plt.show()

        irf_output = np.array(irf.sel(V='y'))

        irf_output = [(x - 1) for x in irf_output]
        output_forecast = irf_output[2]
        print(f"Next period IRF forecast as of {latest_quarter}: {output_forecast:.5f}")

        self.latest_quarters_dict[q_key] = latest_quarter
        latest_quarters.append(latest_quarter)
        self.output_forecasts_dict[q_key] = output_forecast
        output_forecasts.append(output_forecast)

    output_forecasts_ser = pd.Series(output_forecasts, index=latest_quarters) # Next period forecasts, e.g. 2025Q2 is forecast of 2025Q3
    output_forecasts_ser.name = "output_forecast"
    output_forecasts_ser.index.name = prod_index_name
    output_forecasts_df = output_forecasts_ser.to_frame()
    # output_forecasts_df
```

```
# Obtain realised output
gdp_index_name = self.gdp.index.name
window_gdp = self.gdp[(self.gdp.index >= window_gdp_start) & (self.gdp.index <=
window_gdp_end)]
print("\window_gdp:\n")
print(window_gdp)

latest_quarters_gdp = []
outputs_realised = []
for q in window_gdp.index:
    q_key = str(q) + "_" + str(data_length_years) + "y"
    if q_key in self.latest_quarters_gdp_dict.keys():
        latest_quarters_gdp.append(self.latest_quarters_gdp_dict[q_key])
        outputs_realised.append(self.outputs_realised_dict[q_key])
        print(f"Reading realised value as of {q} with {data_length_years} years of
calibration")
    else:
        gdp_data = self.gdp[(self.gdp.index >= self.get_shifted_date(q,
data_length_months)) & (self.gdp.index <= q)]
        assert len(gdp_data.index) == data_length_quarters + 1

        gdp_cycle, gdp_trend = hpfilter(np.log(gdp_data), lamb=1600)
        output_gap = gdp_cycle
        latest_quarter_gdp = output_gap.last_valid_index()
```

```
            output_realised = output_gap.iloc[-1]
            print(f"Current period realised value as of {latest_quarter_gdp}: {output_realised:.5f}")

            self.latest_quarters_gdp_dict[q_key] = latest_quarter_gdp
            latest_quarters_gdp.append(latest_quarter_gdp)
            self.outputs_realised_dict[q_key] = output_realised
            outputs_realised.append(output_realised)

    outputs_realised_ser = pd.Series(outputs_realised, index=latest_quarters_gdp) # Realised GDP growth after HP filter, e.g. 2025Q3 reflects GDP growth in 2025Q3
    outputs_realised_ser.name = "output_realised"
    outputs_realised_ser.index.name = gdp_index_name
    outputs_realised_df = outputs_realised_ser.to_frame()
    # outputs_realised_df

    # Prepare signals
    signals_init = outputs_realised_df.join(output_forecasts_df, how="outer")
    output_forecasts_mod_df = signals_init.shift()[["output_forecast"]].dropna()
    signals = outputs_realised_df.join(output_forecasts_mod_df)

    def determine_pos(row, direction, leverage):
        if row["output_forecast"] > row["output_realised"]:
            pos = 1
        elif row["output_forecast"] < row["output_realised"]:
```

```
            pos = -1
        else:
            pos = 0

        if direction == "mr":
            pos *= -1

        return pos * leverage

    signals["pos_mom"] = signals.apply(determine_pos, axis=1, direction="mom", leverage=leverage)
    signals["pos_mr"] = signals.apply(determine_pos, axis=1, direction="mr", leverage=leverage)

    if double_down == True:
        signals_lagged = signals[["pos_mom", "pos_mr"]].shift().fillna(0)
        signals_lagged = signals_lagged.rename(columns={"pos_mom": "pos_mom_prev", "pos_mr": "pos_mr_prev"})
        signals = signals.join(signals_lagged)

        signals["double_down_scalar_mom"] = signals.apply(lambda row: 2 if row["pos_mom"] == row["pos_mom_prev"] else 1, axis=1)
        signals["double_down_scalar_mr"] = signals.apply(lambda row: 2 if row["pos_mr"] == row["pos_mr_prev"] else 1, axis=1)
    else:
        signals["double_down_scalar_mom"] = 1
```

```
        signals["double_down_scalar_mr"] = 1

    signals["pos_final_mom"] = signals["pos_mom"] * signals["double_down_scalar_mom"]
    signals["pos_final_mr"] = signals["pos_mr"] * signals["double_down_scalar_mr"]

    return signals

def summarise_results(self, pos, pnl, strategy_type, asset_name):
    # fig_pos, ax_pos = plt.subplots(figsize=(12, 6))
    # pos.plot(ax=ax_pos, grid=True)
    # plt.close(fig_pos)

    # fig_pnl_cumsum, ax_pnl_cumsum = plt.subplots(figsize=(12, 6))
    # pnl.cumsum().plot(ax=ax_pnl_cumsum, grid=True)
    # plt.close(fig_pnl_cumsum)

    pnl_sum = pnl.sum()
    pnl_mean = pnl.mean()
    pnl_std = pnl.std()
    sr_daily = pnl_mean / pnl_std
    sr_annual = sr_daily * np.sqrt(252)

    results = {
        # "fig_pos_" + strategy_type + "_" + asset_name: fig_pos,
```

```
        # "fig_pnl_cumsum_" + strategy_type + "_" + asset_name: fig_pnl_cumsum,
        "pos_" + strategy_type + "_" + asset_name: pos,
        "pnl_cumsum_" + strategy_type + "_" + asset_name: pnl.cumsum(),
        "pnl_sum_" + strategy_type + "_" + asset_name: pnl_sum,
        "pnl_mean_" + strategy_type + "_" + asset_name: pnl_mean,
        "pnl_std_" + strategy_type + "_" + asset_name: pnl_std,
        "sr_daily_" + strategy_type + "_" + asset_name: sr_daily,
        "sr_annual_" + strategy_type + "_" + asset_name: sr_annual
    }

    return results

def run_strategy(self, signals, signal_lag, window_gdp_start, window_gdp_end, show_pos=False):
    # Delay signals by a particular number of business days to account for lag in GDP and productivity data availability
    signals_actual = signals[["output_realised", "output_forecast", "pos_final_mom", "pos_final_mr"]]
    signals_actual.index += pd.offsets.CustomBusinessDay(n=signal_lag, holidays=self.holidays)

    date_end_asset = self.get_shifted_date(window_gdp_end, self.months_in_quarter, "forwards")
    results = {"signals_actual": signals_actual}
    for i, asset in enumerate(self.assets):
```

```
asset_ser = asset[(asset.index >= window_gdp_start) & (asset.index <= date_end_asset)]
asset_ret_ser = asset_ser.pct_change()
asset_ret_ser.name = "Return"

asset_ret_df = asset_ret_ser.to_frame()
asset_ret_df = asset_ret_df.join(signals_actual[["pos_final_mom", "pos_final_mr"]]).ffill().fillna(0)
if show_pos:
   asset_ret_df.plot(figsize=(12, 6), grid=True)

asset_ret_lagged_df = asset_ret_df[["pos_final_mom", "pos_final_mr"]].shift()
pnl = asset_ret_df[["Return"]].join(asset_ret_lagged_df).fillna(0)

cost_per_trade = 0.0001 # 1 bp
cost = abs(pnl[["pos_final_mom", "pos_final_mr"]].diff().fillna(0).rename(columns={"pos_final_mom": "cost_mom", "pos_final_mr": "cost_mr"})) * 0.5 * cost_per_trade
pnl_full = pnl.join(cost)

pnl_full["pnlg_mom"] = pnl_full["Return"] * pnl_full["pos_final_mom"]
pnl_full["pnln_mom"] = pnl_full["pnlg_mom"] - pnl_full["cost_mom"]
pnl_full["pnlg_mr"] = pnl_full["Return"] * pnl_full["pos_final_mr"]
pnl_full["pnln_mr"] = pnl_full["pnlg_mr"] - pnl_full["cost_mr"]
```

```
            results_momg = self.summarise_results(asset_ret_df["pos_final_mom"],
pnl_full["pnlg_mom"], "momg", self.asset_names[i])
            results_momn = self.summarise_results(asset_ret_df["pos_final_mom"],
pnl_full["pnln_mom"], "momn", self.asset_names[i])
            results_mrg = self.summarise_results(asset_ret_df["pos_final_mr"],
pnl_full["pnlg_mr"], "mrg", self.asset_names[i])
            results_mrn = self.summarise_results(asset_ret_df["pos_final_mr"],
pnl_full["pnln_mr"], "mrn", self.asset_names[i])
            results = results | results_momg | results_momn | results_mrg | results_mrn

        return results
class BuyAndHold(Modifications):
    def __init__(self, assets, asset_names):
        Modifications.__init__(self)
        self.assets = assets
        self.asset_names = asset_names

    def run_strategy(self, window_start, window_end):
        date_end_asset = self.get_shifted_date(window_end, self.months_in_quarter, "forwards")
        results = {"signals_actual": 1}
        for i, asset in enumerate(self.assets):
            asset_ser = asset[(asset.index >= window_start) & (asset.index <= date_end_asset)]

            asset_ret_ser = asset_ser.pct_change().fillna(0)
            asset_ret_ser.name = "Return"
            pnl = asset_ret_ser
```

```
        # fig_pnl_cumsum, ax_pnl_cumsum = plt.subplots(figsize=(12, 6))
        # pnl.cumsum().plot(ax=ax_pnl_cumsum, grid=True)
        # plt.close(fig_pnl_cumsum)

        pnl_sum = pnl.sum()
        pnl_mean = pnl.mean()
        pnl_std = pnl.std()
        sr_daily = pnl_mean / pnl_std
        sr_annual = sr_daily * np.sqrt(252)

        results_bh = {
            # "fig_pnl_cumsum_" + self.asset_names[i]: fig_pnl_cumsum,
            "pnl_cumsum_" + self.asset_names[i]: pnl.cumsum(),
            "pnl_sum_" + self.asset_names[i]: pnl_sum,
            "pnl_mean_" + self.asset_names[i]: pnl_mean,
            "pnl_std_" + self.asset_names[i]: pnl_std,
            "sr_daily_" + self.asset_names[i]: sr_daily,
            "sr_annual_" + self.asset_names[i]: sr_annual
        }

        results = results | results_bh

    return results
```

```
data_10y_19991231_20250930 = Data(10, "1999-12-31", "2025-09-30")
# rbc = Model(
#     "rbc",
#     data_20y_19991231_20250930.prod,
#     data_20y_19991231_20250930.gdp,
#     data_20y_19991231_20250930.asset,
#     data_20y_19991231_20250930.holidays
# )
# rbc = Model(
#     "rbc",
#     data_10y_19991231_20250930.prod,
#     data_10y_19991231_20250930.gdp,
#     [data_10y_19991231_20250930.asset_sp500],
#     ["SP"],
#     data_10y_19991231_20250930.holidays
# )


# # signals = rbc.generate_signals(10, dt.date(1999, 12, 31), dt.date(2025, 9, 30), 1, False,
show_irf=True)
# signals = rbc.generate_signals(10, dt.date(1999, 12, 31), dt.date(2025, 9, 30), 1, False)
# # strategy = rbc.run_strategy(signals, 10, dt.date(1999, 12, 31), dt.date(2025, 9, 30),
show_pos=True)
# strategy = rbc.run_strategy(signals, 10, dt.date(1999, 12, 31), dt.date(2025, 9, 30))
# signals
# signals["pos_final_mr"].plot()
```

```
# strategy
# strategy.keys()
# strategy["pnl_sum_momg_SP"]
# strategy["pnl_sum_momn_SP"]
# strategy["fig_pnl_cumsum_momg_SP"]
# strategy["fig_pnl_cumsum_momn_SP"]
# plt.plot(strategy["pos_momg_SP"])
# plt.plot(strategy["pnl_cumsum_mrg_SP"])
# plt.plot(strategy["pnl_cumsum_mrn_SP"])
# plt.show()
# plt.close()
model_types = ["rbc"]
data_lengths = [5, 10, 20]
windows = [
    [dt.date(1999, 12, 31), dt.date(2025, 9, 30)],
    [dt.date(1999, 12, 31), dt.date(2007, 12, 31)],
    [dt.date(2007, 12, 31), dt.date(2009, 12, 31)],
    [dt.date(2009, 12, 31), dt.date(2019, 12, 31)],
    [dt.date(2019, 12, 31), dt.date(2022, 12, 31)],
    [dt.date(2022, 12, 31), dt.date(2025, 9, 30)],
]
signal_lags = [10, 20, 30]
leverage_cases = [1, 2]
double_down_cases = [False, True]
```

```
results = {}
for model_type in model_types:
    rbc = Model(
        model_type,
        data_10y_19991231_20250930.prod,
        data_10y_19991231_20250930.gdp,
        [data_10y_19991231_20250930.asset_sp500,
data_10y_19991231_20250930.asset_dow],
        ["SP", "DOW"],
        data_10y_19991231_20250930.holidays
    )

    bh = BuyAndHold(
        [data_10y_19991231_20250930.asset_sp500,
data_10y_19991231_20250930.asset_dow],
        ["SP", "DOW"]
    )

    for window in windows:
        window_str = str(window[0]) + "_" + str(window[1])
        results[window_str] = {}

        for data_length in data_lengths:
            for signal_lag in signal_lags:
```

```
            for leverage in leverage_cases:
                for double_down in double_down_cases:
                    if not ((data_length == 20) & (window[0] < dt.date(2009, 12, 31))): # GDP data is unavailable before 1989-12-29 so avoid sourcing data before then
                        signals = rbc.generate_signals(data_length, window[0], window[1], leverage, double_down)
                        strategy = rbc.run_strategy(signals, signal_lag, window[0], window[1])

                        key = model_type + "_" + str(data_length) + "cal_" + window_str + "_" + str(signal_lag) + "lag"
                        if leverage > 1:
                            key += "_" + str(leverage) + "lev"
                        if double_down == True:
                            key += "_dd"

                        results[window_str][key] = {"signals": signals, "strategy": strategy}

    strategy_bh = bh.run_strategy(window[0], window[1])

    key_bh = "bh_" + window_str
    results[window_str][key_bh] = {"strategy": strategy_bh}
# results
main_results_sorted = {}

for window in windows:
```

```
    window_str = str(window[0]) + "_" + str(window[1])
    main_results_sorted[window_str] = {}

    pnls_spg = []
    pnls_spn = []
    srs_spg = []
    srs_spn = []
    pnls_dowg = []
    pnls_down = []
    srs_dowg = []
    srs_down = []

    for strategy in results[window_str].keys():
        if "bh" in strategy:
            pnls_spg.append([strategy, results[window_str][strategy]["strategy"]["pnl_sum_SP"]])
            pnls_spn.append([strategy,  results[window_str][strategy]["strategy"]["pnl_sum_SP"]])
# Assumed g = n for BH
            srs_spg.append([strategy, results[window_str][strategy]["strategy"]["sr_annual_SP"]])
            srs_spn.append([strategy,  results[window_str][strategy]["strategy"]["sr_annual_SP"]])
# Assumed g = n for BH

            pnls_dowg.append([strategy,
results[window_str][strategy]["strategy"]["pnl_sum_DOW"]])
            pnls_down.append([strategy,
results[window_str][strategy]["strategy"]["pnl_sum_DOW"]]) # Assumed g = n for BH
```

```
            srs_dowg.append([strategy, results[window_str][strategy]["strategy"]["sr_annual_DOW"]])
            srs_down.append([strategy, results[window_str][strategy]["strategy"]["sr_annual_DOW"]]) # Assumed g = n for BH
        else:
            pnls_spg.append([strategy + "_momg", results[window_str][strategy]["strategy"]["pnl_sum_momg_SP"]])
            pnls_spn.append([strategy + "_momn", results[window_str][strategy]["strategy"]["pnl_sum_momn_SP"]])
            pnls_spg.append([strategy + "_mrg", results[window_str][strategy]["strategy"]["pnl_sum_mrg_SP"]])
            pnls_spn.append([strategy + "_mrn", results[window_str][strategy]["strategy"]["pnl_sum_mrn_SP"]])
            srs_spg.append([strategy + "_momg", results[window_str][strategy]["strategy"]["sr_annual_momg_SP"]])
            srs_spn.append([strategy + "_momn", results[window_str][strategy]["strategy"]["sr_annual_momn_SP"]])
            srs_spg.append([strategy + "_mrg", results[window_str][strategy]["strategy"]["sr_annual_mrg_SP"]])
            srs_spn.append([strategy + "_mrn", results[window_str][strategy]["strategy"]["sr_annual_mrn_SP"]])

            pnls_dowg.append([strategy + "_momg", results[window_str][strategy]["strategy"]["pnl_sum_momg_DOW"]])
            pnls_down.append([strategy + "_momn", results[window_str][strategy]["strategy"]["pnl_sum_momn_DOW"]])
            pnls_dowg.append([strategy + "_mrg", results[window_str][strategy]["strategy"]["pnl_sum_mrg_DOW"]])
```

```
        pnls_down.append([strategy + "_mrn", results[window_str][strategy]["strategy"]["pnl_sum_mrn_DOW"]])
        srs_dowg.append([strategy + "_momg", results[window_str][strategy]["strategy"]["sr_annual_momg_DOW"]])
        srs_down.append([strategy + "_momn", results[window_str][strategy]["strategy"]["sr_annual_momn_DOW"]])
        srs_dowg.append([strategy + "_mrg", results[window_str][strategy]["strategy"]["sr_annual_mrg_DOW"]])
        srs_down.append([strategy + "_mrn", results[window_str][strategy]["strategy"]["sr_annual_mrn_DOW"]])

    main_results_sorted[window_str]["pnls_spg_sorted"] = sorted(pnls_spg, key=lambda x: x[1], reverse=True)
    main_results_sorted[window_str]["pnls_spn_sorted"] = sorted(pnls_spn, key=lambda x: x[1], reverse=True)
    main_results_sorted[window_str]["srs_spg_sorted"] = sorted(srs_spg, key=lambda x: x[1], reverse=True)
    main_results_sorted[window_str]["srs_spn_sorted"] = sorted(srs_spn, key=lambda x: x[1], reverse=True)


    main_results_sorted[window_str]["pnls_dowg_sorted"] = sorted(pnls_dowg, key=lambda x: x[1], reverse=True)
    main_results_sorted[window_str]["pnls_down_sorted"] = sorted(pnls_down, key=lambda x: x[1], reverse=True)
    main_results_sorted[window_str]["srs_dowg_sorted"] = sorted(srs_dowg, key=lambda x: x[1], reverse=True)
    main_results_sorted[window_str]["srs_down_sorted"] = sorted(srs_down, key=lambda x: x[1], reverse=True)
```

```
main_results_sorted.keys()

main_results_sorted["1999-12-31_2025-09-30"]["pnls_spn_sorted"]

data_bar = main_results_sorted["1999-12-31_2025-09-30"]["pnls_spn_sorted"]

labels = [row[0] for row in data_bar]

values = [row[1] for row in data_bar]


plt.figure(figsize=(12, 6))

plt.bar(labels, values)


plt.xticks(rotation=90, ha='right')


plt.show

plt.figure(figsize=(12, 6))

plt.plot(

    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-30_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],

    label="rbc_10cal_1999-12-31_2025-09-30_30lag"

)

plt.plot(

    results["1999-12-31_2025-09-30"]["bh_1999-12-31_2025-09-30"]["strategy"]["pnl_cumsum_SP"],

    label="bh_1999-12-31_2025-09-30"

)

plt.grid(True)

plt.legend()
```

```
plt.show()

plt.figure(figsize=(12, 6))

plt.plot(

    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-
30_30lag"]["signals"]["pos_final_mr"],

    label="rbc_10cal_1999-12-31_2025-09-30_30lag"

)

# plt.grid(True)

plt.legend()


plt.show()

results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-30_30lag"]["signals"]

fig, ax1 = plt.subplots(figsize=(12, 6))


# Primary y-axis: PnL

ax1.plot(

    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-
30_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],

    label="rbc_10cal_1999-12-31_2025-09-30_30lag"

)


ax1.plot(

    results["1999-12-31_2025-09-30"]["bh_1999-12-31_2025-09-
30"]["strategy"]["pnl_cumsum_SP"],
```

```
    label="bh_1999-12-31_2025-09-30"
)

ax1.set_ylabel("PnL")
ax1.grid(True)

# Secondary y-axis: positions
ax2 = ax1.twinx()

ax2.plot(
    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-
30_30lag"]["signals"]["pos_final_mr"],
    color="green",
    alpha=0.5,
    label="rbc_10cal_1999-12-31_2025-09-30_30lag_pos"
)

ax2.set_ylabel("Position")

# Combined legend
lines_1, labels_1 = ax1.get_legend_handles_labels()
lines_2, labels_2 = ax2.get_legend_handles_labels()

ax1.legend(lines_1 + lines_2, labels_1 + labels_2, loc="best")
```

```
plt.title("PnL and Position: rbc_10cal_1999-12-31_2025-09-30_30lag vs bh_1999-12-31_2025-09-30")

plt.show()

def parse_strategy(strategy_name, metric, metric_name):

    # Buy-and-hold special case

    if strategy_name.startswith("bh_"):

        parts = strategy_name.split("_")


        return {

            "Model": "BH",

            "Data Length": "N/A",

            "Horizon": f"{parts[1]}_{parts[2]}",

            "Signal Lag": "N/A",

            "Leverage": 1,

            "Double Down": "No",

            "Strategy Type": "N/A",

            metric_name: round(metric, 2),

        }


    parts = strategy_name.split("_")


    model = parts[0].upper()

    data_length = parts[1].replace("cal", "y")

    horizon = f"{parts[2]}_{parts[3]}"
```

```
leverage = 1
double_down = "No"

strategy_type = None
signal_lag = None

for token in parts[4:]:
    if token.endswith("lag"):
        signal_lag = token.replace("lag", "b")
    elif token.endswith("lev"):
        leverage = int(token.replace("lev", ""))
    elif token == "dd":
        double_down = "Yes"
    elif token in ["mrn", "mrg"]:
        strategy_type = "MR"
    elif token in ["momn", "momg"]:
        strategy_type = "MOM"

return {
    "Model": model,
    "Data Length": data_length,
    "Horizon": horizon,
    "Signal Lag": signal_lag,
    "Leverage": leverage,
```

```
        "Double Down": double_down,
        "Strategy Type": strategy_type,
        metric_name: round(metric, 2),
    }
results_full_pnls_spn = main_results_sorted["1999-12-31_2025-09-30"]["pnls_spn_sorted"]
results_full_pnls_spg = main_results_sorted["1999-12-31_2025-09-30"]["pnls_spg_sorted"]
results_full_srs_spn = main_results_sorted["1999-12-31_2025-09-30"]["srs_spn_sorted"]
results_full_srs_spg = main_results_sorted["1999-12-31_2025-09-30"]["srs_spg_sorted"]
results_full_pnls_spn_rows = [parse_strategy(strategy, metric, "Net PnL") for strategy, metric in results_full_pnls_spn]
results_full_pnls_spn_formatted = pd.DataFrame(results_full_pnls_spn_rows).set_index(["Model", "Data Length", "Horizon", "Signal Lag", "Leverage", "Double Down", "Strategy Type"])

results_full_pnls_spg_rows = [parse_strategy(strategy, metric, "Gross PnL") for strategy, metric in results_full_pnls_spg]
results_full_pnls_spg_formatted = pd.DataFrame(results_full_pnls_spg_rows).set_index(["Model", "Data Length", "Horizon", "Signal Lag", "Leverage", "Double Down", "Strategy Type"])

results_full_srs_spn_rows = [parse_strategy(strategy, metric, "Net SR") for strategy, metric in results_full_srs_spn]
results_full_srs_spn_formatted = pd.DataFrame(results_full_srs_spn_rows).set_index(["Model", "Data Length", "Horizon", "Signal Lag", "Leverage", "Double Down", "Strategy Type"])
```

```
results_full_srs_spg_rows = [parse_strategy(strategy, metric, "Gross SR") for strategy, metric in results_full_srs_spg]

results_full_srs_spg_formatted = pd.DataFrame(results_full_srs_spg_rows).set_index(["Model", "Data Length", "Horizon", "Signal Lag", "Leverage", "Double Down", "Strategy Type"])


# Optional: sort by Net PnL descending

# df = df.sort_values("Net PnL", ascending=False).reset_index(drop=True)

results_full_sp = results_full_pnls_spn_formatted.join(results_full_pnls_spg_formatted).join(results_full_srs_spn_formatted).join(results_full_srs_spg_formatted).reset_index()

#results_full_sp

results_comb_dict = {}


for asset in ["sp", "dow"]:

    results_comb_list = []


    for horizon in main_results_sorted.keys():

        results_pnls_n = main_results_sorted[horizon]["pnls_" + asset + "n_sorted"]

        results_pnls_n_rows = [parse_strategy(strategy, metric, "Net PnL") for strategy, metric in results_pnls_n]

        results_pnls_n_formatted = pd.DataFrame(results_pnls_n_rows).set_index(["Model", "Data Length", "Horizon", "Signal Lag", "Leverage", "Double Down", "Strategy Type"])


        results_pnls_g = main_results_sorted[horizon]["pnls_" + asset + "g_sorted"]

        results_pnls_g_rows = [parse_strategy(strategy, metric, "Gross PnL") for strategy, metric in results_pnls_g]
```

```
        results_pnls_g_formatted = pd.DataFrame(results_pnls_g_rows).set_index(["Model", "Data Length", "Horizon", "Signal Lag", "Leverage", "Double Down", "Strategy Type"])


        results_srs_n = main_results_sorted[horizon]["srs_" + asset + "n_sorted"]
        results_srs_n_rows = [parse_strategy(strategy, metric, "Net SR") for strategy, metric in results_srs_n]
        results_srs_n_formatted = pd.DataFrame(results_srs_n_rows).set_index(["Model", "Data Length", "Horizon", "Signal Lag", "Leverage", "Double Down", "Strategy Type"])


        results_srs_g = main_results_sorted[horizon]["srs_" + asset + "g_sorted"]
        results_srs_g_rows = [parse_strategy(strategy, metric, "Gross SR") for strategy, metric in results_srs_g]
        results_srs_g_formatted = pd.DataFrame(results_srs_g_rows).set_index(["Model", "Data Length", "Horizon", "Signal Lag", "Leverage", "Double Down", "Strategy Type"])


        results_comb = results_pnls_n_formatted.join(results_pnls_g_formatted).join(results_srs_n_formatted).join(results_srs_g_formatted).reset_index()
        results_comb_list.append(results_comb)
        results_comb.to_csv("results/summary_" + asset + "_" + horizon + ".csv", index=False)


    results_comb_dict[asset] = results_comb_list
### Gross and Net Performance
fig, ax1 = plt.subplots(figsize=(12, 6))


# Primary y-axis: PnL
```

```
ax1.plot(
    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-
30_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="Net PnL"
)

ax1.plot(
    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-
30_30lag"]["strategy"]["pnl_cumsum_mrg_SP"],
    label="Gross PnL"
)

ax1.set_ylabel("PnL")
ax1.grid(True)

# Secondary y-axis: positions
ax2 = ax1.twinx()

ax2.plot(
    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-
30_30lag"]["signals"]["pos_final_mr"],
    color="green",
    alpha=0.5,
    label="Position"
)
```

```
ax2.set_ylabel("Position")

# Combined legend
lines_1, labels_1 = ax1.get_legend_handles_labels()
lines_2, labels_2 = ax2.get_legend_handles_labels()

ax1.legend(lines_1 + lines_2, labels_1 + labels_2, loc="best")

plt.title("PnL and Position: rbc_10cal_1999-12-31_2025-09-30_30lag")
plt.show()
representative_pos = results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-30_30lag"]["signals"]["pos_final_mr"]
representative_pos_change = representative_pos.diff().fillna(0)
representative_pos_change_bin = representative_pos_change.apply(lambda x: 1 if x == 0 else 0).to_list()
pos_counts = []
pos_count = 0
stop = len(representative_pos_change_bin) - 1
for i, n in enumerate(representative_pos_change_bin):
    if n == 1:
        pos_count += 1
        if i == stop:
            pos_counts.append(pos_count)
    else:
```

```
        if pos_count > 0:
            pos_counts.append(pos_count)
        pos_count = 0
pos_counts
len(representative_pos) # 104 quarters total
sum(pos_counts) # of which 81 in which pos fixed
changes = [n for n in representative_pos_change_bin if n == 0]
len(changes) # of which new pos
pos_counts_all = [int(n) for n in np.ones(len(changes))] + pos_counts
np.mean(pos_counts_all) # mean number of quarters a position is held
position_changes_count_dict = {}
position_changes_perc_dict = {}
for strat in results["1999-12-31_2025-09-30"].keys():
    if "bh" not in strat:
        positions_diff = results["1999-12-31_2025-09-30"][strat]["strategy"]["signals_actual"]["pos_final_mom"].diff().fillna(0)
        position_changes = positions_diff[positions_diff != 0]
        position_changes_count = len(position_changes)
        position_changes_count_dict[strat] = position_changes_count

        daily_positions = results["1999-12-31_2025-09-30"][strat]["strategy"]["pos_momg_SP"]
        positions_total_count = len(daily_positions)

        position_changes_perc = round(position_changes_count / positions_total_count * 100, 2)
        position_changes_perc_dict[strat] = position_changes_perc
```

```
positions_total_count

position_changes_df = pd.DataFrame([position_changes_count_dict, position_changes_perc_dict]).T

position_changes_df.columns = ["Position Changes Count", "Position Changes Percentage"]

position_changes_df["Position Changes Count"] = position_changes_df.apply(lambda row: int(row["Position Changes Count"]), axis=1)

position_changes_df = position_changes_df.sort_values(by="Position Changes Count", ascending=False).reset_index().rename(columns={"index": "Strategy"})

position_changes_df

### Horizon

for asset in ["sp", "dow"]:
    horizons_cmp_list = []
    for results_comb in results_comb_dict[asset]:
        horizons_cmp_list.append(results_comb[
            (results_comb["Data Length"] == "10y") &
            (results_comb["Signal Lag"] == "30b") &
            (results_comb["Leverage"] == 1) &
            (results_comb["Double Down"] == "No") &
            (results_comb["Strategy Type"] == "MR")
        ])
    horizons_cmp = pd.concat(horizons_cmp_list)
    horizons_cmp.to_csv("results/horizon_comparison_" + asset + ".csv", index=False)
fig, ax1 = plt.subplots(figsize=(12, 6))


# Primary y-axis: PnL
```

```
ax1.plot(
   results["1999-12-31_2007-12-31"]["rbc_10cal_1999-12-31_2007-12-
31_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
   label="1999-12-31_2007-12-31"
)

ax1.plot(
   results["2007-12-31_2009-12-31"]["rbc_10cal_2007-12-31_2009-12-
31_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
   label="2007-12-31_2009-12-31"
)

ax1.plot(
   results["2009-12-31_2019-12-31"]["rbc_10cal_2009-12-31_2019-12-
31_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
   label="2009-12-31_2019-12-31"
)

ax1.plot(
   results["2019-12-31_2022-12-31"]["rbc_10cal_2019-12-31_2022-12-
31_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
   label="2019-12-31_2022-12-31"
)

ax1.plot(
```

```
    results["2022-12-31_2025-09-30"]["rbc_10cal_2022-12-31_2025-09-30_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="2022-12-31_2025-09-30"
)

ax1.plot(
    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-30_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="1999-12-31_2025-09-30",
    linewidth=0.2
)

ax1.set_ylabel("PnL")
ax1.grid(True)

# Combined legend
lines_1, labels_1 = ax1.get_legend_handles_labels()

ax1.legend(lines_1, labels_1, loc="best")

plt.title("PnL by Horizon")
plt.show()
hybrid_1_s = results["1999-12-31_2007-12-31"]["rbc_10cal_1999-12-31_2007-12-31_30lag"]["strategy"]["pnl_cumsum_mrn_SP"].copy()
hybrid_2_s = results["2007-12-31_2009-12-31"]["rbc_10cal_2007-12-31_2009-12-31_30lag"]["strategy"]["pnl_cumsum_momn_SP"].copy()
```

```
hybrid_2_s += hybrid_1_s[-1]
hybrid_2_s = hybrid_2_s[hybrid_2_s.index > hybrid_1_s.index[-1]]
hybrid_3_s = results["2009-12-31_2019-12-31"]["rbc_10cal_2009-12-31_2019-12-31_30lag"]["strategy"]["pnl_cumsum_mrn_SP"].copy()
hybrid_3_s += hybrid_2_s[-1]
hybrid_3_s = hybrid_3_s[hybrid_3_s.index > hybrid_2_s.index[-1]]
hybrid_4_s = results["2019-12-31_2022-12-31"]["rbc_10cal_2019-12-31_2022-12-31_30lag"]["strategy"]["pnl_cumsum_momn_SP"].copy()
hybrid_4_s += hybrid_3_s[-1]
hybrid_4_s = hybrid_4_s[hybrid_4_s.index > hybrid_3_s.index[-1]]
hybrid_5_s = results["2022-12-31_2025-09-30"]["rbc_10cal_2022-12-31_2025-09-30_30lag"]["strategy"]["pnl_cumsum_mrn_SP"].copy()
hybrid_5_s += hybrid_4_s[-1]
hybrid_5_s = hybrid_5_s[hybrid_5_s.index > hybrid_4_s.index[-1]]
hybrid_s = pd.concat([
    hybrid_1_s,
    hybrid_2_s,
    hybrid_3_s,
    hybrid_4_s,
    hybrid_5_s
])
hybrid_s
fig, ax1 = plt.subplots(figsize=(12, 6))

# Primary y-axis: PnL
```

```
ax1.plot(
    results["1999-12-31_2007-12-31"]["rbc_10cal_1999-12-31_2007-12-31_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="1999-12-31_2007-12-31 MR"
)

ax1.plot(
    results["2007-12-31_2009-12-31"]["rbc_10cal_2007-12-31_2009-12-31_30lag"]["strategy"]["pnl_cumsum_momn_SP"],
    label="2007-12-31_2009-12-31 MOM"
)

ax1.plot(
    results["2009-12-31_2019-12-31"]["rbc_10cal_2009-12-31_2019-12-31_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="2009-12-31_2019-12-31 MR"
)

ax1.plot(
    results["2019-12-31_2022-12-31"]["rbc_10cal_2019-12-31_2022-12-31_30lag"]["strategy"]["pnl_cumsum_momn_SP"],
    label="2019-12-31_2022-12-31 MOM"
)

ax1.plot(
```

```
    results["2022-12-31_2025-09-30"]["rbc_10cal_2022-12-31_2025-09-30_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="2022-12-31_2025-09-30 MR"
)

ax1.plot(
    hybrid_s,
    label="1999-12-31_2025-09-30 Hybrid MR-MOM",
    linewidth=0.2
)

ax1.set_ylabel("PnL")
ax1.grid(True)

# Combined legend
lines_1, labels_1 = ax1.get_legend_handles_labels()

ax1.legend(lines_1, labels_1, loc="best")

plt.title("PnL by Horizon with MR-MOM Switching")
plt.show()
fig, ax1 = plt.subplots(figsize=(12, 6))

# Primary y-axis: PnL
ax1.plot(
```

```
    results["1999-12-31_2007-12-31"]["rbc_10cal_1999-12-31_2007-12-31_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="1999-12-31_2007-12-31 SP",
    color="blue"
)

ax1.plot(
    results["2007-12-31_2009-12-31"]["rbc_10cal_2007-12-31_2009-12-31_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="2007-12-31_2009-12-31 SP",
    color="blue"
)

ax1.plot(
    results["2009-12-31_2019-12-31"]["rbc_10cal_2009-12-31_2019-12-31_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="2009-12-31_2019-12-31 SP",
    color="blue"
)

ax1.plot(
    results["2019-12-31_2022-12-31"]["rbc_10cal_2019-12-31_2022-12-31_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="2019-12-31_2022-12-31 SP",
    color="blue"
)
```

```
ax1.plot(
    results["2022-12-31_2025-09-30"]["rbc_10cal_2022-12-31_2025-09-30_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="2022-12-31_2025-09-30 SP",
    color="blue"
)

ax1.plot(
    results["1999-12-31_2007-12-31"]["rbc_10cal_1999-12-31_2007-12-31_30lag"]["strategy"]["pnl_cumsum_mrn_DOW"],
    label="1999-12-31_2007-12-31 DOW",
    color="orange"
)

ax1.plot(
    results["2007-12-31_2009-12-31"]["rbc_10cal_2007-12-31_2009-12-31_30lag"]["strategy"]["pnl_cumsum_mrn_DOW"],
    label="2007-12-31_2009-12-31 DOW",
    color="orange"
)

ax1.plot(
    results["2009-12-31_2019-12-31"]["rbc_10cal_2009-12-31_2019-12-31_30lag"]["strategy"]["pnl_cumsum_mrn_DOW"],
    label="2009-12-31_2019-12-31 DOW",
```

```
    color="orange"
)

ax1.plot(
    results["2019-12-31_2022-12-31"]["rbc_10cal_2019-12-31_2022-12-31_30lag"]["strategy"]["pnl_cumsum_mrn_DOW"],
    label="2019-12-31_2022-12-31 DOW",
    color="orange"
)

ax1.plot(
    results["2022-12-31_2025-09-30"]["rbc_10cal_2022-12-31_2025-09-30_30lag"]["strategy"]["pnl_cumsum_mrn_DOW"],
    label="2022-12-31_2025-09-30 DOW",
    color="orange"
)

ax1.set_ylabel("PnL")
ax1.grid(True)

# Combined legend
lines_1, labels_1 = ax1.get_legend_handles_labels()

ax1.legend(lines_1, labels_1, loc="best")
```

```
plt.title("PnL by Horizon: SP vs DOW")
plt.show()
### Strategy Type
data_bar = main_results_sorted["1999-12-31_2025-09-30"]["pnls_spn_sorted"]
data_bar = [d for d in data_bar if "lev" not in d[0] and "dd" not in d[0] and "bh" not in d[0]]
labels = [row[0] for row in data_bar]
values = [row[1] for row in data_bar]


plt.figure(figsize=(12, 6))
plt.bar(labels, values)


plt.xticks(rotation=90, ha='right')
plt.ylabel("Net PnL")
plt.title("PnL: MR vs MOM")
plt.show
data_bar = main_results_sorted["1999-12-31_2025-09-30"]["srs_spn_sorted"]
data_bar = [d for d in data_bar if "lev" not in d[0] and "dd" not in d[0] and "bh" not in d[0]]
labels = [row[0] for row in data_bar]
values = [row[1] for row in data_bar]


plt.figure(figsize=(12, 6))
plt.bar(labels, values)


plt.xticks(rotation=90, ha='right')
```

```
plt.ylabel("Net SR")
plt.title("SR: MR vs MOM")
plt.show
### Data Length
for asset in ["sp"]:
    datalengths_cmp_list = []
    for results_comb in results_comb_dict[asset]:
        datalengths_cmp_list.append(results_comb[
            # (results_comb["Horizon"] == "1999-12-31_2025-09-30") &
            (results_comb["Signal Lag"] == "30b") &
            (results_comb["Leverage"] == 1) &
            (results_comb["Double Down"] == "No") &
            (results_comb["Strategy Type"] == "MR")
        ])
    datalengths_cmp = pd.concat(datalengths_cmp_list)
    datalengths_cmp.to_csv("results/datalength_comparison_" + asset + "_30b.csv", index=False)
datalengths_cmp
for asset in ["sp"]:
    datalengths_cmp_list = []
    for results_comb in results_comb_dict[asset]:
        datalengths_cmp_list.append(results_comb[
            # (results_comb["Horizon"] == "1999-12-31_2025-09-30") &
            (results_comb["Signal Lag"] == "20b") &
            (results_comb["Leverage"] == 1) &
```

```
            (results_comb["Double Down"] == "No") &
            (results_comb["Strategy Type"] == "MR")
        ])
    datalengths_cmp = pd.concat(datalengths_cmp_list)
    datalengths_cmp.to_csv("results/datalength_comparison_" + asset + "_20b.csv", index=False)
datalengths_cmp
for asset in ["sp"]:
    datalengths_cmp_list = []
    for results_comb in results_comb_dict[asset]:
        datalengths_cmp_list.append(results_comb[
            # (results_comb["Horizon"] == "1999-12-31_2025-09-30") &
            (results_comb["Signal Lag"] == "10b") &
            (results_comb["Leverage"] == 1) &
            (results_comb["Double Down"] == "No") &
            (results_comb["Strategy Type"] == "MR")
        ])
    datalengths_cmp = pd.concat(datalengths_cmp_list)
    datalengths_cmp.to_csv("results/datalength_comparison_" + asset + "_10b.csv", index=False)
datalengths_cmp
### Signal Lag
for asset in ["sp"]:
    signallag_cmp_list = []
    for results_comb in results_comb_dict[asset]:
```

```
        signallag_cmp_list.append(results_comb[
            (results_comb["Data Length"] == "10y") &
            # (results_comb["Horizon"] == "1999-12-31_2025-09-30") &
            (results_comb["Leverage"] == 1) &
            (results_comb["Double Down"] == "No") &
            (results_comb["Strategy Type"] == "MR")
        ])
    signallag_cmp = pd.concat(signallag_cmp_list)
    signallag_cmp.to_csv("results/signallag_comparison_" + asset + "_mr.csv", index=False)
signallag_cmp
for asset in ["sp"]:
    signallag_cmp_list = []
    for results_comb in results_comb_dict[asset]:
        signallag_cmp_list.append(results_comb[
            (results_comb["Data Length"] == "10y") &
            # (results_comb["Horizon"] == "1999-12-31_2025-09-30") &
            (results_comb["Leverage"] == 1) &
            (results_comb["Double Down"] == "No") &
            (results_comb["Strategy Type"] == "MOM")
        ])
    signallag_cmp = pd.concat(signallag_cmp_list)
    signallag_cmp.to_csv("results/signallag_comparison_" + asset + "_mom.csv", index=False)
signallag_cmp
### Leverage and Doubling Down
```

```
fig, ax1 = plt.subplots(figsize=(12, 6))

# Primary y-axis: PnL
ax1.plot(
    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-30_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="rbc_10cal_1999-12-31_2025-09-30_30lag"
)

ax1.plot(
    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-30_30lag_dd"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="rbc_10cal_1999-12-31_2025-09-30_30lag_dd"
)

ax1.plot(
    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-30_30lag_2lev"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="rbc_10cal_1999-12-31_2025-09-30_30lag_2lev"
)

ax1.plot(
    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-30_30lag_2lev_dd"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="rbc_10cal_1999-12-31_2025-09-30_30lag_2lev_dd"
)
```

```
ax1.set_ylabel("Net PnL")
ax1.grid(True)

# Combined legend
lines_1, labels_1 = ax1.get_legend_handles_labels()


ax1.legend(lines_1, labels_1, loc="best")


plt.title("Impact of Leverage on PnL")
plt.show()
### BH
fig, ax1 = plt.subplots(figsize=(12, 6))

# Primary y-axis: PnL
ax1.plot(
    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-30_30lag"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="rbc_10cal_1999-12-31_2025-09-30_30lag"
)


ax1.plot(
    results["1999-12-31_2025-09-30"]["rbc_10cal_1999-12-31_2025-09-30_30lag_2lev_dd"]["strategy"]["pnl_cumsum_mrn_SP"],
    label="rbc_10cal_1999-12-31_2025-09-30_30lag_2lev_dd"
```

```
)

ax1.plot(
    results["1999-12-31_2025-09-30"]["bh_1999-12-31_2025-09-
30"]["strategy"]["pnl_cumsum_SP"],
    label="bh_1999-12-31_2025-09-30"
)

ax1.set_ylabel("Net PnL")
ax1.grid(True)

# Combined legend
lines_1, labels_1 = ax1.get_legend_handles_labels()

ax1.legend(lines_1, labels_1, loc="best")

plt.title("DSGE Strategy vs BH")
plt.show()
```

Appendix C: Basic RBC Model YAML Code

name: Real Business Cycle

symbols:

  exogenous: [e_z]
  states: [z, k]

```
  controls: [n, i]

  parameters: [beta, sigma, eta, chi, delta, alpha, rho, zbar, sig_z]


definitions: |

   y[t] = exp(z[t])*k[t]^alpha*n[t]^(1-alpha)

   c[t] = y[t] - i[t]

   rk[t] = alpha*y[t]/k[t]

   w[t] = (1-alpha)*y[t]/n[t]


equations:


   arbitrage: |

      chi*n[t]^eta*c[t]^sigma - w[t]                    ⊥ 0.0 <= n[t] <= inf

      1 - beta*(c[t]/c[t+1])^(sigma)*(1-delta+rk[t+1])   ⊥ -inf <= i[t] <= inf


   transition: |

      z[t] = rho*z[t-1] + e_z

      k[t] = (1-delta)*k[t-1] + i[t-1]


calibration:


   # parameters

   beta : 0.99

   phi: 1
```

```
delta : 0.025

alpha : 0.33

rho : 0.8

sigma: 5

eta: 1

sig_z: 0.016

zbar: 0

chi : w/c^sigma/n^eta

c_i: 1.5

c_y: 0.5

e_z: 0.0

n: 0.33

z: zbar

rk: 1/beta-1+delta

w: (1-alpha)*exp(z)*(k/n)^(alpha)

k: n/(rk/alpha)^(1/(1-alpha))

y: exp(z)*k^alpha*n^(1-alpha)

i: delta*k

c: y - i

V: log(c)/(1-beta)

u: c^(1-sigma)/(1-sigma) - chi*n^(1+eta)/(1+eta)

m: beta/c^sigma*(1-delta+rk)

kss: 10
```

```
exogenous: !UNormal
   sigma: 0.01


domain:
   z: [-2*sig_z/(1-rho^2)^0.5,  2*sig_z/(1-rho^2)^0.5]
   k: [ kss*0.5, kss*1.5]


options:
   grid: !Cartesian
      n: [100, 100]
   discrete_choices: [n]
```

Appendix D: Strategy Results across Horizons

S&P500 12/1999-12/2007

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	1999-12-31_2007-12-31	30b	2	Yes	MR	2.24	2.24	0.47	0.47
RBC	5y	1999-12-31_2007-12-31	20b	2	Yes	MR	2.02	2.02	0.42	0.42
RBC	5y	1999-12-31_2007-12-31	10b	2	Yes	MR	1.71	1.71	0.36	0.36
RBC	5y	1999-12-31_2007-12-31	30b	2	No	MR	1.13	1.13	0.39	0.39
RBC	5y	1999-12-31_2007-12-31	30b	1	Yes	MR	1.12	1.12	0.47	0.47
RBC	10y	1999-12-31_2007-12-31	30b	2	Yes	MR	1.06	1.06	0.22	0.22
RBC	5y	1999-12-31_2007-12-31	20b	1	Yes	MR	1.01	1.01	0.42	0.42
RBC	5y	1999-12-31_2007-12-31	20b	2	No	MR	0.96	0.97	0.33	0.33
RBC	5y	1999-12-31_2007-12-31	10b	1	Yes	MR	0.85	0.85	0.36	0.36
RBC	10y	1999-12-31_2007-12-31	20b	2	Yes	MR	0.78	0.78	0.16	0.16
RBC	5y	1999-12-31_2007-12-31	30b	1	No	MR	0.56	0.57	0.39	0.39
RBC	10y	1999-12-31_2007-12-31	30b	2	No	MR	0.54	0.54	0.19	0.19
RBC	10y	1999-12-31_2007-12-31	30b	1	Yes	MR	0.53	0.53	0.22	0.22
RBC	5y	1999-12-31_2007-12-31	10b	2	No	MR	0.52	0.52	0.18	0.18
RBC	5y	1999-12-31_2007-12-31	20b	1	No	MR	0.48	0.48	0.33	0.33

RBC	10y	1999-12-31_2007-12-31	20b	1	Yes	MR	0.39	0.39	0.16	0.16
RBC	10y	1999-12-31_2007-12-31	20b	2	No	MR	0.32	0.32	0.11	0.11
RBC	10y	1999-12-31_2007-12-31	10b	2	No	MOM	0.32	0.32	0.11	0.11
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.27	0.27	0.19	0.19
RBC	5y	1999-12-31_2007-12-31	10b	1	No	MR	0.26	0.26	0.18	0.18
RBC	10y	1999-12-31_2007-12-31	10b	2	Yes	MR	0.19	0.19	0.04	0.04
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MR	0.16	0.16	0.11	0.11
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MOM	0.16	0.16	0.11	0.11
RBC	10y	1999-12-31_2007-12-31	10b	1	Yes	MR	0.1	0.1	0.04	0.04
BH	N/A	1999-12-31_2007-12-31	N/A	1	No	N/A	0.03	0.03	0.02	0.02
RBC	10y	1999-12-31_2007-12-31	10b	1	Yes	MOM	-0.1	-0.1	-0.04	-0.04
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MR	-0.16	-0.16	-0.11	-0.11
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MOM	-0.16	-0.16	-0.11	-0.11
RBC	10y	1999-12-31_2007-12-31	10b	2	Yes	MOM	-0.19	-0.19	-0.04	-0.04
RBC	5y	1999-12-31_2007-12-31	10b	1	No	MOM	-0.26	-0.26	-0.18	-0.18
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MOM	-0.27	-0.27	-0.19	-0.19
RBC	10y	1999-12-31_2007-12-31	10b	2	No	MR	-0.32	-0.32	-0.11	-0.11
RBC	10y	1999-12-31_2007-12-31	20b	2	No	MOM	-0.32	-0.32	-0.11	-0.11
RBC	10y	1999-12-31_2007-12-31	20b	1	Yes	MOM	-0.39	-0.39	-0.16	-0.16
RBC	5y	1999-12-31_2007-12-31	20b	1	No	MOM	-0.48	-0.48	-0.33	-0.33
RBC	5y	1999-12-31_2007-12-31	10b	2	No	MOM	-0.53	-0.52	-0.18	-0.18
RBC	10y	1999-12-31_2007-12-31	30b	1	Yes	MOM	-0.53	-0.53	-0.22	-0.22
RBC	10y	1999-12-31_2007-12-31	30b	2	No	MOM	-0.54	-0.54	-0.19	-0.19
RBC	5y	1999-12-31_2007-12-31	30b	1	No	MOM	-0.57	-0.57	-0.39	-0.39
RBC	10y	1999-12-31_2007-12-31	20b	2	Yes	MOM	-0.79	-0.78	-0.16	-0.16
RBC	5y	1999-12-31_2007-12-31	10b	1	Yes	MOM	-0.86	-0.85	-0.36	-0.36
RBC	5y	1999-12-31_2007-12-31	20b	2	No	MOM	-0.97	-0.97	-0.33	-0.33
RBC	5y	1999-12-31_2007-12-31	20b	1	Yes	MOM	-1.01	-1.01	-0.42	-0.42
RBC	10y	1999-12-31_2007-12-31	30b	2	Yes	MOM	-1.06	-1.06	-0.22	-0.22
RBC	5y	1999-12-31_2007-12-31	30b	1	Yes	MOM	-1.12	-1.12	-0.47	-0.47
RBC	5y	1999-12-31_2007-12-31	30b	2	No	MOM	-1.13	-1.13	-0.39	-0.39
RBC	5y	1999-12-31_2007-12-31	10b	2	Yes	MOM	-1.71	-1.71	-0.36	-0.36
RBC	5y	1999-12-31_2007-12-31	20b	2	Yes	MOM	-2.03	-2.02	-0.42	-0.42
RBC	5y	1999-12-31_2007-12-31	30b	2	Yes	MOM	-2.25	-2.24	-0.47	-0.47

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR

RBC	5y	1999-12-31_2007-12-31	30b	2	Yes	MR	1.94	1.95	0.42	0.42
RBC	5y	1999-12-31_2007-12-31	20b	2	Yes	MR	1.85	1.86	0.4	0.4
RBC	5y	1999-12-31_2007-12-31	10b	2	Yes	MR	1.37	1.37	0.3	0.3
RBC	5y	1999-12-31_2007-12-31	30b	1	Yes	MR	0.97	0.97	0.42	0.42
RBC	5y	1999-12-31_2007-12-31	30b	2	No	MR	0.94	0.94	0.33	0.33
RBC	5y	1999-12-31_2007-12-31	20b	1	Yes	MR	0.93	0.93	0.4	0.4
RBC	5y	1999-12-31_2007-12-31	20b	2	No	MR	0.92	0.92	0.33	0.33
RBC	10y	1999-12-31_2007-12-31	30b	2	Yes	MR	0.81	0.81	0.17	0.17
RBC	5y	1999-12-31_2007-12-31	10b	1	Yes	MR	0.68	0.68	0.3	0.3
RBC	10y	1999-12-31_2007-12-31	20b	2	Yes	MR	0.64	0.64	0.13	0.13
RBC	5y	1999-12-31_2007-12-31	30b	1	No	MR	0.47	0.47	0.33	0.33
RBC	5y	1999-12-31_2007-12-31	20b	1	No	MR	0.46	0.46	0.33	0.33
RBC	10y	1999-12-31_2007-12-31	30b	1	Yes	MR	0.4	0.4	0.17	0.17
RBC	10y	1999-12-31_2007-12-31	10b	2	No	MOM	0.37	0.37	0.13	0.13
RBC	10y	1999-12-31_2007-12-31	30b	2	No	MR	0.36	0.36	0.13	0.13
RBC	5y	1999-12-31_2007-12-31	10b	2	No	MR	0.34	0.34	0.12	0.12
RBC	10y	1999-12-31_2007-12-31	20b	1	Yes	MR	0.32	0.32	0.13	0.13
RBC	10y	1999-12-31_2007-12-31	20b	2	No	MR	0.29	0.29	0.1	0.1
BH	N/A	1999-12-31_2007-12-31	N/A	1	No	N/A	0.19	0.19	0.13	0.13
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MOM	0.19	0.19	0.13	0.13
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.18	0.18	0.13	0.13
RBC	5y	1999-12-31_2007-12-31	10b	1	No	MR	0.17	0.17	0.12	0.12
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MR	0.14	0.14	0.1	0.1
RBC	10y	1999-12-31_2007-12-31	10b	2	Yes	MOM	0	0.01	0	0
RBC	10y	1999-12-31_2007-12-31	10b	1	Yes	MOM	0	0	0	0
RBC	10y	1999-12-31_2007-12-31	10b	1	Yes	MR	0	0	0	0
RBC	10y	1999-12-31_2007-12-31	10b	2	Yes	MR	-0.01	-0.01	0	0
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MOM	-0.15	-0.14	-0.1	-0.1
RBC	5y	1999-12-31_2007-12-31	10b	1	No	MOM	-0.17	-0.17	-0.12	-0.12
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MOM	-0.18	-0.18	-0.13	-0.13
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MR	-0.19	-0.19	-0.13	-0.13
RBC	10y	1999-12-31_2007-12-31	20b	2	No	MOM	-0.29	-0.29	-0.1	-0.1
RBC	10y	1999-12-31_2007-12-31	20b	1	Yes	MOM	-0.32	-0.32	-0.14	-0.13
RBC	5y	1999-12-31_2007-12-31	10b	2	No	MOM	-0.35	-0.34	-0.12	-0.12
RBC	10y	1999-12-31_2007-12-31	30b	2	No	MOM	-0.37	-0.36	-0.13	-0.13
RBC	10y	1999-12-31_2007-12-31	10b	2	No	MR	-0.38	-0.37	-0.13	-0.13
RBC	10y	1999-12-31_2007-12-31	30b	1	Yes	MOM	-0.41	-0.4	-0.17	-0.17
RBC	5y	1999-12-31_2007-12-31	20b	1	No	MOM	-0.46	-0.46	-0.33	-0.33
RBC	5y	1999-12-31_2007-12-31	30b	1	No	MOM	-0.47	-0.47	-0.34	-0.33

RBC	10y	1999-12-31_2007-12-31	20b	2	Yes	MOM	-0.64	-0.64	-0.14	-0.13
RBC	5y	1999-12-31_2007-12-31	10b	1	Yes	MOM	-0.69	-0.68	-0.3	-0.3
RBC	10y	1999-12-31_2007-12-31	30b	2	Yes	MOM	-0.81	-0.81	-0.17	-0.17
RBC	5y	1999-12-31_2007-12-31	20b	2	No	MOM	-0.92	-0.92	-0.33	-0.33
RBC	5y	1999-12-31_2007-12-31	20b	1	Yes	MOM	-0.93	-0.93	-0.4	-0.4
RBC	5y	1999-12-31_2007-12-31	30b	2	No	MOM	-0.94	-0.94	-0.34	-0.33
RBC	5y	1999-12-31_2007-12-31	30b	1	Yes	MOM	-0.97	-0.97	-0.42	-0.42
RBC	5y	1999-12-31_2007-12-31	10b	2	Yes	MOM	-1.37	-1.37	-0.3	-0.3
RBC	5y	1999-12-31_2007-12-31	20b	2	Yes	MOM	-1.86	-1.86	-0.4	-0.4
RBC	5y	1999-12-31_2007-12-31	30b	2	Yes	MOM	-1.95	-1.95	-0.42	-0.42

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2007-12-31_2009-12-31	10b	2	Yes	MOM	1.8	1.8	0.88	0.88
RBC	10y	2007-12-31_2009-12-31	30b	2	Yes	MOM	1.75	1.75	0.89	0.89
RBC	5y	2007-12-31_2009-12-31	30b	2	Yes	MOM	1.72	1.72	0.84	0.84
RBC	10y	2007-12-31_2009-12-31	20b	2	Yes	MOM	1.62	1.62	0.81	0.81
RBC	5y	2007-12-31_2009-12-31	20b	2	Yes	MOM	1.57	1.57	0.76	0.76
RBC	5y	2007-12-31_2009-12-31	10b	2	Yes	MOM	1.51	1.51	0.71	0.71
RBC	5y	2007-12-31_2009-12-31	30b	2	No	MOM	1.12	1.12	0.76	0.76
RBC	10y	2007-12-31_2009-12-31	10b	2	No	MOM	1.08	1.08	0.72	0.72
RBC	5y	2007-12-31_2009-12-31	20b	2	No	MOM	1	1	0.68	0.68
RBC	10y	2007-12-31_2009-12-31	30b	2	No	MOM	0.99	0.99	0.67	0.67
RBC	5y	2007-12-31_2009-12-31	10b	2	No	MOM	0.95	0.95	0.64	0.64
RBC	10y	2007-12-31_2009-12-31	10b	1	Yes	MOM	0.9	0.9	0.88	0.88
RBC	10y	2007-12-31_2009-12-31	30b	1	Yes	MOM	0.87	0.87	0.89	0.89
RBC	10y	2007-12-31_2009-12-31	20b	2	No	MOM	0.86	0.86	0.58	0.58
RBC	5y	2007-12-31_2009-12-31	30b	1	Yes	MOM	0.86	0.86	0.84	0.84
RBC	10y	2007-12-31_2009-12-31	20b	1	Yes	MOM	0.81	0.81	0.81	0.81
RBC	5y	2007-12-31_2009-12-31	20b	1	Yes	MOM	0.79	0.79	0.76	0.76
RBC	5y	2007-12-31_2009-12-31	10b	1	Yes	MOM	0.75	0.75	0.71	0.71
RBC	5y	2007-12-31_2009-12-31	30b	1	No	MOM	0.56	0.56	0.76	0.76
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MOM	0.54	0.54	0.72	0.72
RBC	5y	2007-12-31_2009-12-31	20b	1	No	MOM	0.5	0.5	0.68	0.68
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MOM	0.5	0.5	0.67	0.67
RBC	5y	2007-12-31_2009-12-31	10b	1	No	MOM	0.47	0.47	0.64	0.64
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MOM	0.43	0.43	0.58	0.58

BH	N/A	2007-12-31_2009-12-31	N/A	1	No	N/A	-0.1	-0.1	-0.14	-0.14
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MR	-0.43	-0.43	-0.58	-0.58
RBC	5y	2007-12-31_2009-12-31	10b	1	No	MR	-0.47	-0.47	-0.64	-0.64
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.5	-0.5	-0.67	-0.67
RBC	5y	2007-12-31_2009-12-31	20b	1	No	MR	-0.5	-0.5	-0.68	-0.68
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MR	-0.54	-0.54	-0.72	-0.72
RBC	5y	2007-12-31_2009-12-31	30b	1	No	MR	-0.56	-0.56	-0.76	-0.76
RBC	5y	2007-12-31_2009-12-31	10b	1	Yes	MR	-0.75	-0.75	-0.71	-0.71
RBC	5y	2007-12-31_2009-12-31	20b	1	Yes	MR	-0.79	-0.79	-0.76	-0.76
RBC	10y	2007-12-31_2009-12-31	20b	1	Yes	MR	-0.81	-0.81	-0.81	-0.81
RBC	5y	2007-12-31_2009-12-31	30b	1	Yes	MR	-0.86	-0.86	-0.84	-0.84
RBC	10y	2007-12-31_2009-12-31	20b	2	No	MR	-0.86	-0.86	-0.58	-0.58
RBC	10y	2007-12-31_2009-12-31	30b	1	Yes	MR	-0.87	-0.87	-0.89	-0.89
RBC	10y	2007-12-31_2009-12-31	10b	1	Yes	MR	-0.9	-0.9	-0.88	-0.88
RBC	5y	2007-12-31_2009-12-31	10b	2	No	MR	-0.95	-0.95	-0.64	-0.64
RBC	10y	2007-12-31_2009-12-31	30b	2	No	MR	-0.99	-0.99	-0.67	-0.67
RBC	5y	2007-12-31_2009-12-31	20b	2	No	MR	-1	-1	-0.68	-0.68
RBC	10y	2007-12-31_2009-12-31	10b	2	No	MR	-1.08	-1.08	-0.72	-0.72
RBC	5y	2007-12-31_2009-12-31	30b	2	No	MR	-1.12	-1.12	-0.76	-0.76
RBC	5y	2007-12-31_2009-12-31	10b	2	Yes	MR	-1.51	-1.51	-0.71	-0.71
RBC	5y	2007-12-31_2009-12-31	20b	2	Yes	MR	-1.58	-1.57	-0.76	-0.76
RBC	10y	2007-12-31_2009-12-31	20b	2	Yes	MR	-1.62	-1.62	-0.81	-0.81
RBC	5y	2007-12-31_2009-12-31	30b	2	Yes	MR	-1.72	-1.72	-0.84	-0.84
RBC	10y	2007-12-31_2009-12-31	30b	2	Yes	MR	-1.75	-1.75	-0.89	-0.89
RBC	10y	2007-12-31_2009-12-31	10b	2	Yes	MR	-1.81	-1.8	-0.88	-0.88

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Model	**Data Length**	**Horizon**	**Signal Lag**	**Leverage**	**Double Down**	**Strategy Type**	**Net PnL**	**Gross PnL**	**Net SR**	**Gross SR**
RBC	10y	2007-12-31_2009-12-31	10b	2	Yes	MOM	1.45	1.45	0.78	0.78
RBC	10y	2007-12-31_2009-12-31	30b	2	Yes	MOM	1.42	1.42	0.8	0.8
RBC	5y	2007-12-31_2009-12-31	30b	2	Yes	MOM	1.35	1.35	0.72	0.72
RBC	10y	2007-12-31_2009-12-31	20b	2	Yes	MOM	1.24	1.24	0.68	0.68
RBC	5y	2007-12-31_2009-12-31	10b	2	Yes	MOM	1.19	1.19	0.62	0.62
RBC	5y	2007-12-31_2009-12-31	20b	2	Yes	MOM	1.16	1.16	0.62	0.62
RBC	5y	2007-12-31_2009-12-31	30b	2	No	MOM	0.84	0.84	0.63	0.63
RBC	10y	2007-12-31_2009-12-31	10b	2	No	MOM	0.83	0.83	0.61	0.62
RBC	5y	2007-12-31_2009-12-31	10b	2	No	MOM	0.79	0.79	0.58	0.58

RBC	5y	2007-12-31_2009-12-31	20b	2	No	MOM	0.73	0.73	0.54	0.54
RBC	10y	2007-12-31_2009-12-31	10b	1	Yes	MOM	0.73	0.73	0.78	0.78
RBC	10y	2007-12-31_2009-12-31	30b	2	No	MOM	0.72	0.72	0.54	0.54
RBC	10y	2007-12-31_2009-12-31	30b	1	Yes	MOM	0.71	0.71	0.8	0.8
RBC	5y	2007-12-31_2009-12-31	30b	1	Yes	MOM	0.67	0.68	0.72	0.72
RBC	10y	2007-12-31_2009-12-31	20b	1	Yes	MOM	0.62	0.62	0.68	0.68
RBC	5y	2007-12-31_2009-12-31	10b	1	Yes	MOM	0.59	0.59	0.62	0.62
RBC	5y	2007-12-31_2009-12-31	20b	1	Yes	MOM	0.58	0.58	0.62	0.62
RBC	10y	2007-12-31_2009-12-31	20b	2	No	MOM	0.57	0.57	0.42	0.42
RBC	5y	2007-12-31_2009-12-31	30b	1	No	MOM	0.42	0.42	0.63	0.63
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MOM	0.42	0.42	0.61	0.62
RBC	5y	2007-12-31_2009-12-31	10b	1	No	MOM	0.4	0.4	0.58	0.58
RBC	5y	2007-12-31_2009-12-31	20b	1	No	MOM	0.36	0.37	0.54	0.54
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MOM	0.36	0.36	0.54	0.54
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MOM	0.29	0.29	0.42	0.42
BH	N/A	2007-12-31_2009-12-31	N/A	1	No	N/A	-0.1	-0.1	-0.14	-0.14
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MR	-0.29	-0.29	-0.43	-0.42
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.36	-0.36	-0.54	-0.54
RBC	5y	2007-12-31_2009-12-31	20b	1	No	MR	-0.37	-0.37	-0.54	-0.54
RBC	5y	2007-12-31_2009-12-31	10b	1	No	MR	-0.4	-0.4	-0.59	-0.58
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MR	-0.42	-0.42	-0.62	-0.62
RBC	5y	2007-12-31_2009-12-31	30b	1	No	MR	-0.42	-0.42	-0.63	-0.63
RBC	10y	2007-12-31_2009-12-31	20b	2	No	MR	-0.57	-0.57	-0.43	-0.42
RBC	5y	2007-12-31_2009-12-31	20b	1	Yes	MR	-0.58	-0.58	-0.62	-0.62
RBC	5y	2007-12-31_2009-12-31	10b	1	Yes	MR	-0.59	-0.59	-0.62	-0.62
RBC	10y	2007-12-31_2009-12-31	20b	1	Yes	MR	-0.62	-0.62	-0.68	-0.68
RBC	5y	2007-12-31_2009-12-31	30b	1	Yes	MR	-0.68	-0.68	-0.72	-0.72
RBC	10y	2007-12-31_2009-12-31	30b	1	Yes	MR	-0.71	-0.71	-0.8	-0.8
RBC	10y	2007-12-31_2009-12-31	30b	2	No	MR	-0.72	-0.72	-0.54	-0.54
RBC	10y	2007-12-31_2009-12-31	10b	1	Yes	MR	-0.73	-0.73	-0.78	-0.78
RBC	5y	2007-12-31_2009-12-31	20b	2	No	MR	-0.73	-0.73	-0.54	-0.54
RBC	5y	2007-12-31_2009-12-31	10b	2	No	MR	-0.79	-0.79	-0.59	-0.58
RBC	10y	2007-12-31_2009-12-31	10b	2	No	MR	-0.84	-0.83	-0.62	-0.62
RBC	5y	2007-12-31_2009-12-31	30b	2	No	MR	-0.84	-0.84	-0.63	-0.63
RBC	5y	2007-12-31_2009-12-31	20b	2	Yes	MR	-1.16	-1.16	-0.62	-0.62
RBC	5y	2007-12-31_2009-12-31	10b	2	Yes	MR	-1.19	-1.19	-0.62	-0.62
RBC	10y	2007-12-31_2009-12-31	20b	2	Yes	MR	-1.24	-1.24	-0.68	-0.68
RBC	5y	2007-12-31_2009-12-31	30b	2	Yes	MR	-1.35	-1.35	-0.72	-0.72
RBC	10y	2007-12-31_2009-12-31	30b	2	Yes	MR	-1.42	-1.42	-0.8	-0.8

RBC	10y	2007-12-31_2009-12-31	10b	2	Yes	MR	-1.45	-1.45	-0.78	-0.78

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2009-12-31_2019-12-31	10b	2	Yes	MR	5.19	5.19	0.85	0.85
RBC	10y	2009-12-31_2019-12-31	30b	2	Yes	MR	5.04	5.04	0.83	0.83
RBC	20y	2009-12-31_2019-12-31	10b	2	Yes	MR	5.02	5.02	0.83	0.83
RBC	10y	2009-12-31_2019-12-31	20b	2	Yes	MR	4.81	4.81	0.79	0.79
RBC	20y	2009-12-31_2019-12-31	20b	2	Yes	MR	4.6	4.6	0.77	0.77
RBC	20y	2009-12-31_2019-12-31	30b	2	Yes	MR	4.43	4.43	0.74	0.74
RBC	5y	2009-12-31_2019-12-31	30b	2	Yes	MR	4.17	4.17	0.7	0.71
RBC	5y	2009-12-31_2019-12-31	10b	2	Yes	MR	4.14	4.14	0.7	0.7
RBC	5y	2009-12-31_2019-12-31	20b	2	Yes	MR	3.95	3.95	0.67	0.67
RBC	20y	2009-12-31_2019-12-31	10b	2	No	MR	2.86	2.86	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	10b	2	No	MR	2.85	2.85	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	30b	2	No	MR	2.79	2.79	0.81	0.81
RBC	10y	2009-12-31_2019-12-31	10b	1	Yes	MR	2.6	2.6	0.85	0.85
RBC	10y	2009-12-31_2019-12-31	20b	2	No	MR	2.59	2.59	0.75	0.75
RBC	20y	2009-12-31_2019-12-31	20b	2	No	MR	2.56	2.56	0.74	0.74
RBC	10y	2009-12-31_2019-12-31	30b	1	Yes	MR	2.52	2.52	0.83	0.83
RBC	20y	2009-12-31_2019-12-31	10b	1	Yes	MR	2.51	2.51	0.83	0.83
RBC	20y	2009-12-31_2019-12-31	30b	2	No	MR	2.5	2.5	0.72	0.72
RBC	10y	2009-12-31_2019-12-31	20b	1	Yes	MR	2.4	2.4	0.79	0.79
RBC	5y	2009-12-31_2019-12-31	30b	2	No	MR	2.36	2.37	0.68	0.68
RBC	20y	2009-12-31_2019-12-31	20b	1	Yes	MR	2.3	2.3	0.77	0.77
RBC	20y	2009-12-31_2019-12-31	30b	1	Yes	MR	2.21	2.21	0.74	0.74
RBC	5y	2009-12-31_2019-12-31	10b	2	No	MR	2.12	2.12	0.61	0.61
RBC	5y	2009-12-31_2019-12-31	20b	2	No	MR	2.1	2.1	0.6	0.6
RBC	5y	2009-12-31_2019-12-31	30b	1	Yes	MR	2.08	2.09	0.7	0.71
RBC	5y	2009-12-31_2019-12-31	10b	1	Yes	MR	2.07	2.07	0.7	0.7
RBC	5y	2009-12-31_2019-12-31	20b	1	Yes	MR	1.98	1.98	0.67	0.67
RBC	20y	2009-12-31_2019-12-31	10b	1	No	MR	1.43	1.43	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MR	1.42	1.42	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.4	1.4	0.81	0.81
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MR	1.3	1.3	0.75	0.75
RBC	20y	2009-12-31_2019-12-31	20b	1	No	MR	1.28	1.28	0.74	0.74
RBC	20y	2009-12-31_2019-12-31	30b	1	No	MR	1.25	1.25	0.72	0.72

RBC	5y	2009-12-31_2019-12-31	30b	1	No	MR	1.18	1.18	0.68	0.68
RBC	5y	2009-12-31_2019-12-31	10b	1	No	MR	1.06	1.06	0.61	0.61
RBC	5y	2009-12-31_2019-12-31	20b	1	No	MR	1.05	1.05	0.6	0.6
BH	N/A	2009-12-31_2019-12-31	N/A	1	No	N/A	0.99	0.99	0.57	0.57
RBC	5y	2009-12-31_2019-12-31	20b	1	No	MOM	-1.05	-1.05	-0.6	-0.6
RBC	5y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.06	-1.06	-0.61	-0.61
RBC	5y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.18	-1.18	-0.68	-0.68
RBC	20y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.25	-1.25	-0.72	-0.72
RBC	20y	2009-12-31_2019-12-31	20b	1	No	MOM	-1.28	-1.28	-0.74	-0.74
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MOM	-1.3	-1.3	-0.75	-0.75
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.4	-1.4	-0.81	-0.81
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.42	-1.42	-0.82	-0.82
RBC	20y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.43	-1.43	-0.82	-0.82
RBC	5y	2009-12-31_2019-12-31	20b	1	Yes	MOM	-1.98	-1.98	-0.67	-0.67
RBC	5y	2009-12-31_2019-12-31	10b	1	Yes	MOM	-2.07	-2.07	-0.7	-0.7
RBC	5y	2009-12-31_2019-12-31	30b	1	Yes	MOM	-2.09	-2.09	-0.71	-0.71
RBC	5y	2009-12-31_2019-12-31	20b	2	No	MOM	-2.1	-2.1	-0.6	-0.6
RBC	5y	2009-12-31_2019-12-31	10b	2	No	MOM	-2.12	-2.12	-0.61	-0.61
RBC	20y	2009-12-31_2019-12-31	30b	1	Yes	MOM	-2.22	-2.21	-0.74	-0.74
RBC	20y	2009-12-31_2019-12-31	20b	1	Yes	MOM	-2.3	-2.3	-0.77	-0.77
RBC	5y	2009-12-31_2019-12-31	30b	2	No	MOM	-2.37	-2.37	-0.68	-0.68
RBC	10y	2009-12-31_2019-12-31	20b	1	Yes	MOM	-2.4	-2.4	-0.79	-0.79
RBC	20y	2009-12-31_2019-12-31	30b	2	No	MOM	-2.5	-2.5	-0.72	-0.72
RBC	20y	2009-12-31_2019-12-31	10b	1	Yes	MOM	-2.51	-2.51	-0.84	-0.83
RBC	10y	2009-12-31_2019-12-31	30b	1	Yes	MOM	-2.52	-2.52	-0.83	-0.83
RBC	20y	2009-12-31_2019-12-31	20b	2	No	MOM	-2.56	-2.56	-0.74	-0.74
RBC	10y	2009-12-31_2019-12-31	20b	2	No	MOM	-2.59	-2.59	-0.75	-0.75
RBC	10y	2009-12-31_2019-12-31	10b	1	Yes	MOM	-2.6	-2.6	-0.85	-0.85
RBC	10y	2009-12-31_2019-12-31	30b	2	No	MOM	-2.79	-2.79	-0.81	-0.81
RBC	10y	2009-12-31_2019-12-31	10b	2	No	MOM	-2.85	-2.85	-0.82	-0.82
RBC	20y	2009-12-31_2019-12-31	10b	2	No	MOM	-2.86	-2.86	-0.82	-0.82
RBC	5y	2009-12-31_2019-12-31	20b	2	Yes	MOM	-3.96	-3.95	-0.67	-0.67
RBC	5y	2009-12-31_2019-12-31	10b	2	Yes	MOM	-4.14	-4.14	-0.7	-0.7
RBC	5y	2009-12-31_2019-12-31	30b	2	Yes	MOM	-4.18	-4.17	-0.71	-0.71
RBC	20y	2009-12-31_2019-12-31	30b	2	Yes	MOM	-4.43	-4.43	-0.74	-0.74
RBC	20y	2009-12-31_2019-12-31	20b	2	Yes	MOM	-4.6	-4.6	-0.77	-0.77
RBC	10y	2009-12-31_2019-12-31	20b	2	Yes	MOM	-4.81	-4.81	-0.79	-0.79
RBC	20y	2009-12-31_2019-12-31	10b	2	Yes	MOM	-5.02	-5.02	-0.84	-0.83
RBC	10y	2009-12-31_2019-12-31	30b	2	Yes	MOM	-5.04	-5.04	-0.83	-0.83

RBC	10y	2009-12-31_2019-12-31	10b	2	Yes	MOM	-5.2	-5.19	-0.85	-0.85

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2009-12-31_2019-12-31	10b	2	Yes	MR	4.86	4.86	0.82	0.82
RBC	20y	2009-12-31_2019-12-31	10b	2	Yes	MR	4.77	4.77	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	30b	2	Yes	MR	4.51	4.52	0.77	0.77
RBC	10y	2009-12-31_2019-12-31	20b	2	Yes	MR	4.37	4.37	0.74	0.74
RBC	20y	2009-12-31_2019-12-31	20b	2	Yes	MR	4.25	4.26	0.73	0.73
RBC	20y	2009-12-31_2019-12-31	30b	2	Yes	MR	3.96	3.96	0.69	0.69
RBC	5y	2009-12-31_2019-12-31	10b	2	Yes	MR	3.86	3.87	0.67	0.67
RBC	5y	2009-12-31_2019-12-31	30b	2	Yes	MR	3.78	3.78	0.66	0.66
RBC	5y	2009-12-31_2019-12-31	20b	2	Yes	MR	3.62	3.62	0.63	0.63
RBC	20y	2009-12-31_2019-12-31	10b	2	No	MR	2.78	2.78	0.81	0.81
RBC	10y	2009-12-31_2019-12-31	10b	2	No	MR	2.74	2.74	0.8	0.8
RBC	10y	2009-12-31_2019-12-31	30b	2	No	MR	2.54	2.54	0.74	0.74
RBC	10y	2009-12-31_2019-12-31	10b	1	Yes	MR	2.43	2.43	0.82	0.82
RBC	20y	2009-12-31_2019-12-31	20b	2	No	MR	2.42	2.43	0.71	0.71
RBC	10y	2009-12-31_2019-12-31	20b	2	No	MR	2.41	2.42	0.7	0.71
RBC	20y	2009-12-31_2019-12-31	10b	1	Yes	MR	2.38	2.38	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	30b	1	Yes	MR	2.26	2.26	0.77	0.77
RBC	20y	2009-12-31_2019-12-31	30b	2	No	MR	2.25	2.25	0.66	0.66
RBC	5y	2009-12-31_2019-12-31	30b	2	No	MR	2.2	2.2	0.64	0.64
RBC	10y	2009-12-31_2019-12-31	20b	1	Yes	MR	2.19	2.19	0.74	0.74
RBC	20y	2009-12-31_2019-12-31	20b	1	Yes	MR	2.13	2.13	0.73	0.73
RBC	5y	2009-12-31_2019-12-31	10b	2	No	MR	2.06	2.06	0.6	0.6
RBC	20y	2009-12-31_2019-12-31	30b	1	Yes	MR	1.98	1.98	0.69	0.69
RBC	5y	2009-12-31_2019-12-31	20b	2	No	MR	1.97	1.97	0.57	0.58
RBC	5y	2009-12-31_2019-12-31	10b	1	Yes	MR	1.93	1.93	0.67	0.67
RBC	5y	2009-12-31_2019-12-31	30b	1	Yes	MR	1.89	1.89	0.66	0.66
RBC	5y	2009-12-31_2019-12-31	20b	1	Yes	MR	1.81	1.81	0.63	0.63
RBC	20y	2009-12-31_2019-12-31	10b	1	No	MR	1.39	1.39	0.81	0.81
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MR	1.37	1.37	0.8	0.8
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.27	1.27	0.74	0.74
RBC	20y	2009-12-31_2019-12-31	20b	1	No	MR	1.21	1.21	0.71	0.71
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MR	1.21	1.21	0.7	0.71
RBC	20y	2009-12-31_2019-12-31	30b	1	No	MR	1.12	1.12	0.66	0.66

RBC	5y	2009-12-31_2019-12-31	30b	1	No	MR	1.1	1.1	0.64	0.64
RBC	5y	2009-12-31_2019-12-31	10b	1	No	MR	1.03	1.03	0.6	0.6
RBC	5y	2009-12-31_2019-12-31	20b	1	No	MR	0.98	0.99	0.57	0.58
BH	N/A	2009-12-31_2019-12-31	N/A	1	No	N/A	0.89	0.89	0.52	0.52
RBC	5y	2009-12-31_2019-12-31	20b	1	No	MOM	-0.99	-0.99	-0.58	-0.58
RBC	5y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.03	-1.03	-0.6	-0.6
RBC	5y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.1	-1.1	-0.65	-0.64
RBC	20y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.13	-1.12	-0.66	-0.66
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MOM	-1.21	-1.21	-0.71	-0.71
RBC	20y	2009-12-31_2019-12-31	20b	1	No	MOM	-1.21	-1.21	-0.71	-0.71
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.27	-1.27	-0.74	-0.74
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.37	-1.37	-0.8	-0.8
RBC	20y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.39	-1.39	-0.81	-0.81
RBC	5y	2009-12-31_2019-12-31	20b	1	Yes	MOM	-1.81	-1.81	-0.63	-0.63
RBC	5y	2009-12-31_2019-12-31	30b	1	Yes	MOM	-1.89	-1.89	-0.66	-0.66
RBC	5y	2009-12-31_2019-12-31	10b	1	Yes	MOM	-1.93	-1.93	-0.67	-0.67
RBC	5y	2009-12-31_2019-12-31	20b	2	No	MOM	-1.97	-1.97	-0.58	-0.58
RBC	20y	2009-12-31_2019-12-31	30b	1	Yes	MOM	-1.98	-1.98	-0.69	-0.69
RBC	5y	2009-12-31_2019-12-31	10b	2	No	MOM	-2.06	-2.06	-0.6	-0.6
RBC	20y	2009-12-31_2019-12-31	20b	1	Yes	MOM	-2.13	-2.13	-0.73	-0.73
RBC	10y	2009-12-31_2019-12-31	20b	1	Yes	MOM	-2.19	-2.19	-0.74	-0.74
RBC	5y	2009-12-31_2019-12-31	30b	2	No	MOM	-2.21	-2.2	-0.65	-0.64
RBC	20y	2009-12-31_2019-12-31	30b	2	No	MOM	-2.25	-2.25	-0.66	-0.66
RBC	10y	2009-12-31_2019-12-31	30b	1	Yes	MOM	-2.26	-2.26	-0.77	-0.77
RBC	20y	2009-12-31_2019-12-31	10b	1	Yes	MOM	-2.38	-2.38	-0.82	-0.82
RBC	10y	2009-12-31_2019-12-31	20b	2	No	MOM	-2.42	-2.42	-0.71	-0.71
RBC	20y	2009-12-31_2019-12-31	20b	2	No	MOM	-2.43	-2.43	-0.71	-0.71
RBC	10y	2009-12-31_2019-12-31	10b	1	Yes	MOM	-2.43	-2.43	-0.82	-0.82
RBC	10y	2009-12-31_2019-12-31	30b	2	No	MOM	-2.54	-2.54	-0.74	-0.74
RBC	10y	2009-12-31_2019-12-31	10b	2	No	MOM	-2.74	-2.74	-0.8	-0.8
RBC	20y	2009-12-31_2019-12-31	10b	2	No	MOM	-2.78	-2.78	-0.81	-0.81
RBC	5y	2009-12-31_2019-12-31	20b	2	Yes	MOM	-3.63	-3.62	-0.63	-0.63
RBC	5y	2009-12-31_2019-12-31	30b	2	Yes	MOM	-3.78	-3.78	-0.66	-0.66
RBC	5y	2009-12-31_2019-12-31	10b	2	Yes	MOM	-3.87	-3.87	-0.67	-0.67
RBC	20y	2009-12-31_2019-12-31	30b	2	Yes	MOM	-3.96	-3.96	-0.69	-0.69
RBC	20y	2009-12-31_2019-12-31	20b	2	Yes	MOM	-4.26	-4.26	-0.73	-0.73
RBC	10y	2009-12-31_2019-12-31	20b	2	Yes	MOM	-4.37	-4.37	-0.74	-0.74
RBC	10y	2009-12-31_2019-12-31	30b	2	Yes	MOM	-4.52	-4.52	-0.77	-0.77
RBC	20y	2009-12-31_2019-12-31	10b	2	Yes	MOM	-4.77	-4.77	-0.82	-0.82

RBC	10y	2009-12-31_2019-12-31	10b	2	Yes	MOM	-4.86	-4.86	-0.82	-0.82

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Model	**Data Length**	**Horizon**	**Signal Lag**	**Leverage**	**Double Down**	**Strategy Type**	**Net PnL**	**Gross PnL**	**Net SR**	**Gross SR**
RBC	5y	2019-12-31_2022-12-31	20b	2	Yes	MOM	1.43	1.44	0.55	0.55
RBC	10y	2019-12-31_2022-12-31	20b	2	Yes	MOM	1.43	1.44	0.55	0.55
RBC	20y	2019-12-31_2022-12-31	20b	2	Yes	MOM	1.43	1.44	0.55	0.55
RBC	5y	2019-12-31_2022-12-31	10b	2	Yes	MOM	1.33	1.33	0.51	0.51
RBC	10y	2019-12-31_2022-12-31	10b	2	Yes	MOM	1.33	1.33	0.51	0.51
RBC	20y	2019-12-31_2022-12-31	10b	2	Yes	MOM	1.33	1.33	0.51	0.51
RBC	5y	2019-12-31_2022-12-31	30b	2	Yes	MOM	1.22	1.22	0.47	0.47
RBC	10y	2019-12-31_2022-12-31	30b	2	Yes	MOM	1.22	1.22	0.47	0.47
RBC	20y	2019-12-31_2022-12-31	30b	2	Yes	MOM	1.22	1.22	0.47	0.47
RBC	5y	2019-12-31_2022-12-31	20b	1	Yes	MOM	0.72	0.72	0.55	0.55
RBC	10y	2019-12-31_2022-12-31	20b	1	Yes	MOM	0.72	0.72	0.55	0.55
RBC	20y	2019-12-31_2022-12-31	20b	1	Yes	MOM	0.72	0.72	0.55	0.55
RBC	5y	2019-12-31_2022-12-31	10b	1	Yes	MOM	0.67	0.67	0.51	0.51
RBC	10y	2019-12-31_2022-12-31	10b	1	Yes	MOM	0.67	0.67	0.51	0.51
RBC	20y	2019-12-31_2022-12-31	10b	1	Yes	MOM	0.67	0.67	0.51	0.51
RBC	5y	2019-12-31_2022-12-31	30b	1	Yes	MOM	0.61	0.61	0.47	0.47
RBC	10y	2019-12-31_2022-12-31	30b	1	Yes	MOM	0.61	0.61	0.47	0.47
RBC	20y	2019-12-31_2022-12-31	30b	1	Yes	MOM	0.61	0.61	0.47	0.47
RBC	5y	2019-12-31_2022-12-31	20b	2	No	MOM	0.57	0.57	0.35	0.35
RBC	10y	2019-12-31_2022-12-31	20b	2	No	MOM	0.57	0.57	0.35	0.35
RBC	20y	2019-12-31_2022-12-31	20b	2	No	MOM	0.57	0.57	0.35	0.35
RBC	5y	2019-12-31_2022-12-31	10b	2	No	MOM	0.51	0.51	0.32	0.32
RBC	10y	2019-12-31_2022-12-31	10b	2	No	MOM	0.51	0.51	0.32	0.32
RBC	20y	2019-12-31_2022-12-31	10b	2	No	MOM	0.51	0.51	0.32	0.32
RBC	5y	2019-12-31_2022-12-31	30b	2	No	MOM	0.4	0.4	0.25	0.25
RBC	10y	2019-12-31_2022-12-31	30b	2	No	MOM	0.4	0.4	0.25	0.25
RBC	20y	2019-12-31_2022-12-31	30b	2	No	MOM	0.4	0.4	0.25	0.25
BH	N/A	2019-12-31_2022-12-31	N/A	1	No	N/A	0.34	0.34	0.42	0.42
RBC	5y	2019-12-31_2022-12-31	20b	1	No	MOM	0.29	0.29	0.35	0.35
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MOM	0.29	0.29	0.35	0.35
RBC	20y	2019-12-31_2022-12-31	20b	1	No	MOM	0.29	0.29	0.35	0.35
RBC	5y	2019-12-31_2022-12-31	10b	1	No	MOM	0.25	0.26	0.32	0.32
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MOM	0.25	0.26	0.32	0.32

RBC	20y	2019-12-31_2022-12-31	10b	1	No	MOM	0.25	0.26	0.32	0.32
RBC	5y	2019-12-31_2022-12-31	30b	1	No	MOM	0.2	0.2	0.25	0.25
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MOM	0.2	0.2	0.25	0.25
RBC	20y	2019-12-31_2022-12-31	30b	1	No	MOM	0.2	0.2	0.25	0.25
RBC	5y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	20y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	5y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32
RBC	20y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32
RBC	5y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35
RBC	20y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35
RBC	5y	2019-12-31_2022-12-31	30b	2	No	MR	-0.4	-0.4	-0.25	-0.25
RBC	10y	2019-12-31_2022-12-31	30b	2	No	MR	-0.4	-0.4	-0.25	-0.25
RBC	20y	2019-12-31_2022-12-31	30b	2	No	MR	-0.4	-0.4	-0.25	-0.25
RBC	5y	2019-12-31_2022-12-31	10b	2	No	MR	-0.51	-0.51	-0.32	-0.32
RBC	10y	2019-12-31_2022-12-31	10b	2	No	MR	-0.51	-0.51	-0.32	-0.32
RBC	20y	2019-12-31_2022-12-31	10b	2	No	MR	-0.51	-0.51	-0.32	-0.32
RBC	5y	2019-12-31_2022-12-31	20b	2	No	MR	-0.57	-0.57	-0.35	-0.35
RBC	10y	2019-12-31_2022-12-31	20b	2	No	MR	-0.57	-0.57	-0.35	-0.35
RBC	20y	2019-12-31_2022-12-31	20b	2	No	MR	-0.57	-0.57	-0.35	-0.35
RBC	5y	2019-12-31_2022-12-31	30b	1	Yes	MR	-0.61	-0.61	-0.47	-0.47
RBC	10y	2019-12-31_2022-12-31	30b	1	Yes	MR	-0.61	-0.61	-0.47	-0.47
RBC	20y	2019-12-31_2022-12-31	30b	1	Yes	MR	-0.61	-0.61	-0.47	-0.47
RBC	5y	2019-12-31_2022-12-31	10b	1	Yes	MR	-0.67	-0.67	-0.51	-0.51
RBC	10y	2019-12-31_2022-12-31	10b	1	Yes	MR	-0.67	-0.67	-0.51	-0.51
RBC	20y	2019-12-31_2022-12-31	10b	1	Yes	MR	-0.67	-0.67	-0.51	-0.51
RBC	5y	2019-12-31_2022-12-31	20b	1	Yes	MR	-0.72	-0.72	-0.55	-0.55
RBC	10y	2019-12-31_2022-12-31	20b	1	Yes	MR	-0.72	-0.72	-0.55	-0.55
RBC	20y	2019-12-31_2022-12-31	20b	1	Yes	MR	-0.72	-0.72	-0.55	-0.55
RBC	5y	2019-12-31_2022-12-31	30b	2	Yes	MR	-1.22	-1.22	-0.47	-0.47
RBC	10y	2019-12-31_2022-12-31	30b	2	Yes	MR	-1.22	-1.22	-0.47	-0.47
RBC	20y	2019-12-31_2022-12-31	30b	2	Yes	MR	-1.22	-1.22	-0.47	-0.47
RBC	5y	2019-12-31_2022-12-31	10b	2	Yes	MR	-1.33	-1.33	-0.51	-0.51
RBC	10y	2019-12-31_2022-12-31	10b	2	Yes	MR	-1.33	-1.33	-0.51	-0.51
RBC	20y	2019-12-31_2022-12-31	10b	2	Yes	MR	-1.33	-1.33	-0.51	-0.51
RBC	5y	2019-12-31_2022-12-31	20b	2	Yes	MR	-1.44	-1.44	-0.55	-0.55
RBC	10y	2019-12-31_2022-12-31	20b	2	Yes	MR	-1.44	-1.44	-0.55	-0.55

RBC	20y	2019-12-31_2022-12-31	20b	2	Yes	MR	-1.44	-1.44	-0.55	-0.55

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	2019-12-31_2022-12-31	20b	2	Yes	MOM	1.2	1.2	0.5	0.5
RBC	10y	2019-12-31_2022-12-31	20b	2	Yes	MOM	1.2	1.2	0.5	0.5
RBC	20y	2019-12-31_2022-12-31	20b	2	Yes	MOM	1.2	1.2	0.5	0.5
RBC	5y	2019-12-31_2022-12-31	10b	2	Yes	MOM	1.18	1.19	0.48	0.48
RBC	10y	2019-12-31_2022-12-31	10b	2	Yes	MOM	1.18	1.19	0.48	0.48
RBC	20y	2019-12-31_2022-12-31	10b	2	Yes	MOM	1.18	1.19	0.48	0.48
RBC	5y	2019-12-31_2022-12-31	30b	2	Yes	MOM	1.17	1.18	0.49	0.49
RBC	10y	2019-12-31_2022-12-31	30b	2	Yes	MOM	1.17	1.18	0.49	0.49
RBC	20y	2019-12-31_2022-12-31	30b	2	Yes	MOM	1.17	1.18	0.49	0.49
RBC	5y	2019-12-31_2022-12-31	20b	1	Yes	MOM	0.6	0.6	0.5	0.5
RBC	10y	2019-12-31_2022-12-31	20b	1	Yes	MOM	0.6	0.6	0.5	0.5
RBC	20y	2019-12-31_2022-12-31	20b	1	Yes	MOM	0.6	0.6	0.5	0.5
RBC	5y	2019-12-31_2022-12-31	10b	1	Yes	MOM	0.59	0.59	0.48	0.48
RBC	10y	2019-12-31_2022-12-31	10b	1	Yes	MOM	0.59	0.59	0.48	0.48
RBC	20y	2019-12-31_2022-12-31	10b	1	Yes	MOM	0.59	0.59	0.48	0.48
RBC	5y	2019-12-31_2022-12-31	30b	1	Yes	MOM	0.59	0.59	0.49	0.49
RBC	10y	2019-12-31_2022-12-31	30b	1	Yes	MOM	0.59	0.59	0.49	0.49
RBC	20y	2019-12-31_2022-12-31	30b	1	Yes	MOM	0.59	0.59	0.49	0.49
RBC	5y	2019-12-31_2022-12-31	20b	2	No	MOM	0.44	0.44	0.27	0.27
RBC	10y	2019-12-31_2022-12-31	20b	2	No	MOM	0.44	0.44	0.27	0.27
RBC	20y	2019-12-31_2022-12-31	20b	2	No	MOM	0.44	0.44	0.27	0.27
RBC	5y	2019-12-31_2022-12-31	10b	2	No	MOM	0.41	0.41	0.26	0.26
RBC	10y	2019-12-31_2022-12-31	10b	2	No	MOM	0.41	0.41	0.26	0.26
RBC	20y	2019-12-31_2022-12-31	10b	2	No	MOM	0.41	0.41	0.26	0.26
RBC	5y	2019-12-31_2022-12-31	30b	2	No	MOM	0.37	0.37	0.23	0.23
RBC	10y	2019-12-31_2022-12-31	30b	2	No	MOM	0.37	0.37	0.23	0.23
RBC	20y	2019-12-31_2022-12-31	30b	2	No	MOM	0.37	0.37	0.23	0.23
BH	N/A	2019-12-31_2022-12-31	N/A	1	No	N/A	0.25	0.25	0.32	0.32
RBC	5y	2019-12-31_2022-12-31	20b	1	No	MOM	0.22	0.22	0.27	0.27
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MOM	0.22	0.22	0.27	0.27
RBC	20y	2019-12-31_2022-12-31	20b	1	No	MOM	0.22	0.22	0.27	0.27
RBC	5y	2019-12-31_2022-12-31	10b	1	No	MOM	0.21	0.21	0.26	0.26
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MOM	0.21	0.21	0.26	0.26

RBC	20y	2019-12-31_2022-12-31	10b	1	No	MOM	0.21	0.21	0.26	0.26
RBC	5y	2019-12-31_2022-12-31	30b	1	No	MOM	0.19	0.19	0.23	0.23
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MOM	0.19	0.19	0.23	0.23
RBC	20y	2019-12-31_2022-12-31	30b	1	No	MOM	0.19	0.19	0.23	0.23
RBC	5y	2019-12-31_2022-12-31	30b	1	No	MR	-0.19	-0.19	-0.23	-0.23
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.19	-0.19	-0.23	-0.23
RBC	20y	2019-12-31_2022-12-31	30b	1	No	MR	-0.19	-0.19	-0.23	-0.23
RBC	5y	2019-12-31_2022-12-31	10b	1	No	MR	-0.21	-0.21	-0.26	-0.26
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MR	-0.21	-0.21	-0.26	-0.26
RBC	20y	2019-12-31_2022-12-31	10b	1	No	MR	-0.21	-0.21	-0.26	-0.26
RBC	5y	2019-12-31_2022-12-31	20b	1	No	MR	-0.22	-0.22	-0.27	-0.27
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MR	-0.22	-0.22	-0.27	-0.27
RBC	20y	2019-12-31_2022-12-31	20b	1	No	MR	-0.22	-0.22	-0.27	-0.27
RBC	5y	2019-12-31_2022-12-31	30b	2	No	MR	-0.37	-0.37	-0.23	-0.23
RBC	10y	2019-12-31_2022-12-31	30b	2	No	MR	-0.37	-0.37	-0.23	-0.23
RBC	20y	2019-12-31_2022-12-31	30b	2	No	MR	-0.37	-0.37	-0.23	-0.23
RBC	5y	2019-12-31_2022-12-31	10b	2	No	MR	-0.41	-0.41	-0.26	-0.26
RBC	10y	2019-12-31_2022-12-31	10b	2	No	MR	-0.41	-0.41	-0.26	-0.26
RBC	20y	2019-12-31_2022-12-31	10b	2	No	MR	-0.41	-0.41	-0.26	-0.26
RBC	5y	2019-12-31_2022-12-31	20b	2	No	MR	-0.44	-0.44	-0.27	-0.27
RBC	10y	2019-12-31_2022-12-31	20b	2	No	MR	-0.44	-0.44	-0.27	-0.27
RBC	20y	2019-12-31_2022-12-31	20b	2	No	MR	-0.44	-0.44	-0.27	-0.27
RBC	5y	2019-12-31_2022-12-31	30b	1	Yes	MR	-0.59	-0.59	-0.49	-0.49
RBC	10y	2019-12-31_2022-12-31	30b	1	Yes	MR	-0.59	-0.59	-0.49	-0.49
RBC	20y	2019-12-31_2022-12-31	30b	1	Yes	MR	-0.59	-0.59	-0.49	-0.49
RBC	5y	2019-12-31_2022-12-31	10b	1	Yes	MR	-0.59	-0.59	-0.48	-0.48
RBC	10y	2019-12-31_2022-12-31	10b	1	Yes	MR	-0.59	-0.59	-0.48	-0.48
RBC	20y	2019-12-31_2022-12-31	10b	1	Yes	MR	-0.59	-0.59	-0.48	-0.48
RBC	5y	2019-12-31_2022-12-31	20b	1	Yes	MR	-0.6	-0.6	-0.5	-0.5
RBC	10y	2019-12-31_2022-12-31	20b	1	Yes	MR	-0.6	-0.6	-0.5	-0.5
RBC	20y	2019-12-31_2022-12-31	20b	1	Yes	MR	-0.6	-0.6	-0.5	-0.5
RBC	5y	2019-12-31_2022-12-31	30b	2	Yes	MR	-1.18	-1.18	-0.49	-0.49
RBC	10y	2019-12-31_2022-12-31	30b	2	Yes	MR	-1.18	-1.18	-0.49	-0.49
RBC	20y	2019-12-31_2022-12-31	30b	2	Yes	MR	-1.18	-1.18	-0.49	-0.49
RBC	5y	2019-12-31_2022-12-31	10b	2	Yes	MR	-1.19	-1.19	-0.48	-0.48
RBC	10y	2019-12-31_2022-12-31	10b	2	Yes	MR	-1.19	-1.19	-0.48	-0.48
RBC	20y	2019-12-31_2022-12-31	10b	2	Yes	MR	-1.19	-1.19	-0.48	-0.48
RBC	5y	2019-12-31_2022-12-31	20b	2	Yes	MR	-1.21	-1.2	-0.5	-0.5
RBC	10y	2019-12-31_2022-12-31	20b	2	Yes	MR	-1.21	-1.2	-0.5	-0.5

RBC	20y	2019-12-31_2022-12-31	20b	2	Yes	MR	-1.21	-1.2	-0.5	-0.5

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2022-12-31_2025-09-30	30b	2	Yes	MR	0.8	0.8	0.69	0.69
RBC	20y	2022-12-31_2025-09-30	30b	2	Yes	MR	0.8	0.8	0.69	0.69
RBC	10y	2022-12-31_2025-09-30	30b	2	No	MR	0.64	0.64	0.76	0.76
RBC	20y	2022-12-31_2025-09-30	30b	2	No	MR	0.64	0.64	0.76	0.76
BH	N/A	2022-12-31_2025-09-30	N/A	1	No	N/A	0.62	0.62	1.37	1.37
RBC	10y	2022-12-31_2025-09-30	20b	2	Yes	MR	0.56	0.56	0.48	0.48
RBC	20y	2022-12-31_2025-09-30	20b	2	Yes	MR	0.56	0.56	0.48	0.48
RBC	10y	2022-12-31_2025-09-30	20b	2	No	MR	0.54	0.54	0.64	0.64
RBC	20y	2022-12-31_2025-09-30	20b	2	No	MR	0.54	0.54	0.64	0.64
RBC	5y	2022-12-31_2025-09-30	30b	2	Yes	MR	0.41	0.42	0.27	0.28
RBC	10y	2022-12-31_2025-09-30	30b	1	Yes	MR	0.4	0.4	0.69	0.69
RBC	20y	2022-12-31_2025-09-30	30b	1	Yes	MR	0.4	0.4	0.69	0.69
RBC	5y	2022-12-31_2025-09-30	30b	2	No	MR	0.36	0.36	0.43	0.43
RBC	10y	2022-12-31_2025-09-30	10b	2	Yes	MR	0.34	0.34	0.28	0.28
RBC	20y	2022-12-31_2025-09-30	10b	2	Yes	MR	0.34	0.34	0.28	0.28
RBC	10y	2022-12-31_2025-09-30	10b	2	No	MR	0.33	0.33	0.38	0.38
RBC	20y	2022-12-31_2025-09-30	10b	2	No	MR	0.33	0.33	0.38	0.38
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.32	0.76	0.76
RBC	20y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.32	0.76	0.76
RBC	10y	2022-12-31_2025-09-30	20b	1	Yes	MR	0.28	0.28	0.48	0.48
RBC	20y	2022-12-31_2025-09-30	20b	1	Yes	MR	0.28	0.28	0.48	0.48
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MR	0.27	0.27	0.64	0.64
RBC	20y	2022-12-31_2025-09-30	20b	1	No	MR	0.27	0.27	0.64	0.64
RBC	5y	2022-12-31_2025-09-30	30b	1	Yes	MR	0.21	0.21	0.27	0.28
RBC	5y	2022-12-31_2025-09-30	30b	1	No	MR	0.18	0.18	0.43	0.43
RBC	10y	2022-12-31_2025-09-30	10b	1	Yes	MR	0.17	0.17	0.28	0.28
RBC	20y	2022-12-31_2025-09-30	10b	1	Yes	MR	0.17	0.17	0.28	0.28
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MR	0.16	0.16	0.38	0.38
RBC	20y	2022-12-31_2025-09-30	10b	1	No	MR	0.16	0.16	0.38	0.38
RBC	5y	2022-12-31_2025-09-30	10b	2	Yes	MOM	0.1	0.1	0.06	0.06
RBC	5y	2022-12-31_2025-09-30	20b	2	No	MR	0.05	0.05	0.06	0.06
RBC	5y	2022-12-31_2025-09-30	10b	1	Yes	MOM	0.05	0.05	0.06	0.06
RBC	5y	2022-12-31_2025-09-30	10b	2	No	MOM	0.04	0.04	0.05	0.05

RBC	5y	2022-12-31_2025-09-30	20b	1	No	MR	0.03	0.03	0.06	0.06
RBC	5y	2022-12-31_2025-09-30	10b	1	No	MOM	0.02	0.02	0.05	0.05
RBC	5y	2022-12-31_2025-09-30	20b	2	Yes	MR	0	0	0	0
RBC	5y	2022-12-31_2025-09-30	20b	1	Yes	MR	0	0	0	0
RBC	5y	2022-12-31_2025-09-30	20b	1	Yes	MOM	0	0	0	0
RBC	5y	2022-12-31_2025-09-30	20b	2	Yes	MOM	0	0	0	0
RBC	5y	2022-12-31_2025-09-30	10b	1	No	MR	-0.02	-0.02	-0.05	-0.05
RBC	5y	2022-12-31_2025-09-30	20b	1	No	MOM	-0.03	-0.03	-0.06	-0.06
RBC	5y	2022-12-31_2025-09-30	10b	2	No	MR	-0.04	-0.04	-0.05	-0.05
RBC	5y	2022-12-31_2025-09-30	10b	1	Yes	MR	-0.05	-0.05	-0.07	-0.06
RBC	5y	2022-12-31_2025-09-30	20b	2	No	MOM	-0.05	-0.05	-0.06	-0.06
RBC	5y	2022-12-31_2025-09-30	10b	2	Yes	MR	-0.1	-0.1	-0.07	-0.06
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MOM	-0.16	-0.16	-0.38	-0.38
RBC	20y	2022-12-31_2025-09-30	10b	1	No	MOM	-0.16	-0.16	-0.38	-0.38
RBC	10y	2022-12-31_2025-09-30	10b	1	Yes	MOM	-0.17	-0.17	-0.29	-0.28
RBC	20y	2022-12-31_2025-09-30	10b	1	Yes	MOM	-0.17	-0.17	-0.29	-0.28
RBC	5y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.18	-0.18	-0.43	-0.43
RBC	5y	2022-12-31_2025-09-30	30b	1	Yes	MOM	-0.21	-0.21	-0.28	-0.28
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MOM	-0.27	-0.27	-0.64	-0.64
RBC	20y	2022-12-31_2025-09-30	20b	1	No	MOM	-0.27	-0.27	-0.64	-0.64
RBC	10y	2022-12-31_2025-09-30	20b	1	Yes	MOM	-0.28	-0.28	-0.48	-0.48
RBC	20y	2022-12-31_2025-09-30	20b	1	Yes	MOM	-0.28	-0.28	-0.48	-0.48
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.32	-0.32	-0.76	-0.76
RBC	20y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.32	-0.32	-0.76	-0.76
RBC	10y	2022-12-31_2025-09-30	10b	2	No	MOM	-0.33	-0.33	-0.38	-0.38
RBC	20y	2022-12-31_2025-09-30	10b	2	No	MOM	-0.33	-0.33	-0.38	-0.38
RBC	10y	2022-12-31_2025-09-30	10b	2	Yes	MOM	-0.34	-0.34	-0.29	-0.28
RBC	20y	2022-12-31_2025-09-30	10b	2	Yes	MOM	-0.34	-0.34	-0.29	-0.28
RBC	5y	2022-12-31_2025-09-30	30b	2	No	MOM	-0.36	-0.36	-0.43	-0.43
RBC	10y	2022-12-31_2025-09-30	30b	1	Yes	MOM	-0.4	-0.4	-0.69	-0.69
RBC	20y	2022-12-31_2025-09-30	30b	1	Yes	MOM	-0.4	-0.4	-0.69	-0.69
RBC	5y	2022-12-31_2025-09-30	30b	2	Yes	MOM	-0.42	-0.42	-0.28	-0.28
RBC	10y	2022-12-31_2025-09-30	20b	2	No	MOM	-0.55	-0.54	-0.64	-0.64
RBC	20y	2022-12-31_2025-09-30	20b	2	No	MOM	-0.55	-0.54	-0.64	-0.64
RBC	10y	2022-12-31_2025-09-30	20b	2	Yes	MOM	-0.56	-0.56	-0.48	-0.48
RBC	20y	2022-12-31_2025-09-30	20b	2	Yes	MOM	-0.56	-0.56	-0.48	-0.48
RBC	10y	2022-12-31_2025-09-30	30b	2	No	MOM	-0.64	-0.64	-0.76	-0.76
RBC	20y	2022-12-31_2025-09-30	30b	2	No	MOM	-0.64	-0.64	-0.76	-0.76
RBC	10y	2022-12-31_2025-09-30	30b	2	Yes	MOM	-0.8	-0.8	-0.69	-0.69

RBC	20y	2022-12-31_2025-09-30	30b	2	Yes	MOM	-0.8	-0.8	-0.69	-0.69

Dow Jones Industrial Average 12/2022-12/2025

Model	**Data Length**	**Horizon**	**Signal Lag**	**Leverage**	**Double Down**	**Strategy Type**	**Net PnL**	**Gross PnL**	**Net SR**	**Gross SR**
RBC	10y	2022-12-31_2025-09-30	30b	2	Yes	MR	0.88	0.89	0.86	0.86
RBC	20y	2022-12-31_2025-09-30	30b	2	Yes	MR	0.88	0.89	0.86	0.86
RBC	10y	2022-12-31_2025-09-30	30b	2	No	MR	0.71	0.71	0.95	0.95
RBC	20y	2022-12-31_2025-09-30	30b	2	No	MR	0.71	0.71	0.95	0.95
RBC	10y	2022-12-31_2025-09-30	20b	2	Yes	MR	0.5	0.5	0.47	0.47
RBC	20y	2022-12-31_2025-09-30	20b	2	Yes	MR	0.5	0.5	0.47	0.47
RBC	10y	2022-12-31_2025-09-30	30b	1	Yes	MR	0.44	0.44	0.86	0.86
RBC	20y	2022-12-31_2025-09-30	30b	1	Yes	MR	0.44	0.44	0.86	0.86
RBC	5y	2022-12-31_2025-09-30	30b	2	Yes	MR	0.43	0.44	0.33	0.33
RBC	10y	2022-12-31_2025-09-30	20b	2	No	MR	0.43	0.43	0.57	0.58
RBC	20y	2022-12-31_2025-09-30	20b	2	No	MR	0.43	0.43	0.57	0.58
BH	N/A	2022-12-31_2025-09-30	N/A	1	No	N/A	0.4	0.4	1	1
RBC	5y	2022-12-31_2025-09-30	30b	2	No	MR	0.37	0.37	0.5	0.5
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.36	0.36	0.95	0.95
RBC	20y	2022-12-31_2025-09-30	30b	1	No	MR	0.36	0.36	0.95	0.95
RBC	10y	2022-12-31_2025-09-30	10b	2	Yes	MR	0.29	0.29	0.26	0.26
RBC	20y	2022-12-31_2025-09-30	10b	2	Yes	MR	0.29	0.29	0.26	0.26
RBC	10y	2022-12-31_2025-09-30	20b	1	Yes	MR	0.25	0.25	0.47	0.47
RBC	20y	2022-12-31_2025-09-30	20b	1	Yes	MR	0.25	0.25	0.47	0.47
RBC	10y	2022-12-31_2025-09-30	10b	2	No	MR	0.24	0.24	0.31	0.32
RBC	20y	2022-12-31_2025-09-30	10b	2	No	MR	0.24	0.24	0.31	0.32
RBC	5y	2022-12-31_2025-09-30	30b	1	Yes	MR	0.22	0.22	0.33	0.33
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MR	0.22	0.22	0.57	0.58
RBC	20y	2022-12-31_2025-09-30	20b	1	No	MR	0.22	0.22	0.57	0.58
RBC	5y	2022-12-31_2025-09-30	30b	1	No	MR	0.19	0.19	0.5	0.5
RBC	10y	2022-12-31_2025-09-30	10b	1	Yes	MR	0.14	0.14	0.26	0.26
RBC	20y	2022-12-31_2025-09-30	10b	1	Yes	MR	0.14	0.14	0.26	0.26
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MR	0.12	0.12	0.31	0.32
RBC	20y	2022-12-31_2025-09-30	10b	1	No	MR	0.12	0.12	0.31	0.32
RBC	5y	2022-12-31_2025-09-30	20b	2	Yes	MOM	0.09	0.1	0.07	0.07
RBC	5y	2022-12-31_2025-09-30	20b	2	No	MOM	0.08	0.08	0.11	0.11
RBC	5y	2022-12-31_2025-09-30	10b	2	Yes	MOM	0.06	0.06	0.05	0.05
RBC	5y	2022-12-31_2025-09-30	20b	1	Yes	MOM	0.05	0.05	0.07	0.07

RBC	5y	2022-12-31_2025-09-30	20b	1	No	MOM	0.04	0.04	0.11	0.11
RBC	5y	2022-12-31_2025-09-30	10b	1	Yes	MOM	0.03	0.03	0.05	0.05
RBC	5y	2022-12-31_2025-09-30	10b	2	No	MOM	0.02	0.02	0.02	0.02
RBC	5y	2022-12-31_2025-09-30	10b	1	No	MOM	0.01	0.01	0.02	0.02
RBC	5y	2022-12-31_2025-09-30	10b	1	No	MR	-0.01	-0.01	-0.02	-0.02
RBC	5y	2022-12-31_2025-09-30	10b	2	No	MR	-0.02	-0.02	-0.02	-0.02
RBC	5y	2022-12-31_2025-09-30	10b	1	Yes	MR	-0.03	-0.03	-0.05	-0.05
RBC	5y	2022-12-31_2025-09-30	20b	1	No	MR	-0.04	-0.04	-0.11	-0.11
RBC	5y	2022-12-31_2025-09-30	20b	1	Yes	MR	-0.05	-0.05	-0.07	-0.07
RBC	5y	2022-12-31_2025-09-30	10b	2	Yes	MR	-0.06	-0.06	-0.05	-0.05
RBC	5y	2022-12-31_2025-09-30	20b	2	No	MR	-0.09	-0.08	-0.11	-0.11
RBC	5y	2022-12-31_2025-09-30	20b	2	Yes	MR	-0.1	-0.1	-0.07	-0.07
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MOM	-0.12	-0.12	-0.32	-0.32
RBC	20y	2022-12-31_2025-09-30	10b	1	No	MOM	-0.12	-0.12	-0.32	-0.32
RBC	10y	2022-12-31_2025-09-30	10b	1	Yes	MOM	-0.15	-0.14	-0.27	-0.26
RBC	20y	2022-12-31_2025-09-30	10b	1	Yes	MOM	-0.15	-0.14	-0.27	-0.26
RBC	5y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.19	-0.19	-0.5	-0.5
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MOM	-0.22	-0.22	-0.58	-0.58
RBC	20y	2022-12-31_2025-09-30	20b	1	No	MOM	-0.22	-0.22	-0.58	-0.58
RBC	5y	2022-12-31_2025-09-30	30b	1	Yes	MOM	-0.22	-0.22	-0.33	-0.33
RBC	10y	2022-12-31_2025-09-30	10b	2	No	MOM	-0.24	-0.24	-0.32	-0.32
RBC	20y	2022-12-31_2025-09-30	10b	2	No	MOM	-0.24	-0.24	-0.32	-0.32
RBC	10y	2022-12-31_2025-09-30	20b	1	Yes	MOM	-0.25	-0.25	-0.47	-0.47
RBC	20y	2022-12-31_2025-09-30	20b	1	Yes	MOM	-0.25	-0.25	-0.47	-0.47
RBC	10y	2022-12-31_2025-09-30	10b	2	Yes	MOM	-0.29	-0.29	-0.27	-0.26
RBC	20y	2022-12-31_2025-09-30	10b	2	Yes	MOM	-0.29	-0.29	-0.27	-0.26
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.36	-0.36	-0.96	-0.95
RBC	20y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.36	-0.36	-0.96	-0.95
RBC	5y	2022-12-31_2025-09-30	30b	2	No	MOM	-0.37	-0.37	-0.5	-0.5
RBC	10y	2022-12-31_2025-09-30	20b	2	No	MOM	-0.43	-0.43	-0.58	-0.58
RBC	20y	2022-12-31_2025-09-30	20b	2	No	MOM	-0.43	-0.43	-0.58	-0.58
RBC	5y	2022-12-31_2025-09-30	30b	2	Yes	MOM	-0.44	-0.44	-0.33	-0.33
RBC	10y	2022-12-31_2025-09-30	30b	1	Yes	MOM	-0.44	-0.44	-0.86	-0.86
RBC	20y	2022-12-31_2025-09-30	30b	1	Yes	MOM	-0.44	-0.44	-0.86	-0.86
RBC	10y	2022-12-31_2025-09-30	20b	2	Yes	MOM	-0.5	-0.5	-0.47	-0.47
RBC	20y	2022-12-31_2025-09-30	20b	2	Yes	MOM	-0.5	-0.5	-0.47	-0.47
RBC	10y	2022-12-31_2025-09-30	30b	2	No	MOM	-0.72	-0.71	-0.96	-0.95
RBC	20y	2022-12-31_2025-09-30	30b	2	No	MOM	-0.72	-0.71	-0.96	-0.95
RBC	10y	2022-12-31_2025-09-30	30b	2	Yes	MOM	-0.89	-0.89	-0.86	-0.86

RBC	20y	2022-12-31_2025-09-30	30b	2	Yes	MOM	-0.89	-0.89	-0.86	-0.86

Appendix E: Strategy Results across Data Lengths

S&P500 30b Signal Lag All Horizons

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.16	1.16	0.23	0.23
RBC	5y	1999-12-31_2025-09-30	30b	1	No	MR	0.98	0.98	0.2	0.2
RBC	5y	1999-12-31_2007-12-31	30b	1	No	MR	0.56	0.57	0.39	0.39
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.27	0.27	0.19	0.19
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.5	-0.5	-0.67	-0.67
RBC	5y	2007-12-31_2009-12-31	30b	1	No	MR	-0.56	-0.56	-0.76	-0.76
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.4	1.4	0.81	0.81
RBC	20y	2009-12-31_2019-12-31	30b	1	No	MR	1.25	1.25	0.72	0.72
RBC	5y	2009-12-31_2019-12-31	30b	1	No	MR	1.18	1.18	0.68	0.68
RBC	5y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	20y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.32	0.76	0.76
RBC	20y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.32	0.76	0.76
RBC	5y	2022-12-31_2025-09-30	30b	1	No	MR	0.18	0.18	0.43	0.43

S&P500 20b Signal Lag All Horizons

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MR	0.92	0.92	0.18	0.18
RBC	5y	1999-12-31_2025-09-30	20b	1	No	MR	0.64	0.64	0.13	0.13
RBC	5y	1999-12-31_2007-12-31	20b	1	No	MR	0.48	0.48	0.33	0.33
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MR	0.16	0.16	0.11	0.11
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MR	-0.43	-0.43	-0.58	-0.58
RBC	5y	2007-12-31_2009-12-31	20b	1	No	MR	-0.5	-0.5	-0.68	-0.68
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MR	1.3	1.3	0.75	0.75
RBC	20y	2009-12-31_2019-12-31	20b	1	No	MR	1.28	1.28	0.74	0.74
RBC	5y	2009-12-31_2019-12-31	20b	1	No	MR	1.05	1.05	0.6	0.6
RBC	5y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35

RBC	20y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MR	0.27	0.27	0.64	0.64
RBC	20y	2022-12-31_2025-09-30	20b	1	No	MR	0.27	0.27	0.64	0.64
RBC	5y	2022-12-31_2025-09-30	20b	1	No	MR	0.03	0.03	0.06	0.06

S&P500 10b Signal Lag All Horizons

Model	**Data Length**	**Horizon**	**Signal Lag**	**Leverage**	**Double Down**	**Strategy Type**	**Net PnL**	**Gross PnL**	**Net SR**	**Gross SR**
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MR	0.51	0.51	0.1	0.1
RBC	5y	1999-12-31_2025-09-30	10b	1	No	MR	0.37	0.37	0.07	0.07
RBC	5y	1999-12-31_2007-12-31	10b	1	No	MR	0.26	0.26	0.18	0.18
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MR	-0.16	-0.16	-0.11	-0.11
RBC	5y	2007-12-31_2009-12-31	10b	1	No	MR	-0.47	-0.47	-0.64	-0.64
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MR	-0.54	-0.54	-0.72	-0.72
RBC	20y	2009-12-31_2019-12-31	10b	1	No	MR	1.43	1.43	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MR	1.42	1.42	0.82	0.82
RBC	5y	2009-12-31_2019-12-31	10b	1	No	MR	1.06	1.06	0.61	0.61
RBC	5y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32
RBC	20y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MR	0.16	0.16	0.38	0.38
RBC	20y	2022-12-31_2025-09-30	10b	1	No	MR	0.16	0.16	0.38	0.38
RBC	5y	2022-12-31_2025-09-30	10b	1	No	MR	-0.02	-0.02	-0.05	-0.05